

The New AMETRIC Boundary Endpoint of Black Holes

The Unresolved Horizon-Isolated Residue,
Einstein-Projection Boundedness, and No Inheritance under AASC

An AASC-Internal Einstein-Projection and Governance-Depth Closure Theorem

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Series position. This extension gives the AASC endpoint-normal-form analysis of *Black Holes as Licensed Limit Regimes* and *Standingless Interiors and Residual Coherence* for the unresolved horizon-isolated residue at maximum governance-depth. It imports the fixed-domain kernel, the independently constructed AMETRIC boundary normal form, the licensed Einstein metric projection, and the structural inadmissibility of the null regime. It proves that maximum governance-depth has New AMETRIC Boundary closure-status under the stated endpoint antecedent: AASC first licenses the metric regime, imports the minimal Einstein-form projection under the declared first-response same-scope metric-projection predicate, and then classifies the boundary at which that projection no longer has governance authority.

Abstract

This manuscript is an AASC-internal extension theorem that gives the endpoint-normal-form analysis of the *unresolved horizon-isolated residue* of a black-hole endpoint at maximum governance-depth. It imports as binding dependencies the fixed-domain kernel, the independently constructed AMETRIC boundary normal form, the AASC Einstein-projection theorem, horizon isolation and no-totalization, standingless-interior and residual-coherence results, matter-energy residualization, and the structural inadmissibility of the null regime. It does not rederive AASC from first principles. Its burden is fidelity to those results and exhaustion of same-type endpoint alternatives.

The Einstein-projection theorem supplies the metric bridge. General relativity is not treated as an independently primitive background that AASC later criticizes. Under the declared minimal first-response same-scope metric-projection predicate, the licensed field-equation form is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The endpoint analysis therefore classifies not an external failure of GR, but the boundary at which the licensed Einstein projection no longer has governance authority for the horizon-isolated residue [Maley, 2026c]. The bridge fixes the status of the metric regime; the present manuscript's independent burden remains endpoint classification under the maximum-governance-depth antecedent.

The theorem is not centered on literal horizon permanence or primitive information. In AASC, bare information-language is skin unless supported by reference, standing, carrier, witness, quotient identity, and admissible reconstruction. The conventional black-hole information paradox is therefore retyped as an exterior reference-standing reconstruction problem. Exterior radiation, Page-curve behavior, holographic descriptions, boundary records, and diagrams matter only at the role they actually carry. Skin remains useful for licensed exterior description but is role-demoted at endpoint standing. Genuine standing-bearing exterior witness relations resolve only the burdens they re-fix. The theorem classifies the unresolved horizon-isolated residue that remains.

Let B_H be the horizon-generated reference-standing burden, let $W_{\text{ext}}(B_H)$ be the portion resolved by standing-bearing exterior witness relations, and let

$$R_H = B_H \setminus W_{\text{ext}}(B_H)$$

be the unresolved horizon-isolated residue. Here R_H is role-ledger notation for unresolved reference-standing burden, not a primitive set of information tokens. If $R_H \neq \emptyset$ and residual support for the licensed metric projection exhausts at maximum governance-depth, the endpoint cannot coherently be classified as a classical singularity, remnant, information reservoir, dynamical bounce, wormhole passage, energy-transfer site, or admissible null endpoint. A singularity is a diagnosis internal to a still-operative metric articulation; maximum governance-depth is the loss of the residual role-support under which metric articulation can diagnose a limit. Storage and transfer require standing-bearing carrier identity and admissible continuation. Bounce and passage require temporal, causal, metric, and operator structure. Total vacuity cannot be an admissible closure state.

The endpoint therefore satisfies the independently fixed AMETRIC boundary normal form and, together with null-regime closure, the combined AASC endpoint-normal form. The closure-status of R_H is a New AMETRIC Boundary. This boundary is not the geometric horizon, not a spatial surface, and not a dynamical event. It is the structural endpoint exposed when the unresolved residue cannot continue, totalize, repair, support a singularity diagnostic, or terminate as admissible

nullity.

Nothing old-scope crosses the New AMETRIC Boundary. No standing, reference, energy or source ledger, matter, field, particle identity, metric structure, law, causal chain, memory, branch label, or global-state coordinate persists into a post-boundary domain as the same standing-bearing content. Parent-exterior ledgers remain licensed within their declared parent scope; what fails is their cross-AMetric extension. This is neither an information-destruction event nor an energy-creation, energy-destruction, or energy-nonconservation event. Those predicates require temporal, causal, metric, and ledger structure that has lost its cross-boundary conditions of application.

Because nothing crosses, AASC endpoint analysis of D_0 cannot establish the contents, laws, population, metricity, or concrete instantiation of any post-boundary domain. It can derive the endpoint status itself from exhaustion of old-scope role-support and establish origin-condition analogy: the New AMETRIC Boundary has the same structural role-profile as the AMETRIC starting condition of the realized physical interior. Here *existence-status* means structural domainhood under a domain's own governance; *populated-existence* means concrete internal contents, laws, fields, metricity, or physical occupancy. The result explains existence-status without inheritance and origin-condition homology without transmissive proof; it does not prove populated-existence beyond the boundary.

Main theorem and claim discipline

Principal theorem. Let D_0 be a fixed parent exterior domain whose AASC gravitational architecture is faithfully represented by the licensed Einstein projection

$$\Pi_g(A_G) = (M_4, g_{\mu\nu}), \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Let B_H be the reference-standing burden generated by horizon isolation, let $W_{\text{ext}}(B_H)$ be the portion resolved by standing-bearing exterior witness relations, and let $R_H = B_H \setminus W_{\text{ext}}(B_H)$ be the unresolved horizon-isolated residue. Suppose $R_H \neq \emptyset$, retained organization is role-indexed residual coherence, no independent governance is declared for the residue before metric-role exhaustion, and residual metric-comparison successor support exhausts at maximum governance-depth. Then the closure-status of R_H is a New AMETRIC Boundary rather than singularity, remnant, dynamical transition, storage, wormhole passage, transfer endpoint, or admissible nullity. No old-scope standing-bearing structure or Einstein source/energy ledger crosses that boundary. AASC endpoint analysis of D_0 derives the boundary status and its analogy to the AMETRIC starting condition of the realized physical interior, but cannot establish the internal contents or populated-existence of a post-boundary domain. This is endpoint classification under the stated exhaustion antecedent, not an empirical attainment theorem for every astrophysical black hole.

The result has seven logically distinct layers.

1. **Einstein-projection layer.** AASC first licenses metricity and imports the minimal Einstein-form projection under the declared first-response same-scope predicate. The endpoint theorem bounds this projection rather than rejecting general relativity from outside.
2. **Exterior role-typing layer.** Skin retains representational and calculational use; ledgers

retain their declared exterior roles; genuine witness relations resolve only the reference-standing burdens they re-fix.

3. **Residue layer.** The theorem object is R_H , the nonempty burden remaining after exterior resolution. Horizon duration is not the target variable.
4. **Endpoint layer.** When residual metric comparison exhausts, singularity, nullity, storage, bounce, throat, destruction, and cross-boundary conservation classifications fail because each requires a role whose domain of application has ended.
5. **Boundary layer.** The remaining endpoint profile satisfies the independently fixed AMETRIC boundary normal form. The New AMETRIC Boundary is structurally necessary, not dynamically formed.
6. **Derivability layer.** Endpoint-status is derivable by AASC analysis from old-scope role exhaustion; post-boundary contents are not derivable because doing so would require the crossing relation the boundary excludes.
7. **Origin-analogy layer.** The endpoint condition is structurally analogous to the AMETRIC starting condition of the realized physical interior. Any positive post-boundary domain, if independently admitted, would be newly governed rather than inherited.

AASC-internal status

This manuscript is an AASC-internal extension theorem. It imports the kernel, AMETRIC boundary theorem, Einstein-projection theorem, horizon-obstruction result, residual-coherence theorem, matter-energy residualization, and null-regime closure as binding dependencies. It does not undertake their rederivation. Its burden is endpoint synthesis: to type exterior roles correctly, define the unresolved residue, and derive the endpoint-normal form without scope drift, selector import, skin laundering, metric backflow, or post-hoc repair.

Terminology lock

Term	Use in this manuscript	Excluded reading
Unresolved horizon-isolated residue	$R_H = B_H \setminus W_{\text{ext}}(B_H)$, the reference-standing burden remaining after genuine exterior witness resolution	a claim that the horizon must last forever
Skin	useful representational or calculational presentation whose variation changes no standing-bearing content	useless or disposable description
Role-demotion of skin	withdrawal of endpoint-governing authority while descriptive utility remains	elimination of exterior dynamics or diagrams
Exterior witness relation	admissible carrier relation that re-fixes a declared reference-standing class	rhetoric, entropy curves, or radiation flux alone
Exterior witness resolution	removal of the re-fixed burden from R_H	repair or transformation of the interior
Information	derivative carrier/encoding/reference/witness package	primitive cargo or substance

Term	Use in this manuscript	Excluded reading
Reference-standing reconstruction problem	AASC-native form of the conventional information paradox	a primitive information-storage puzzle
New AMETRIC Boundary	structural closure-status at maximum governance-depth	dynamical formation event or spatial shell
Endpoint	closure classification of the unresolved residue	final metric time or place
No information persists	no old-scope standing-bearing carrier package is inherited across B_A^*	dynamical erasure event
Existence-status	structural domainhood under fresh governance	substance transported through the boundary or concrete population
Populated-existence	concrete internal occupants, laws, fields, or metric realization	something established by endpoint classification

Modal-status ledger

Statement	Status	Basis
The geometric horizon is not an AMETRIC boundary.	Proved/imported.	horizon identity is downstream of metric governance
Horizon isolation generates a reference-standing burden B_H .	Proved/imported.	horizon isolation and no-totalization
Exterior skin or ledgers resolve B_H by themselves.	Denied.	no skin promotion; witness map required
A genuine exterior witness relation may resolve part of B_H .	Conditional.	target-fixing carrier and quotient-preserving map
The theorem object is the nonempty residue R_H .	Definition.	exterior role typing and witness resolution
Residual metric support exhausts at maximum governance-depth.	Endpoint antecedent.	empty residual metric successor support
A singularity is unavailable there.	Proved conditionally.	singularity requires metric diagnostic authority
An admissible null endpoint is unavailable.	Proved/imported.	null-regime closure
The residue endpoint is a New AMETRIC Boundary.	Main theorem.	route exhaustion plus normal-form satisfaction
Nothing old-scope crosses.	Proved.	AMETRIC non-transmissivity and operator undefinedness
Every astrophysical black hole produces a nonempty residue.	Not proved.	exterior witness relations and physical attainment are model-dependent
AASC endpoint analysis establishes post-boundary contents.	Denied.	such a result requires cross-boundary standing
The endpoint is analogous to the AMETRIC origin condition.	Proved structurally.	shared boundary normal form, not shared content

Claim-status ledger

Claim	Status	Discipline
Einstein dynamics is a primitive background independent of AASC.	Denied.	It is the licensed minimal first-response metric projection of the AASC gravity role.
The endpoint theorem rejects GR in its licensed domain.	Denied.	It classifies the loss of endpoint authority of the licensed projection.
The theorem depends on literal horizon permanence.	Denied.	Its target is the unresolved horizon-isolated residue.
Information is primitive.	Denied.	Bare information-language lacks standing without a carrier package.
Exterior radiation repairs the interior.	Denied.	It is an exterior ledger unless paired with a witness relation; a witness resolves a burden outside the residue.
A Page curve is a reconstruction witness.	Denied by itself.	It is an entropy profile unless paired with a target-fixing map.
Holographic language is a witness.	Denied by itself.	The map, not the slogan, carries standing.
Skin is useless.	Denied.	It remains useful but is role-demoted at endpoint standing.
Crossed content remains ordinary same-scope GR matter inside.	Denied.	It has residual-coherence status absent new governance.
A classical singularity remains coherent after metricity exhausts.	Denied.	Singularity is a metric diagnosis.
The endpoint is a New AMETRIC Boundary.	Main theorem.	It satisfies the independently fixed endpoint normal form.
The paper proves populated-existence beyond B_A^* .	Denied.	Endpoint-status does not cross into interior cosmography.

Nonclaim ledger

This manuscript does not claim that Einstein dynamics is false in its declared parent metric scope, that every astrophysical black hole generates a nonempty residue, or that a measurable surface marks the New AMETRIC Boundary. It does not assign a proper time, radius, curvature threshold, or affine parameter to maximum governance-depth. It does not derive a post-boundary metric, matter spectrum, energy inventory, law package, or populated universe.

The manuscript does not deny the utility of exterior radiation calculations, entropy profiles, Page curves, Penrose diagrams, holographic models, island prescriptions, remnants, fuzzballs, final-state proposals, or baby-universe comparators. It types them. Skin and ledgers retain their licensed roles. Standing-bearing exterior witness relations resolve only the burdens they actually re-fix. The remaining nonempty residue is the theorem object.

Disciplinary orientation

For readers approaching from general relativity, the decisive distinction is between a licensed metric representative and the governance conditions under which metric and source comparison carry

standing. A singularity is a metric diagnosis inside that regime; the New AMETRIC Boundary theorem concerns exhaustion of the residual authority of the regime itself.

For readers approaching from thermodynamics, the decisive distinction is between a valid parent-domain ledger and illicit cross-AMetric backflow. The theorem does not assert energy nonconservation; it classifies the loss of the conservation statement's cross-boundary conditions of application.

For readers approaching from quantum information, the decisive correction is that information-language is not primitive cargo. The conventional paradox is retyped as a reference-standing reconstruction problem. Radiation, Page curves, boundary algebras, and global states matter only at the carrier and witness roles they actually supply.

For readers approaching from cosmology, the positive result is origin-condition analogy, not an internal-universe proof. The endpoint has the same structural profile as the AMETRIC starting condition of the realized physical interior, while post-boundary contents remain unavailable to parent-side endpoint analysis.

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Part I

Target, Central Correction, and Series Completion

1 The missing endpoint in the black-hole sequence

The present manuscript is an endpoint-normal-form extension of the AASC black-hole sequence. The exterior theorem identified the reference-standing burden created when declared parent-scope distinctions become dependent on a horizon-isolated carrier. The interior theorem then asked what crossed content can be when old-scope standing is not transported into that carrier and is not recreated there. Its answer was residual coherence: inherited role-organization without standing. Those papers correctly stopped before identifying the deepest closure-status of the residual carrier.

The present manuscript sharpens the theorem object. It does not center literal horizon permanence or primitive information. It begins by typing exterior structures. Skin remains useful for representation and calculation but has no endpoint-governing authority. Ledgers carry the exterior quantities they are licensed to carry. A genuine standing-bearing exterior witness relation may resolve a declared reference-standing burden in the parent scope. The endpoint theorem concerns whatever remains unresolved after those role distinctions are made.

The missing question appears only after six results are jointly available:

1. Einstein dynamics has been licensed as the unique minimal first-response same-scope metric projection of the AASC gravity role;
2. the black-hole horizon is a derivative metric boundary inside that projection, not an AMETRIC licensing boundary;
3. horizon isolation generates a parent reference-standing burden B_H ;
4. standing-bearing exterior witness relations may resolve only the burdens they actually re-fix;
5. the unresolved remainder $R_H = B_H \setminus W_{\text{ext}}(B_H)$ has residual-coherence status rather than ordinary same-scope interior standing; and
6. total vacuity cannot be the admissible closure state of admissibility-bearing evaluability.

Once $R_H \neq \emptyset$ and residual metric comparison exhausts, the old metric vocabulary cannot identify a singularity, place, later moment, causal transition, storage carrier, or dynamical repair for the residue. Once the null route is also closed, the endpoint cannot be admitted as total absence. The resulting status has the exact structural profile of a New AMETRIC Boundary: non-metric, non-eventful, non-operatorial, non-transmissive, and prior to any possible new governance.

Proposition 1.1 (Series completion). *The earlier black-hole manuscripts underclassify the endpoint of a nonempty unresolved horizon-isolated residue until the AMETRIC boundary theorem, Einstein-projection bridge, exterior role-typing discipline, and null-regime closure theorem are jointly applied. After those imports, the closure-status is no longer an open metric endpoint classification.*

Proof. The Einstein-projection theorem determines the status of the metric regime being bounded. Horizon isolation determines the threatened parent burden. Exterior role typing distinguishes skin

and ledgers from genuine witness relations and defines the unresolved remainder. The standingless-interior theorem determines the post-break role-status of that remainder while inherited organization persists. The AMETRIC theorem supplies the boundary normal form, and the null-regime theorem excludes admissible vacuity. Their conjunction closes a question each source paper correctly left open in isolation. $\square$

The resulting manuscript changes the center of gravity of the black-hole series. Its primary result is not a baby-universe mechanism and not a thesis about primitive information storage. It is a New AMETRIC Boundary endpoint theorem for the unresolved horizon-isolated residue. The old domain's deepest significance is not a singular object or hidden archive. It is the exposure of the origin architecture that metricity normally conceals.

2 Endpoint classification, not empirical attainment

The endpoint theorem is conditional on two target antecedents: $R_H \neq \emptyset$ and residual metric-comparison successor support has exhausted at maximum governance-depth. The manuscript does not prove that every astrophysical black hole leaves a nonempty residue, nor that every black hole reaches the metric-support exhaustion condition. It does not translate governance-depth into observational time, radius, curvature scale, or universal collapse history.

Proposition 2.1 (Endpoint-classification discipline). *If a nonempty unresolved horizon-isolated residue reaches residual metric-comparison exhaustion, its endpoint is classified by the New AMETRIC Boundary theorem. This is an endpoint normal-form theorem, not an empirical attainment theorem for all black holes.*

Proof. The antecedent fixes the theorem object and the role exhaustion being classified. The conclusion exhausts the same-type endpoint alternatives. Establishing the physical production of a nonempty residue or universal attainment of maximum governance-depth would require additional physical bridge theorems. None is silently imported here. $\square$

This limitation does not weaken the word “must.” Once the residue and exhaustion antecedents obtain, no same-type fixed-domain alternative survives. The nontrivial step is not that metric descriptors fail when metric support exhausts. It is that all rival endpoint categories require roles unavailable at that status, while nullity is not admissible.

3 AASC-internal theorem status and dependency lock

This manuscript is not a foundational reconstruction of AASC. It is an AASC-internal endpoint theorem. Its local proof obligation is to derive the endpoint from the imported kernel, boundary, Einstein-projection, horizon, residual-interior, matter-energy, exterior role-typing, and null-regime results without selector import, scope drift, role laundering, metric backflow, or post-hoc repair.

Derivational standard. Given the imported AASC results, does a nonempty unresolved horizon-isolated residue lose residual governance support at maximum governance-depth, does that endpoint satisfy the AMETRIC boundary normal form, and do no-crossing, no interior-instantiation proof, conservation-ledger residualization, and origin-condition analogy follow? This is the central derivational question.

Binding dependency	Work performed here	Where a critic must challenge it
Kernel {Adm, Stand, Ref, Irr}	fixes standing-bearing construction, reference, and no-repair	kernel source manuscript
AMETRIC boundary theorem	fixes non-action, non-transmissivity, no-selector, no-parameter, and target gating	AMETRIC source manuscript
Einstein-projection theorem	licenses the minimal Einstein-form metric projection and fixes $T_{\mu\nu}^{\text{closed}}$	Einstein-projection source manuscript
Horizon isolation/no-totalization	generates B_H and fixes the partial standing-transport problem	exterior black-hole theorem
Standingless interior/residual coherence	fixes post-break role status and governance-depth for R_H	standingless-interior theorem
Exterior role typing	distinguishes skin, ledgers, witness relations, and the unresolved residue	present manuscript
Null-regime closure	excludes total vacuity as admitted endpoint	null-regime theorem
Matter-energy residualization	fixes energy and conservation language as governance-dependent ledgers	standingless-interior theorem

A denial of one of these dependencies belongs at the source theorem. A criticism of the present manuscript must identify a false inference in the residue construction or endpoint chain, or exhibit a same-type endpoint alternative that survives the imported closures.

4 For non-AASC readers: the target in ordinary language

The manuscript asks a conventional claim-status question in a more exact form. A black-hole model may contain exterior radiation, entropy curves, diagrams, holographic descriptions, or other records. Those items do not all perform the same role. Some are representational skin, some are exterior ledgers, and some may be components of a genuine reconstruction map. The theorem first separates those roles. It then asks what remains of the interior burden after any real exterior reconstruction has done its work.

The resulting object is not “all information that fell in.” It is the unresolved reference-standing burden: the distinctions the model declared must remain identifiable but for which no exterior witness relation has re-fixed them. If no such burden remains, the endpoint theorem has no residue to classify. If a burden remains and the inherited metric role eventually loses all residual successor support, the theorem asks which endpoint description still has the roles needed to apply.

This is stronger than saying that classical GR becomes unreliable near a singularity. It asks whether singularity, storage, transfer, bounce, or nullity remain coherent classifications once the metric role itself no longer has authority.

5 The target question

Fix a parent domain D_0 and a black-hole reconstruction scope

$$S_{\text{BH}} = (E_{\text{ext}}, B_H, \mathcal{A}_{\text{wit}}, Q_{\text{ext}}, \text{Cont}, \text{Env}).$$

Here B_H is the horizon-generated reference-standing burden, not the boundary itself. Let $W_{\text{ext}}(B_H)$ be the portion resolved by standing-bearing exterior witness relations and let

$$R_H = B_H \setminus W_{\text{ext}}(B_H).$$

This is role-ledger notation: B_H , $W_{\text{ext}}(B_H)$, and R_H record burdens, exterior witness resolution, and the unresolved remainder under admissible reconstruction discipline. They are not primitive information-token sets.

The target question is:

If $R_H \neq \emptyset$ and the residual metric-comparison role has no successor support at maximum governance-depth, what endpoint status remains admissible for R_H ?

The theorem does not ask whether every exterior description is useful, or whether every model supplies the same W_{ext} . It asks how to classify a nonempty residue after those exterior questions have been answered within their own scopes.

6 False endpoint preview

Table 5: Candidate endpoint roles and their unmet requirements.

Candidate endpoint	Required roles	Failure at maximum governance-depth
Ordinary GR interior	metric comparison, source ledger, time, causality, local continuation	unavailable as full same-scope standing-bearing interior
Classical singularity	metric/affine diagnostic authority	metric diagnostic role exhausted
Remnant or archive	carrier identity and retrieval map	no standing-bearing residue map
Information-transfer endpoint	primitive information cargo or carrier transport	information-language is derivative on carrier roles; transport unavailable
Energy-transfer endpoint	common energy/source ledger	no cross-AMetric ledger identity
Dynamical bounce	before/after order and evolution operator	excluded at a non-eventful boundary
Wormhole throat	shared geometry and topology	larger-scope metric theory, not AMetric closure
Admissible nullity	positive-side admission with no governance	contradictory
New AMetric Boundary	no inherited metric, process, transport, or null closure	surviving endpoint normal form

Each candidate must carry the roles its own description requires. A singularity is a metric diagnostic. A bounce is a temporal and dynamical structure. Bare information-language has no endpoint standing absent a carrier package. A baby universe ordinarily imports history, causality, or continuity. Once those burdens are explicit, the endpoint alternatives can be tested rather than chosen by imagery.

7 Why the conclusion is stronger than interior inaccessibility

A standard epistemic reading says that the interior remains an ordinary physical region but is inaccessible. The standingless-interior result is stronger. Inaccessibility leaves ordinary interior ontology intact and merely limits knowledge. Standing failure removes the old-scope governance relation by which crossed content remains the same standing-bearing target. Residual coherence may preserve patterns, but patterns are not standing.

Suppose a probe p is prepared outside the horizon with a stable reference record. Before horizon dependence,

$$\text{Stand}_{D_0}(p) = 1.$$

If an exterior witness relation re-fixes the declared probe class, that burden belongs to $W_{\text{ext}}(B_H)$ and not to R_H . If no such relation exists for p , then the old standing-transport assignment is

undefined:

$$T_H(p) \uparrow.$$

The claim that “the same probe continues inside” then requires new governance or an unsupported identity assignment. Exterior skin does not supply that identity, and an exterior witness relation resolves the burden outside the residue rather than repairing the interior.

Residual coherence explains why a local metric representation may remain accurate near the horizon without reinstating old identity. A bound system can retain organized field and source patterns. A freely falling description may remain smooth. None of that makes the old reference class standing-bearing in the residue. Thus the argument predicts neither local chaos nor mere ignorance. It identifies a governance break whose representations can remain locally regular for a residual interval.

8 Penrose diagrams and endpoint authority

A Penrose diagram belongs to the skin and metric-representational layers. It summarizes causal structure within a declared spacetime model and can be indispensable for classical analysis. It does not determine whether every displayed region carries standing-bearing physical reference in the same reconstruction scope.

Governance-depth is not conformal distance, causal distance, or any diagrammatic coordinate. Maximum governance-depth may be represented by a mark, line, or omitted region, but those marks do not constitute the theorem object. The theorem object is the exhaustion of residual successor support for the metric-comparison role:

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{metric}) = \emptyset.$$

A diagram may still be drawn. Its continuation beyond that status remains useful as a representation of the classical projection, but it is non-governing with respect to the residue’s endpoint.

Proposition 8.1 (Penrose-diagram scope under the Einstein bridge). *A Penrose diagram may accurately represent the conformal causal structure of the licensed classical metric envelope, but it cannot represent the governed endpoint after the residual authority of that envelope has exhausted. It remains useful skin of the projection, not an endpoint witness.*

9 Method: role typing, residue construction, and fixed-domain closure

The proof method is eliminative and role-theoretic. It does not solve field equations in an unknown high-curvature regime. It fixes the parent scope, types exterior structures, constructs the unresolved residue, follows residual role dependencies, and tests every proposed endpoint against the roles it would need to perform.

1. **Scope lock.** Fix D_0 , its witness algebra, standing quotient, continuation relation, and declared target burdens.
2. **Exterior role typing.** Distinguish skin, ledgers, and standing-bearing witness relations.

3. **Residue construction.** Form $R_H = B_H \setminus W_{\text{ext}}(B_H)$.
4. **Standing-break import.** Establish that standing transport is undefined for burdens left in R_H .
5. **Residual-role analysis.** Distinguish retained organization from standing and order support by role ancestry.
6. **Endpoint exhaustion.** Test singularity, nullity, storage, bounce, transfer, and continuation claims at metric-support exhaustion.
7. **Boundary classification.** Identify the surviving endpoint role as AMETRIC boundary status.

The residue formulation does not weaken the theorem. It makes the target exact. A critic can challenge the construction by supplying a genuine witness relation for a burden placed in R_H , by denying metric-support exhaustion, or by exhibiting a same-type endpoint category whose roles remain available. What cannot count as a counterexample is a diagram, flux, entropy curve, or slogan treated as if it already performed witness work.

10 Exposure is not production

Saying that a black hole “creates” an AMETRIC boundary can suggest a temporal process in which metric dynamics produce a pre-metric boundary. That reverses the dependency order. Dynamics is downstream of metricity; metricity is downstream of governance; an AMETRIC boundary is upstream of both.

The correct language is *exposure*, *closure-status*, or *structural obtaining*. The unresolved residual regime reaches the limit of its role-support. At that limit, the remaining endpoint profile is the boundary normal form. From an external theorem perspective, the black hole exposes an origin boundary. Internally, no event of exposure occurs because eventhood itself is unavailable there.

Definition 10.1 (AMetric exposure). An AMETRIC exposure is the structural classification of a regime limit at which metric, temporal, causal, and operator roles lose support and the remaining admissibility-relevant status is pre-metric boundary status. It is not a transformation performed by the old regime.

Part II

Imported Architecture and Fixed-Domain Apparatus

11 The four-role kernel

The AASC kernel is not a list of physical substances. It is the minimal governance work required for identity-preserving construction or description on a fixed domain. Write

$$K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D).$$

Admissibility distinguishes licensed from unlicensed use. Standing marks eligibility for later commitment, continuation, or reuse. Reference fixes what is being preserved as the same target. Irreversibility prevents later intervention from converting a failed act into the same successful act. The kernel theorem establishes that any non-degenerate regime purporting to support determinate target identity, evaluable steps, act-time finality, and faithful redescription already performs these roles, whether or not it uses the vocabulary of AASC [Maley, 2026d].

For the black-hole problem, the roles are instantiated as follows.

- Adm_D determines whether a claimed continuation, reconstruction, witness, or interior assignment is licensed in the declared scope.
- Stand_D determines whether a preparation record, parameter class, correlation class, or object identity remains available for later standing-bearing use.
- Ref_D fixes the threatened class: the infalling probe, purifier relation, source history, or other target that is claimed to persist.
- Irr_D blocks the claim that a later reconstruction device retroactively made an earlier undefined transport assignment defined in the same act and scope.

Theorem 11.1 (Kernel necessity). *Any same-scope black-hole claim that treats an exterior preparation, horizon history, interior continuation, endpoint, or recovered information class as a determinate object of admissible comparison instantiates the four kernel roles or a governance-equivalent package.*

Proof. A determinate black-hole target requires reference. A claimed continuation requires a distinction between licensed and unlicensed continuation, hence admissibility. Later use of the target requires standing. A distinction between continuation and after-the-fact repair requires irreversibility. Removing any role changes the target from a standing-bearing reconstruction claim into raw notation, unconstrained model extension, or bookkeeping. $\square$

A useful derived identity is that standing is reuse-stable admissibility. On the evaluated fragment of a fixed scope,

$$\text{Stand}_D(x) = 1 \iff x \text{ is admissible and remains admissible under licensed same-scope reuse.}$$

Standing is therefore not a hidden energy or informational quantity. It does not gradually drain into a black hole. It is bivalent relative to the fixed scope, even though downstream reliability, approximation, entropy, or residual coherence may be graded.

12 The AMetric boundary package

The AMETRIC boundary is the structural interface between pre-admissible classification and post-admissible licensed operation [Maley, 2026a]. It is “AMetric” not merely because no spacetime metric is present, but because no admissibility-relevant ranking, order, selector, probability, typicality measure, canonical representative, discrete index, or graded distance can carry boundary authority. Before target fixation, such structures would privilege representatives without a licensed witness.

Definition 12.1 (AMetric boundary). An AMETRIC boundary B_A is an admissibility-generating interface at which standing may be fixed on the evaluated fragment but at which no admissible transformation, process, or operator is defined. It is prior to the metric, temporal, causal, and dynamical roles of any admitted interior.

The package has six consequences that are load-bearing in the black-hole extension.

1. **Boundary non-action.** No action, execution, process, or transformation occurs at B_A .
2. **Operator undefinedness.** An operator is a licensed transformation rule; before admission its license conditions have no domain of application.
3. **Non-transmissivity.** No operator, license, redescription, metric, causal rule, temporal order, or admissible structure is transported across B_A as inherited authority.
4. **Bivalence on the evaluated fragment.** A candidate is admitted or not admitted. Undefinedness is non-membership in the evaluated fragment, not a third standing value.
5. **No same-act repair.** Standing-negative evaluation cannot be revised into standing-positive status while preserving the same evaluated act.
6. **Independent target gating.** A new domain does not inherit standing from an old domain. It must instantiate its own gate and domain-of-application closure.

Lemma 12.2 (No boundary transport). *Let D_0 and D_1 be distinct domains separated by an AMETRIC boundary. There is no same-role transport map*

$$\tau_A : \mathcal{G}(D_0) \longrightarrow \mathcal{G}(D_1)$$

that carries the admissibility-bearing governance content $\mathcal{G}(D_0)$ into D_1 as inherited standing.

Proof. Such a map would have to be an admissible operator. Operatorhood requires a domain of licensed transformation. But the AMETRIC boundary is where standing and domain eligibility are first fixed; operatorhood cannot be presupposed there. If τ_A is defined by D_0 , it illegitimately extends old authority into a target domain without independent gating. If it is defined by D_1 , it presupposes D_1 before the boundary that licenses it. If defined by a larger domain, it is an external comparison or richer-scope theory, not internal inheritance between D_0 and D_1 . $\square$

The phrase “nothing crosses” will always mean this governance-theoretic non-transport. It does not picture a substance striking an impermeable membrane. It states that the very types needed to formulate crossing—objects, trajectories, times, laws, and transport operators—are not available at the boundary as inherited roles.

13 The independently fixed AMetric boundary normal form and the combined AASC endpoint normal form

The endpoint theorem must not define “AMetric” as whatever remains after metric failure. The strict boundary profile is fixed independently in the AMetric source manuscript. For later use, define the AMetric normal-form predicate $\text{ANF}(B)$ by the conjunction of the following clauses:

- (N1) B carries no admissibility-relevant metric, rank, order, probability, typicality, parameter, or canonical representative.
- (N2) No action, event, transition, execution, or process occurs at B .
- (N3) No admissible operator has B as an inherited transformation domain.
- (N4) No prior-domain license, standing, reference, or redescription rule is transported through B .
- (N5) Standing is bivalent only on the evaluated fragment; undefinedness is non-membership, not an intermediate status.
- (N6) No standing-negative act is repaired into the same standing-positive act.
- (N7) Any target domain requires independent gating and domain-of-application closure.
- (N8) No metric, temporal, causal, modal, probabilistic, or process structure authorizes its own admission.

Definition 13.1 (AMetric boundary normal form). A boundary profile B has AMetric boundary status exactly when $\text{ANF}(B)$ holds within the declared comparison class.

The black-hole endpoint theorem also imports a separate closure result: total AASC vacuity cannot occupy the admitted endpoint role. Define the combined AASC endpoint-normal form by

$$\text{AENF}(B) := \text{ANF}(B) \wedge \neg \text{NullAdm}(B) \wedge \text{EndpointRouteClosure}(B).$$

The last conjunct records that any alleged successor relation that changes no standing-bearing fact is bookkeeping, while any relation that changes such a fact supplies new governance or changes scope.

Definition 13.2 (AASC endpoint-normal form). A profile satisfies the AASC endpoint-normal form when it satisfies the independently fixed AMetric boundary normal form together with null-regime closure and same-type endpoint-route exhaustion.

Definition 13.3 (New AMetric Boundary). A New AMETRIC Boundary is AMetric boundary status exposed at the closure of a downstream governed regime and not identical with that regime's original admissibility-generating boundary. "New" is relational to the parent domain's governance architecture. It does not mean dynamically manufactured, temporally later at the boundary, spatially internal, or populated beyond the boundary.

Remark 13.4 (Theorem-order vocabulary). Terms such as "before," "after," "post-boundary," "downstream," and "endpoint" are role-dependency and proof-order terms unless a governed domain independently supplies temporal order. They are not temporal predicates at B_A^* .

Lemma 13.5 (Boundary-status realism). *An AASC boundary is a real boundary-status when it fixes an admissibility-role discontinuity, even if it is not a metric surface, temporal event, causal separator, or proven interface to a populated successor domain. Its reality is role-theoretic: it determines definedness, operator availability, standing transport, target gating, and endpoint-route closure.*

Proof. Boundary work in AASC is performed by changes in definedness, standing, operator domain, transport authority, and target gating. These are anchor-level facts. The reality of the boundary is therefore role-theoretic: it determines which operations are defined, which identities can be transported, and which targets require fresh gating. Spatial embedding is unnecessary. If the endpoint satisfies the independently fixed boundary normal form, it performs boundary work regardless of whether a successor interior is instantiated or parent-scope observable. $\square$

Proposition 13.6 (Independence of the endpoint application). *The predicates ANF and AENF are fixed prior to the black-hole endpoint application: ANF by the AMetric boundary theorem, and the additional endpoint clauses by null-regime closure and fixed-domain route exhaustion. Accordingly, proving that maximum governance-depth satisfies AENF is a normal-form satisfaction theorem, not a stipulative renaming of metric failure.*

Proof. The strict AMetric clauses are imported from pre-fixational relabeling invariance, boundary non-action, operator undefinedness, non-transmissivity, independent target gating, bivalence, and no-repair. The additional endpoint clauses are imported from null-regime closure and fixed-domain route exhaustion. None mentions black holes, horizons, residual metricity, or maximum governance-depth. Their later satisfaction by the black-hole endpoint is therefore an application of independently specified profiles. $\square$

14 Metric projection as a downstream governance object

The black-hole reconstruction object is evaluated inside a licensed metric projection, not against a primitive spacetime container. The Einstein-projection bridge is developed in the next Part; its imported conclusion is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

as the unique minimal first-response same-scope metric projection of the AASC gravity role [Maley, 2026c]. This result adds three disciplines to the endpoint apparatus.

1. Metric and causal descriptors are projected tensor content downstream of governance.
2. The source tensor is a metric-facing closed source ledger, not primitive energy substance.
3. High-curvature or horizon corrections require declared load, a standing-bearing exterior witness relation, a lawful leakage channel, non-sink quotient, or richer scope; they cannot silently repair the original projection.

The endpoint theorem will therefore ask when this licensed projection loses residual role authority for crossed content. It will not ask whether an independently primitive metric universe happens to stop being calculable.

15 The black-hole reconstruction object

The parent black-hole scope is fixed as

$$S_{\text{BH}} = (E_{\text{ext}}, H_{\text{BH}}, R_{\text{ext}}, W_{\text{ext}}, \text{Cont}, \text{Env}, \text{Param}, \text{Therm}).$$

Here E_{ext} is the exterior admissible description domain, H_{BH} the horizon typing, R_{ext} the standing-bearing exterior reference family, W_{ext} the exterior witness family, Cont the continuation relation, Env the physical model envelope, Param the mass/charge/angular-momentum ledger, and Therm the thermodynamic ledger [Maley, 2026b].

For $r, r' \in R_{\text{ext}}$, define the exterior equivalence relation by equality of witness, continuation, boundary-access, and admissibility profiles. The quotient

$$Q_{\text{ext}} = R_{\text{ext}} / \sim_{\text{ext}}$$

records every difference relevant to standing while removing coordinate, chart, gauge, and other presentation surplus. The exterior witness algebra is schematically

$$\mathcal{A}_{\text{wit}} = \text{Alg}(W_{\text{ext}}, \mathcal{A}_{\partial H}^{\text{adm}}, \text{Cont}).$$

It may contain detector records, asymptotic fields, radiation correlations, memory observables, boundary operators, and reconstruction maps. Their role is not fixed by sophistication or nonlocality. Skin and ledgers retain their declared uses; a standing-bearing witness requires a target-fixing, quotient-preserving relation.

Let the horizon carrier be

$$X_H = E_{\text{ext}} \cup E_{\partial H} \cup E_{\text{int}},$$

and let

$$T_H : X_H \rightharpoonup R_{\text{ext}}$$

be the partial standing-transport assignment. Let $B_H \subseteq R_{\text{ext}}$ be the family of declared reference-standing burdens threatened by horizon isolation. For every burden $b \in B_H$ not re-fixed by a standing-bearing exterior witness relation,

$$T_H(b) \uparrow,$$

where $\uparrow$ denotes non-membership in the admissible domain rather than practical ignorance.

Theorem 15.1 (Horizon isolation, imported burden form). *If a standing-bearing exterior reference class becomes dependent on a horizon-isolated carrier and no standing-bearing exterior witness relation re-fixes it in the declared parent scope, then the corresponding burden has no admissible image in the old exterior witness algebra.*

The associated no-totalization theorem states that no map

$$T_H^* : X_H \rightarrow R_{\text{ext}}$$

can define the old undefined assignments while preserving the old quotient, witness algebra, continuation relation, and ledgers. A purported definition is either already present in the exterior witness algebra, a new witness relation, a scope change, a selector, a repair, carrier laundering, or bookkeeping.

The present manuscript inherits this result rather than reproving it. Its use is residue-specific: exterior role typing determines which burdens are genuinely re-fixed; the unresolved remainder is $R_H = B_H \setminus W_{\text{ext}}(B_H)$. The endpoint theorem applies to that remainder, not to every black-hole

reference class and not to a claim about literal horizon duration.

16 Standingless interiors and residual coherence

The horizon obstruction fixes what does not continue. The standingless-interior theorem classifies what may remain describable. Let $\mathcal{R}_D$ be the relevant role family,

$$\mathcal{R}_D = \{\text{skin, record, gauge, tensor, metric, measurement, field, particle, source, time, causal}\}.$$

For role $r \in \mathcal{R}_D$, write

$$\text{ResCoh}_D^r(x)$$

for the inherited organization of type r that remains after x has lost standing, provided the organization is non-nullly representable, traceable to a pre-break role, and insufficient for full role licensing [Maley, 2026f].

Residual coherence is neither a third standing value nor a hidden information substance. It is role-indexed and gradable because different roles depend on different support chains. Skin is usually weakest. Tensor-like and source-like organization may persist more robustly. Metric comparison may remain an excellent residual approximation near a large horizon. None of those facts re-establishes the standing of the crossed class.

Definition 16.1 (Residual successor support). For a crossed item x and role r , $\text{Succ}_D^{\text{res}}(x, r)$ is the class of post-break continuations that retain enough inherited organization for r to remain descriptively usable without restoring standing.

Governance-depth is an order on the thinning of these successor classes:

$$\text{GDepth}_D(x, r) \preceq \text{GDepth}_D(y, r) \iff \text{Succ}_D^{\text{res}}(y, r) \subseteq \text{Succ}_D^{\text{res}}(x, r),$$

up to residual-equivalent redescription. It is not a physical scalar. Near the horizon, radial or proper-time language may approximate this order because residual metric coherence is high. At greater governance-depth the approximation can fail. At maximum governance-depth for the metric role,

$$\text{Succ}_D^{\text{res}}(x, \text{metric}) = \emptyset.$$

Then metric depth is no longer a licensed residual descriptor.

Theorem 16.2 (Residual metric exhaustion). *If residual metric-comparison successor support is empty for a crossed class, no metric descriptor—including depth, curvature locus, geodesic continuation, or metric endpoint—retains residual role authority for that class in the old scope.*

Proof. Each descriptor requires metric comparison. After standing loss, metric comparison is available only through residual successor support. Empty support means no continuation preserves the role even residually. A formula may remain syntactically writable, but writability is skin once the comparison role it presents has exhausted. $\square$

The theorem does not say that a coordinate cannot be printed or that a classical solution cannot be studied as mathematics. It says that such notation cannot be promoted into a governed

description of the exhausted carrier without new role support.

17 The null-regime closure result

One might attempt to avoid a new boundary classification by saying that the residual regime simply becomes nothing. The null-regime theorem closes only a precise version of that route: total AASC vacuity proposed as an admissible same-type endpoint [Maley, 2026e].

Definition 17.1 (AASC vacuity). A candidate N has AASC-vacuous content when it has no admissibility-relevant invariant bundle, identity carrier, transport rule, quotient identity, anchor, tensor, gate authority, licensed continuation condition, or standing-bearing role.

Definition 17.2 (Null admissible regime). A null admissible regime is a candidate N such that

$$\text{Adm}_D(N) = 1 \quad \text{and} \quad C(N) = \text{Vac}.$$

The contradiction is immediate once positive-side identity collapse is imported. Admission entails standing, evaluable identity, transport consistency, quotient identity, and classificatory legitimacy. Vacuity denies every one. Thus no null admissible endpoint exists.

The theorem does not turn silence into ontology. Completely mute nullity performs no classification and has no same-type counterexample force. Its status is AMETRIC pre-admissibility: non-selective, non-transportive, non-identifying, and non-comparative. If mute nullity is used to classify the endpoint, it ceases to be mute and incurs governance obligations. This is exactly why residual metric exhaustion has an AMETRIC boundary profile rather than a positively admissible null profile.

Corollary 17.3 (No admissible terminal vacuity). *The closure of the horizon-isolated residual carrier cannot be both admissibility-relevant and totally vacuous. If it is entirely mute, it is pre-admissible boundary status rather than an admitted terminal regime.*

This corollary is the missing hinge between residual exhaustion and the present paper's closure theorem. It eliminates the apparent third option between singular metric endpoint and the New AMetric Boundary endpoint. Mute nullity does not compete with that boundary; it is the pre-admissible boundary profile under another description.

18 Anchor–tensor–skin discipline at the endpoint

Every descriptive element in a fixed domain is typed as anchor, tensor, or skin. Anchors determine definedness, boundary typing, or domain identity. Tensors carry quotient-level invariant content that changes admissible continuation under fixed anchors. Skin is representational surplus that changes neither classification nor continuation.

At maximum governance-depth, old metric notation is often misread because a formula that once presented tensor content can remain writable after its role-support has failed. The same symbol $g_{\mu\nu}$ may therefore occupy different roles at different stages:

- in the licensed parent metric regime, it presents metric tensor content;

- in the near-horizon residual regime, it presents residual metric organization;
- after residual metric successor support exhausts, it is skin relative to the crossed target unless a new domain re-licenses metricity.

The transition among these classifications is not a dynamic conversion of a tensor into skin. It is a change in role authority under the fixed-domain analysis. This is essential for the singularity result: a formally written curvature divergence after metric-role exhaustion is not a residual singularity. It is a symbolic continuation unsupported by the old carrier.

Lemma 18.1 (Endpoint role exhaustion). *At maximum governance-depth, no old-scope descriptive item can retain anchor or tensor authority merely by syntactic persistence. Any retained notation either receives new governance, belongs to a richer external scope, or is skin.*

Proof. Anchor authority would change boundary or domain identity and must be declared as new governance. Tensor authority would alter admissible continuation and requires residual or new standing-bearing support. With residual metric support exhausted and no new scope declared, neither role is available. The only remaining same-scope role is skin. $\square$

Part III

The Einstein-Projection Bridge

Bridge thesis. AASC does not encounter general relativity as an independently primitive theory and then comment on its failure. It first licenses metricity, imports the AASC gravity role, and imports the minimal Einstein-form projection under the declared first-response same-scope metric-projection predicate. The black-hole endpoint theorem then classifies the boundary at which this licensed projection no longer has governance authority for the crossed carrier.

19 From AASC gravity to licensed metric dynamics

The endpoint argument requires a precise claim order. The AASC kernel fixes admissibility, standing, reference, irreversibility, quotient identity, lawful redescription, and continuation. The imported gravity result then identifies the gravitational role as the constraint architecture required by an admissible physical interior: standing surface, observable quotient, admissible redescription, first-class closure, and multiplier-type Hamiltonian presentation. Metric curvature is not the source of that role. It is a faithful presentation of it once metric comparison has been licensed.

The Einstein-projection theorem supplies the next layer [Maley, 2026c]. Let A_G denote an AASC gravitational architecture and let

$$\Pi_g(A_G) = (M_4, g_{\mu\nu})$$

be a faithful four-dimensional Lorentzian metric projection. Faithfulness requires the metric to remain downstream of governance, coordinates and frames to remain skin, metric distinctions with

physical force to be anchor or tensor content, source continuation to remain role-fixed, and no hidden connection, field, boundary, or source structure to be imported.

The proof order is therefore

$$\begin{aligned}
& \text{AASC kernel and quotient discipline} \\
& \implies \text{AASC gravity role} \\
& \implies \text{faithful Lorentzian metric projection} \\
& \implies \text{closed metric/source continuation} \\
& \implies \text{minimal first-response curvature exhaustion} \\
& \implies G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.
\end{aligned}$$

Standard general-relativistic texts begin after the metric regime has already been admitted and develop the geometry, dynamics, and source relations internal to that regime [Misner et al., 1973, Wald, 1984, Carroll, 2004]. The AASC bridge does not contest that mathematics. It fixes the upstream status of the metric regime itself.

20 Same-scope first-response curvature exhaustion

The Einstein-projection theorem defines a deliberately narrow target class. Write

$$\text{SameScopeMetric}_1(E_{\mu\nu})$$

when a candidate geometric response tensor is internal to the licensed metric projection, tensorial under lawful redescription, symmetric rank two, local, linear in the first admitted curvature obstruction, compatible with a closed source ledger, and free of undeclared anchor motion, higher-response curvature products, curvature derivatives, extra fields, torsion, metric-affine nonmetricity, nonlocal operators, boundary/global data, representative privilege, source laundering, and hidden horizon repair.

This predicate does not deny modified gravity. It fixes the minimal same-scope first-response class. Higher curvature, torsion, independent connection structure, scalar/vector/tensor additions, nonlocality, semiclassical expectation values, and horizon-specific responses remain available as declared higher-response, effective, mixed, boundary-witness-bearing, or richer-scope structures.

Within the target class, the first-response geometric slots are exhausted by

$$R_{\mu\nu}, \quad Rg_{\mu\nu}, \quad g_{\mu\nu}.$$

Hence

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed same-scope source continuation requires

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0,$$

and therefore

$$\nabla^\mu E_{\mu\nu} = 0.$$

Using metric compatibility and the contracted curvature identity,

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R, \quad \nabla^\mu g_{\mu\nu} = 0,$$

one obtains $b = -a/2$ and therefore

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

After coupling normalization,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Theorem 20.1 (Licensed Einstein Projection Theorem). *Within a faithful four-dimensional Lorentzian metric projection of the AASC gravity role, and under the minimal first-response same-scope predicate SameScopeMetric_1 , Einstein dynamics is the unique continuation-compatible metric field-equation form:*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

It has standing as a licensed projection theorem, not as primitive metric ontology or external postulate.

Proof sketch. The imported gravity theorem supplies the constraint role. Faithful metric projection fixes the Lorentzian metric envelope without metric backflow. First-response curvature exhaustion yields the Ricci, scalar-trace, and homogeneous metric slots. Closed source continuation imposes divergence compatibility, which fixes the Einstein combination. The remaining coefficients are projection- and calibration-level quantities rather than AMETRIC defaults. The full proof is given in the Einstein-projection manuscript [Maley, 2026c]. $\square$

20.1 Relation to metric uniqueness classifications

The bridge is consistent with familiar metric uniqueness results but does not import them as proof authority. Lovelock-style uniqueness results are comparator orientation only; the AASC bridge imports projection-standing, source-ledger, and endpoint-scope discipline, not merely metric uniqueness. In four dimensions, Lovelock's classification identifies the metric and Einstein tensors as the relevant symmetric divergence-free second-order metric concomitants under its assumptions [Lovelock, 1971]. The AASC burden is prior and differently typed: it asks how metricity obtains standing, how the projection envelope is fixed, why source continuation is closed, and why undeclared higher-response or boundary content does not count as same-scope first response. Lovelock's theorem is therefore a metric-internal comparator, not the source of the AASC projection result.

21 The source tensor as an AASC metric-facing ledger

The bridge also fixes the role of the source side. The tensor

$$T_{\mu\nu}^{\text{closed}}$$

is not primitive mass-energy substance. It is the metric-facing closed ledger of standing-bearing deformation response, field-carrier mediation, localized persistence classes, inertia and bounded

transport, interactions, restoration burden, and the admitted vacuum baseline. Standard field-theoretic and algebraic formalisms remain useful comparators for source and field structure [Weinberg, 1995, Haag, 1996]; they do not supply the AASC governance role.

The cosmological term is likewise role-typed. The combination $\Lambda g_{\mu\nu}$ is the quotient-stable homogeneous metric-facing ledger. Its numerical value is not generated at the AMETRIC boundary and requires scale, extraction, and cosmological calibration anchors.

Proposition 21.1 (Einstein source-ledger scope). *The standing of $T_{\mu\nu}^{\text{closed}}$ is bounded by the licensed metric/source projection in which its carrier, identity, closure, and comparison roles are fixed. Tensor writability beyond that role-support does not export parent energy, source identity, or conservation authority.*

Proof. A stress-energy tensor is metric-facing extraction from an already admitted source ledger. Its covariance and divergence condition presuppose the metric, connection, source identity, quotient, and continuation envelope. When those supports are absent or merely residual, the same symbol cannot carry more authority than the role package that licensed it. Export would require a separate standing-transport map, which the tensor equation does not provide. $\square$

22 High curvature, horizons, and no hidden repair

The Einstein-projection theorem does not treat high curvature as permission to alter the old scope silently. A high-curvature or horizon response with standing-bearing force must be classified as one of the following:

1. declared higher-response or effective tensor load;
2. lawful leakage or radiation channel;
3. independently licensed endpoint witness relation;
4. boundary-to-boundary reconstruction witness relation;
5. non-sink quotient;
6. declared richer-scope theory.

Anything else is skin, source incompleteness, global-retention laundering, hidden selector import, remnant compression, or post-hoc repair. Thus the endpoint manuscript does not infer the New AMETRIC Boundary because Einstein dynamics was too weak to invent a correction. It infers the boundary because no undeclared same-scope correction can acquire governance authority after the old projection's support has exhausted.

Theorem 22.1 (No GR self-certification at the endpoint). *The licensed Einstein projection cannot certify its own continuation, repair, or re-licensing at maximum governance-depth.*

Proof. The Einstein equation is downstream of the governance architecture that licenses metricity, source identity, and continuation. Using the equation to restore those roles after their standing has failed would make a downstream projection authorize its own admissibility conditions. That is metric backflow. Endpoint continuation therefore requires an independently declared standing-bearing

relation, lawful leakage route, non-sink quotient, high-curvature/effective load, or richer governance. In their absence, the projection cannot self-certify. $\square$

23 Three levels of failure

The bridge separates three endpoint levels that ordinary singularity language can otherwise conflate.

Level	Failure type	AASC classification
Equation-level	curvature divergence, inextendibility, geodesic incompleteness, high-curvature breakdown, or incomplete source response inside the metric regime	metric-internal diagnostic; may motivate declared higher-response or richer-scope structure
Projection-level	the licensed metric projection no longer retains residual comparison, source, continuation, or measurement support for crossed content	governance-depth exhaustion of the metric role
Boundary-level	no old-scope metric, operator, causal, temporal, source-ledger, or transport authority remains, and admissible nullity is unavailable	New AMETRIC Boundary closure-status

Classical GR already recognizes equation-level singular behavior. The additional AASC result is projection-level and boundary-level classification. The singularity is the metric regime's internal signature of its limit; the New AMETRIC Boundary is the governance-level normal form once the residual authority of the metric regime itself has exhausted.

Theorem 23.1 (Licensed Projection Failure Is Not Primitive GR Failure). *At maximum governance-depth, the failure under analysis is not the breakdown of an independently primitive GR ontology. It is exhaustion of the residual governance support of the AASC-licensed Einstein projection for the crossed carrier.*

Proof. Einstein dynamics obtains standing only after the AASC gravity role and faithful metric projection are fixed. The geometric horizon lies inside that projection. For burdens retained in the unresolved horizon-isolated residue, old-scope standing does not continue; the carrier is residual; and at maximum governance-depth residual metric comparison has no successor support. The endpoint therefore concerns the authority of the licensed projection, not merely an equation evaluated at a pathological coordinate or curvature locus. $\square$

24 GR is preserved and bounded

The bridge preserves general relativity more strongly than an external failure diagnosis would. AASC does not deny the Einstein equation in its licensed scope. It imports the result that the minimal Einstein-form equation is the unique same-scope first-response metric projection of AASC

gravity under the stated predicate. The New AMETRIC Boundary theorem then bounds that projection: GR has authority where the metric/source projection retains governance support and has no old-scope endpoint authority where residual metric comparison has exhausted.

Licensed and bounded. AASC licenses the Einstein metric projection and then bounds that projection. It neither treats Einstein dynamics as primitive ontology nor rejects it within its declared projection scope.

This is the sense in which AASC is governance-complete relative to classical GR's unclassified boundary assumptions. It does not claim all gravitational dynamics or supply a numerical quantum-gravity completion. It supplies the upstream admissibility architecture that licenses metricity, imports the minimal Einstein-form projection under the declared predicate, types its source and homogeneous vacuum ledgers, and classifies the boundary at which that projected description loses authority.

Part IV

Exterior Role Typing and the Unresolved Horizon-Isolated Residue

Residue thesis. The endpoint theorem is not a thesis about literal horizon permanence, primitive information, or exterior phenomena repairing an interior. Its target is the *unresolved horizon-isolated residue*: the reference-standing burden that remains after exterior skin has been assigned its proper descriptive role and after genuine standing-bearing exterior witness relations have resolved the burdens they actually carry. Skin describes; witnesses resolve; residue closes.

25 From the older horizon qualifier to the residue target

Earlier AASC black-hole manuscripts used the phrase “uncertified permanent horizon” to isolate a case in which threatened old-scope standing had no admissible exterior image and no same-scope successor. That phrase served a local proof purpose, but it can invite two misleading readings: that the endpoint theorem depends on a horizon lasting forever in exterior time, or that physics requires an administrative act of certification. Neither reading is the target of the present manuscript.

The sharpened theorem object is the burden that remains unresolved after exterior structures have been role-typed. Horizon duration is not the governing variable. Nor is the mere existence of radiation, diagrams, entropy profiles, boundary descriptions, or global-state labels. The relevant question is whether an exterior structure supplies a standing-bearing relation that re-fixes a declared reference-standing class. If it does, that burden is resolved outside the residue. If it does not, the exterior structure remains skin, ledger content, comparator structure, or bookkeeping relative to the endpoint burden.

Definition 25.1 (Horizon-generated standing burden). Let B_H denote the family of old-scope reference-standing burdens threatened when their declared continuation becomes dependent on

the horizon-isolated carrier. Membership in B_H is target-relative: only distinctions the parent reconstruction problem declares standing-bearing belong to the burden.

Definition 25.2 (Standing-bearing exterior witness relation). A standing-bearing exterior witness relation for a burden $b \in B_H$ is an admissible relation W that supplies all of the following:

1. a fixed target reference-standing class;
2. an exterior carrier, record family, or witness algebra;
3. a quotient-preserving map from the exterior carrier to the target class;
4. stability under lawful redescription;
5. explicit success and failure conditions;
6. continuation stability in the declared exterior scope;
7. no reliance on hidden interior storage as the source of identity.

A slogan, diagram, entropy curve, flux measurement, or untyped correlation is not such a relation merely by being associated with a recovery narrative.

Definition 25.3 (Unresolved horizon-isolated residue). Let $W_{\text{ext}}(B_H)$ denote the portion of B_H resolved by standing-bearing exterior witness relations. The unresolved horizon-isolated residue is

$$R_H := B_H \setminus W_{\text{ext}}(B_H).$$

It is the reference-standing burden that remains horizon-isolated after exterior structures have been role-typed and genuine exterior witness relations have resolved the burdens they actually carry. It is not defined by literal temporal permanence. It is defined by remaining unresolved under admissible exterior reference-standing reconstruction.

The notation is role-ledger notation for reference-standing burdens under quotient-preserving exterior resolution. It is not a primitive set of information tokens, not a hidden inventory of interior objects, and not a claim that information is an ontological substance.

Remark 25.4 (Historical terminology). Where the earlier phrase “uncertified permanent horizon” is quoted or discussed, it should now be read as an older description of the nonempty residue condition $R_H \neq \emptyset$ together with horizon isolation. The residue formulation is more exact because it separates exterior duration, exterior dynamics, and endpoint standing.

26 Skin in licensed dynamics and endpoint standing

Skin is not discarded or treated as useless. In a licensed exterior or metric-dynamical regime, skin may be indispensable for representation, calculation, visualization, coordinate control, detector description, radiation bookkeeping, and comparison. Coordinates, foliations, Penrose diagrams, gauge representatives, radiation plots, asymptotic labels, and exterior time descriptions can organize lawful analysis.

Its role is diminished only when the question becomes endpoint standing. Skin does not by itself fix a target reference class, preserve quotient identity, transport standing, repair horizon isolation, or define an endpoint. It remains descriptive and calculational; it is not endpoint-governing.

Principle 26.1 (Skin role-demotion). At the horizon-interior endpoint, skin retains its licensed representational and calculational uses but is role-demoted from endpoint-governing authority. This is a change in role relevance, not a declaration of uselessness.

Theorem 26.2 (No Skin Promotion). *Exterior skin does not become a standing-bearing witness through reinterpretation, rhetorical emphasis, diagrammatic refinement, entropy bookkeeping, or added descriptive detail. A standing-bearing exterior witness exists only when a distinct admissible carrier relation is supplied.*

Proof. Skin is defined by the fact that its variation changes no standing-bearing content. A witness relation, by contrast, must fix a target, carry quotient-stable reference, and distinguish successful from failed re-fixation. If no new carrier relation is added, nothing standing-bearing changes and the item remains skin or ledger presentation. If a carrier relation is added, that relation—not the prior skin—does the witness work. Hence skin is not promoted; additional anchor or tensor-level witness structure is supplied. $\square$

Table 7: Exterior role typing before residue formation.

Exterior structure	AASC role	Endpoint effect
Coordinates, labels, Penrose diagrams	skin / metric representative	describe the licensed exterior or classical metric envelope; do not govern R_H
Radiation flux	exterior energy or thermodynamic ledger	changes parent flux accounting; resolves no reference-standing burden without a map
Page curve	entropy/ledger profile	constrains exterior entropy behavior; is not itself a witness
Holographic or island language	skin or comparator unless mapped	has no residue effect without a role-fixed reconstruction relation
Boundary operator algebra with target map	possible exterior witness carrier	may resolve the target burden it re-fixes
Radiation algebra with stable decoding map	possible exterior witness carrier	may resolve the target burden it re-fixes
Remnant label without exterior distinguishability	hidden-storage skin or unsupported load	resolves no burden
Remnant-to-witness map	possible non-sink exterior relation	may resolve the mapped burden
Remaining unresolved burden	horizon-isolated residue R_H	target of the New AMetric Boundary theorem

27 Information language retyped as reference-standing reconstruction

The conventional black-hole information paradox is useful comparator language, but information is not primitive in AASC. Bare information-language lacks standing unless supported by a carrier, encoding, reference class, witness condition, quotient identity, and admissible reconstruction. A scalar entropy, Hilbert-space dimension, state label, or correlation pattern does not independently supply those roles.

Lemma 27.1 (Information is not primitive). *An information claim has standing only through an admissible carrier package*

$$(c, e, \sim, r, w, u),$$

where c is a carrier, e an encoding, $\sim$ a sameness relation, r a reference class, w a witness relation, and u a licensed use condition. Without that package, information-language is skin or bookkeeping relative to the reference-standing problem.

Corollary 27.2 (Retyping of the conventional paradox). *The conventional black-hole information paradox is retyped as an exterior reference-standing reconstruction problem: does any exterior carrier relation re-fix the threatened reference-standing class in the declared parent scope?*

Table 8: Orthodox information language and AASC-native retyping.

Orthodox phrase	AASC-native formulation
Information paradox	exterior reference-standing reconstruction problem
Information loss	failure of reference-standing re-fixation
Information escapes in radiation	radiation supports a standing-bearing exterior witness relation, if a map is supplied
Information is hidden inside	unsupported interior-storage skin unless a carrier relation is declared
Information is preserved somewhere	the standing-bearing carrier and identity map must be shown
Radiation contains information	radiation is a ledger or carrier candidate; witness status requires an admissible map

28 Exterior witness resolution

A standing-bearing exterior witness does not repair or redescribe the interior. It resolves a burden in the exterior reconstruction problem. The distinction is decisive: exterior resolution changes the composition of R_H but not the endpoint status of whatever remains in R_H .

Theorem 28.1 (Exterior Witness Resolution). *If an exterior structure supplies a genuine standing-bearing witness relation for a burden $b \in B_H$, then $b \in W_{\text{ext}}(B_H)$ and $b \notin R_H$. The witness resolves that burden outside the residue. It does not repair, describe, or re-govern the interior endpoint.*

Proof. By definition, the exterior witness re-fixes b in the declared exterior witness algebra and preserves its standing quotient under lawful continuation. Therefore b no longer belongs to the unresolved remainder. The map lands in the exterior reconstruction structure; it supplies no operator on the horizon-isolated carrier and no cross-AMetric transport. Its effect is resolution of the burden, not modification of the residue's interior role-status. $\square$

Theorem 28.2 (Exterior Non-Repair of the Interior Endpoint). *Exterior skin, exterior ledgers, radiation records, Page-curve behavior, diagrams, and reconstruction language do not alter the endpoint status of R_H . A standing-bearing exterior witness relation may reduce R_H by resolving the burden it carries; it does not repair the remaining residue.*

Proof. Skin and ledgers have no endpoint-governing authority by themselves. A genuine witness relation acts only on the burden it re-fixes in the exterior reconstruction scope. The residual carrier and its governance-depth are unchanged for every burden left in R_H . Thus exterior structures either leave R_H unchanged or reduce its membership; they do not transform the endpoint-normal form of the nonempty remainder. $\square$

Control rule. **Skin describes; witnesses resolve; residue closes.** Skin retains its licensed exterior and metric-dynamical roles. Genuine exterior witness relations resolve only the burdens they re-fix. If $R_H \neq \emptyset$ and residual metric authority exhausts at maximum governance-depth, the endpoint theorem classifies R_H as New AMetric Boundary status.

29 Page curves, holography, islands, and diagrams at their proper roles

A Page curve may support a claim about entropy behavior, exterior unitarity, or coarse-grained reconstruction structure, but it is not itself a standing-bearing witness. It affects R_H only when paired with a role-fixed witness algebra or reconstruction map for a declared reference-standing burden.

Holographic and island programs likewise divide by role. The phrase “encoded on the boundary” is skin unless a target-fixing map is supplied. A boundary operator algebra or entanglement-wedge reconstruction map may function as a witness carrier when its domain, quotient, and target are explicit. An island formula used only as an entropy result remains a ledger or comparator. The map, not the slogan, carries standing.

Penrose diagrams remain useful conformal representations of a classical metric envelope. They are skin-level representatives of the licensed Einstein projection. They cannot become endpoint witnesses. At maximum governance-depth, the metric and causal structures that make the diagram meaningful no longer carry endpoint authority.

Table 9: Exterior comparators and possible witness structures.

Item	AASC endpoint classification
Page curve without target map	entropy/ledger profile; no burden resolved
Radiation flux without decoding map	exterior energy ledger; no burden resolved
“Encoded on the boundary”	skin unless a role-fixed map is supplied
Boundary operator algebra with target map	possible exterior witness carrier
Entanglement-wedge reconstruction map	possible exterior witness relation
Island formula as entropy result	ledger/comparator
Island reconstruction with target map	possible exterior witness relation
Penrose diagram	useful metric skin; never an endpoint witness

30 Residue scope lock

The residue formulation does not weaken the endpoint theorem. It fixes the theorem object after exterior roles have been correctly typed. If $R_H = \emptyset$, no unresolved interior burden remains for this theorem to classify. If $R_H \neq \emptyset$, the theorem concerns the standingless horizon-isolated remainder itself. Exterior dynamics may continue, radiation may be emitted, and exterior witnesses may resolve other burdens; none of those facts supplies metric, reference, or operator authority to the nonempty residue.

The endpoint implication is therefore

$$R_H \neq \emptyset \quad \wedge \quad \text{Succ}_{D_0}^{\text{res}}(R_H, \text{metric}) = \emptyset \quad \implies \quad \text{Status}(R_H) = B_A^*,$$

subject to the imported null-regime and endpoint-normal-form conditions proved below.

Part V

Maximum Governance-Depth and Exposure of the New AMetric Boundary

31 The geometric horizon is not the origin boundary

The black-hole horizon and the AMETRIC boundary occupy opposite positions in the dependency order. The AMETRIC boundary is prior to metricity. The horizon is a derivative structure inside the licensed Einstein projection

$$\Pi_g(A_G) = (M_4, g_{\mu\nu}), \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

It is identified only after metric comparison, causal typing, exterior/interior role separation, source-ledger structure, and a reconstruction scope are available. It therefore cannot be the same-scope

origin of the roles it presupposes.

Theorem 31.1 (Horizon non-origin theorem). *Let H_{BH} be a horizon role inside the licensed Einstein metric projection of D_0 . Then H_{BH} is a derivative metric boundary and cannot function as the AMETRIC licensing boundary for the same domain.*

Proof. To identify H_{BH} one needs at minimum a metric or metric projection, causal or trapping structure, a distinction between exterior and interior roles, and enough continuation discipline to distinguish a horizon from arbitrary representational skin. Those roles are downstream of the governance kernel. An AMETRIC boundary, by contrast, is prior to metricity and licenses the first admissibility-bearing regime. If H_{BH} is retyped as a licensing boundary, either a new domain is declared, a standing-bearing relation maps a new domain to the old one, a richer-scope theory is introduced, or hidden governance has been imported. None is ordinary same-scope horizon crossing. $\square$

This theorem is compatible with the claim that the black-hole interior eventually exposes a distinct AMETRIC boundary. The geometric horizon initiates the standing break. It does not itself perform origin work. The two boundaries must be kept separate:

Table 10: Boundary-role comparison

Feature	Geometric horizon	New AMetric Boundary endpoint
Dependency level	downstream of metric and causal structure	upstream of any admitted metric or causal structure
Primary role	obstructs or filters old-scope standing transport	determines eligibility for a new admissibility-bearing domain
Spatial status	representable in a metric model	not a metric location or hypersurface
Dynamics	local metric dynamics may remain regular	no operator or process is defined
Reference-standing burden	may isolate old-scope reference	no old-scope carrier package can cross
Scope effect	standing break inside D_0	boundary of any possible D_1

The distinction prevents two opposite errors. Treating the horizon as the origin boundary introduces new governance too early and contradicts the no-reset theorem. Refusing any later origin boundary because the horizon is not one ignores the exhaustion of residual metricity. The present result occupies the middle: the horizon starts the residual problem; maximum governance-depth exposes the New AMETRIC Boundary closure-status.

32 Three residual regimes

The standingless-interior analysis divides the horizon-isolated carrier into structural regimes rather than exact radial shells.

32.1 Near-horizon residual regime

Immediately after the old-scope standing break, residual metric, field, source, and bound-system organization may be high. The licensed Einstein projection may remain an excellent local description, a freely falling account can remain approximately ordinary, curvature may be modest, and the closed source ledger may remain writable. The equivalence-principle expectation of no local drama at a large horizon is therefore not contradicted. None of these local facts supplies old-scope standing transport.

Yet the status has changed for every burden retained in the unresolved horizon-isolated residue. The old exterior reference class no longer possesses standing in the parent continuation scope. The smooth local description is residual metric and source coherence, not proof of ordinary same-scope interior identity. Skin-level descriptions are already vulnerable because they depend entirely on the higher-load structures they present.

32.2 Intermediate governance-deficient regime

As residual successor support thins, different roles lose stability at different rates. Coordinate depth, foliation-relative simultaneity, and representative labels may cease to track stable structure. Particle identity becomes less secure as localization, field-carrier, and gauge support weaken. Mass and energy persist only as residual deformation and source patterns. Time and causality remain useful only while retained order and dependency are supported.

This regime is not a universal dynamical decay law. Governance-depth is role-relative. A robust source pattern may retain residual organization where a weak label does not. A particular quantum-gravity model could replace the carrier before the metric role exhausts. The theorem concerns the nonempty unresolved residue route, not every physical model.

32.3 Maximum governance-depth

Maximum governance-depth for the metric role is defined by

$$\text{Succ}_{D_0}^{\text{res}}(x, \text{metric}) = \emptyset.$$

The condition says that no post-break continuation of the crossed target preserves enough inherited organization for the licensed metric projection to remain a residual descriptor. It is not the largest value of a depth coordinate. It is the failure of the role that makes the Einstein metric, its source ledger, and any such coordinate physically comparative for the crossed carrier.

Definition 32.1 (Maximum governance-depth). For a role r , maximum governance-depth is the structural status at which residual successor support for r is empty. The black-hole closure theorem uses maximum governance-depth for metric comparison and the role families that depend on it.

Proposition 32.2 (Projection-bounded maximum governance-depth). *Maximum governance-depth is the exhaustion of residual support for the licensed Einstein metric projection as applied to the crossed carrier. It is not an equation-level claim that the Einstein tensor has become algebraically false, and it is not a numerical curvature threshold.*

Proof. The Einstein equation has standing as a projected theorem object only while metric com-

parison, source identity, quotient stability, and continuation retain role-support. Empty residual metric successor support removes that application domain for the crossed target. Formal equations may remain true in the classical representative, but the representative can no longer certify its own physical continuation. This is projection-level exhaustion, not contradiction of the equation inside its licensed envelope. $\square$

33 Endpoint classification rather than an attainment law

Maximum governance-depth is the endpoint antecedent of this manuscript. The theorem classifies the closure-status if and when the residual metric successor set is empty. It does not derive a universal astrophysical law that every black hole reaches this antecedent, and it does not identify a dynamical trajectory by which the antecedent is attained.

Proposition 33.1 (No universal attainment claim). *The New AMETRIC Boundary theorem is universal over endpoint candidates satisfying the maximum-governance-depth antecedent. It is not universal over all astrophysical black holes without that antecedent.*

The distinction is type-level rather than conciliatory. A physical model that prevents residual metric exhaustion exits the endpoint class. A model that reaches the endpoint antecedent remains subject to the full endpoint theorem.

At this status, phrases such as “further inside,” “later along the infalling trajectory,” “closer to the singularity,” and “at higher curvature” are not automatically false statements inside a metric model. They are unsupported as descriptions of the governed crossed carrier. Their residual role has ended.

34 Structural language discipline at the endpoint

Because maximum governance-depth removes the residual authority of metric time, causal process, and operator evolution, the endpoint cannot be described coherently as something that dynamically forms. The relevant necessity is modal and structural: all same-type fixed-domain alternatives have been exhausted.

Proposition 34.1 (No dynamic formation of the New AMetric Boundary). *The New AMetric Boundary does not form, emerge, evolve, nucleate, or occur as an event in the old residual metric regime. Its status obtains structurally as the closure classification of that regime.*

Proof. Formation, emergence, evolution, and occurrence require an admitted before/after order, a process carrier, and an operator or causal relation connecting stages. At maximum governance-depth those old-scope roles have no residual successor support. Using them to explain the endpoint would therefore presuppose the structures whose exhaustion defines the endpoint. The permissible claim is only that the endpoint profile satisfies the independently fixed AMetric normal form. $\square$

Table 11: Endpoint language discipline.

Avoid	Use
The boundary forms dynamically.	Boundary status obtains structurally.
The black hole creates a new universe.	Black-hole residual closure exposes the New AMetric Boundary endpoint.
Information enters the new domain.	No old-scope information is inherited; a new domain has its own information roles.
The interior transitions through the boundary.	The old interior role architecture exhausts; no transition operator is defined.
The endpoint occurs at a finite proper time.	No old-scope proper-time localization is licensed.

35 The domain-stack diagram

Figure 1 records the proof architecture. Arrows above the AMETRIC boundary are not transport arrows into a new domain; they are the order of classifications inside the parent-domain analysis. The dashed separation below B_A^* indicates no inherited map.

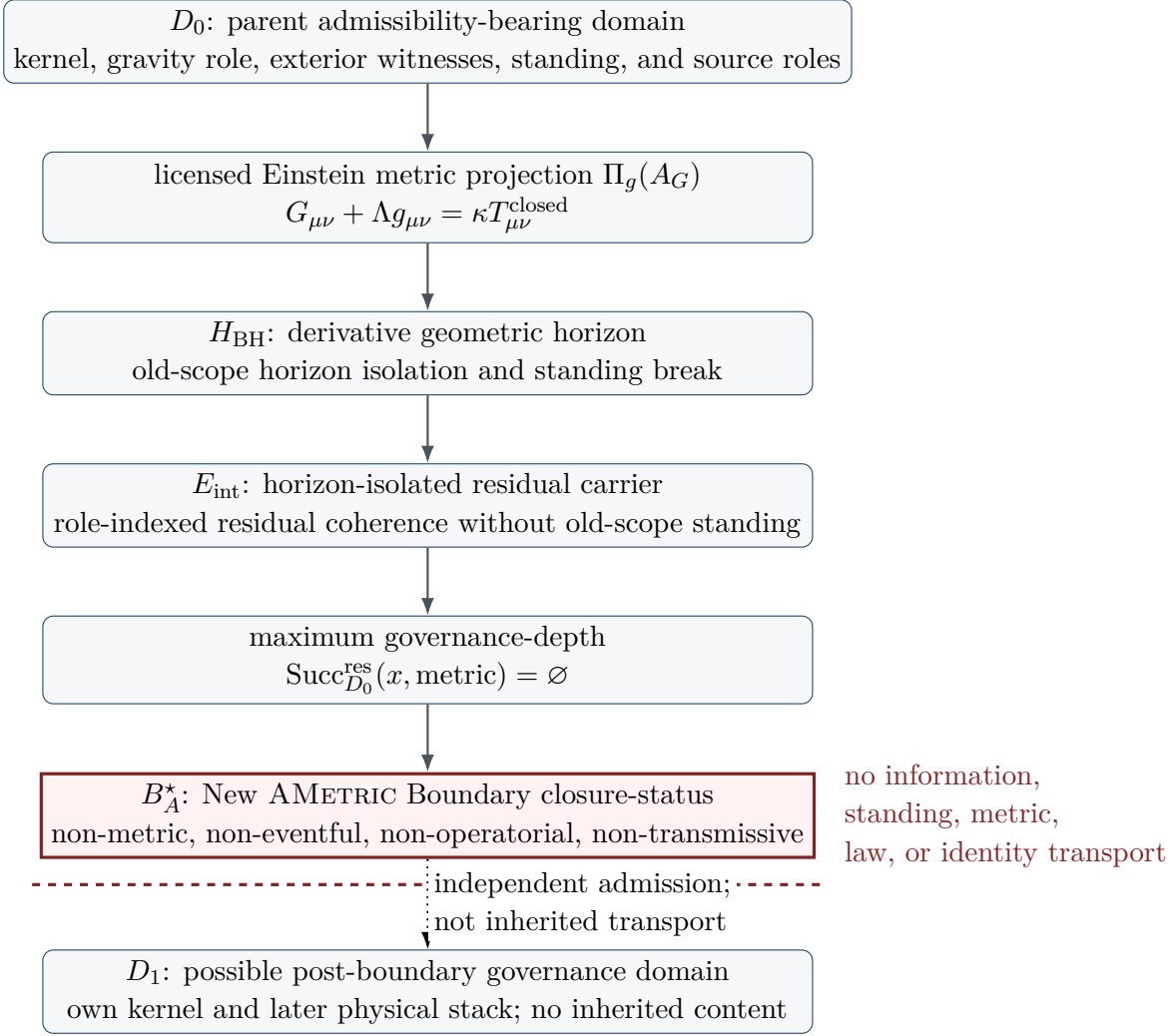


Figure 1: Structural order from the AASC gravity role through the licensed Einstein projection and parent-domain black-hole reconstruction to the New AMETRIC Boundary endpoint. The final dotted arrow represents independent post-boundary admission, not a process, crossing, or inherited map.

The figure is an external theorem diagram. It must not be read as depicting a worldline or a sequence of events occurring at the boundary. In particular, the dotted arrow to D_1 cannot carry labels from D_0 . Its role is only to show the type order: if a new domain is admitted, its governance is logically downstream of its own boundary status.

36 Why metric writability does not rescue the endpoint

A common objection begins with a correct observation: the licensed Einstein projection admits Schwarzschild or Kerr representatives on interior coordinate domains, and curvature invariants can be computed. The objection then infers that a metric singularity remains a coherent feature regardless of standing. That inference mistakes syntax for role authority.

Writability is a property of an external formal system. Metricity in AASC is quotient-stable

comparison inside a governed domain. The expression

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

remains the licensed minimal metric projection in its declared envelope [Maley, 2026c]. It does not certify that every carrier represented by the extended coordinates retains the standing, comparison, source identity, or measurement roles required for same-scope physical articulation. When those roles are residual, the equation is a residual projection. When residual metric support is empty, the equation may still be copied but cannot function as a metric description of the crossed target in the old scope.

Theorem 36.1 (No metric-projection self-certification). *The persistence of Einstein metric notation, source equations, or curvature identities beyond maximum governance-depth cannot certify residual or standing-bearing metricity for the crossed target.*

Proof. If the written Einstein equation alone certified metricity, a downstream projected tensor relation would regenerate the governance conditions that licensed it, and skin would generate tensor authority. But anchor–tensor–skin closure excludes such promotion. To restore tensor authority, a new metric role, comparison quotient, and governance domain must be declared. That is re-governance, not persistence of the old metric. $\square$

This theorem is the exact point at which the present manuscript goes beyond cautious language that singularities remain mathematical features of classical representatives. They remain features of the formal representative. They do not remain features of the governed residual carrier after the representative’s metric role-support has exhausted. The distinction is not optional once governance is granted priority over representation.

37 Endpoint alternative normal form

The endpoint exhaustion is not an enumeration of favored metaphors. Every same-type endpoint claim must carry at least one of six governance profiles.

Lemma 37.1 (Endpoint Alternative Exhaustion). *Any admissibility-relevant endpoint candidate for the old residual carrier is exactly one of:*

1. *metric-bearing;*
2. *operator-, process-, temporal-, or causal-bearing;*
3. *information-, reference-, carrier-, or conservation-ledger-bearing;*
4. *null or vacuous;*
5. *boundary/pre-admissible;*
6. *external third-domain comparison or bookkeeping.*

No seventh same-type endpoint profile remains inside the fixed comparison class.

Proof. An endpoint candidate either continues old descriptive roles or does not. If it continues metric comparison it is (1). If it continues transformation, order, or witnessed dependence it is (2).

If it continues identity-bearing content, storage, source balance, or transport it is (3). If it claims no role content it is (4). If it supplies no old role but fixes new eligibility or definedness it is (5). If it belongs only to a larger comparison language or changes no old classification, it is (6). Any proposed additional profile must change classification or continuation, reducing to one of (1)–(5), or change neither, reducing to (6). $\square$

At maximum governance-depth, (1) fails by residual metric exhaustion; (2) fails because time, causality, process, and operatorhood lack old-scope support; (3) fails by horizon standing break and absence of a common carrier, transport, or closed ledger; (4) fails as admitted endpoint by null-regime closure and otherwise collapses to mute pre-admissibility; and (6) is not an endpoint internal to the old black-hole regime. The only surviving same-type profile is (5), AMetric boundary status.

38 Endpoint route exhaustion

Before null-regime closure is imported, maximum governance-depth appears to present three endpoint routes:

1. metric endpoint, such as singularity or bounce;
2. total exhaustion into nullity;
3. AMETRIC boundary status.

The first branch fails because metricity has exhausted. The second branch divides into mute and admissible nullity. Mute nullity performs no endpoint classification and is itself AMETRIC pre-admissibility. Admissible nullity is contradictory. Thus the route set collapses to the third branch, which is the New AMetric Boundary endpoint.

Theorem 38.1 (Endpoint route-exhaustion theorem). *At maximum governance-depth in the target regime, no same-type endpoint classification remains except New AMETRIC Boundary status.*

Proof. A metric endpoint requires residual metric comparison, contrary to the maximum-depth condition. A dynamical endpoint additionally requires time and operators. An admissible null endpoint contradicts null-regime closure. A mute endpoint has no same-type classificatory force and is AMETRIC pre-admissibility. Therefore the remaining structural classification is AMETRIC boundary status. $\square$

The theorem derives a boundary classification, not a positive physical occupant. Positive post-boundary domain claims enter only after this result and require their own admission discipline.

Part VI

Exhaustion of False Endpoint Classes

39 No singularity at maximum governance-depth

In classical general relativity, singularity language is already disciplined. A spacetime singularity is not ordinarily a point belonging to the manifold in the same manner as regular events. The decisive formulations concern geodesic incompleteness, inextendibility, or curvature pathology within a metric-geometric framework [Penrose, 1965, Hawking and Ellis, 1973, Wald, 1984]. The AASC classification adds a prior question: does the metric-geometric framework retain role authority for the crossed carrier at the endpoint being described?

Lemma 39.1 (Singularity diagnostic versus AMetric closure). *A singularity is a failure diagnosed inside a still-operative metric-geometric regime. AMetric closure is the failure of the residual support required for metric-geometric diagnosis. Therefore AMetric closure is not a singularity, including a generalized singularity, unless a new metric-governed scope is declared.*

Proof. Curvature blow-up, geodesic incompleteness, inextendibility, and singular loci are predicates over metric, affine, or causal structures. They diagnose how such a structure fails or ends. Maximum governance-depth, by contrast, is defined by the absence of residual successor support for metric comparison itself. A predicate cannot remain authoritative after its target category has ceased to be instantiated for the crossed carrier. Calling that loss a singularity merely reimports the metric category whose authority has ended. $\square$

Decisive distinction. A singularity is a limit diagnosis internal to metric articulation; a New AMETRIC Boundary is the loss of the conditions under which metric articulation can diagnose a limit.

Theorem 39.2 (Singularity as the Metric-Internal Shadow of Projection Exhaustion). *A classical black-hole singularity is a diagnostic internal to the licensed Einstein metric projection. At maximum governance-depth, where residual support for that projection exhausts for crossed content, the singularity diagnostic no longer has the conditions of application. The endpoint-normal form is New AMETRIC Boundary status.*

Proof. The Einstein-projection bridge first fixes the metric regime as a licensed theorem object rather than primitive ontology. Curvature divergence, geodesic incompleteness, inextendibility, and singular loci are predicates internal to that projection. They can diagnose equation-level or metric-level failure only while enough metric and affine authority remains to state the diagnosis. Maximum governance-depth removes residual successor support for that role. The licensed projection cannot self-certify continuation beyond its own application domain, and a formally retained singular locus is then skin relative to the crossed target. The singularity is therefore the metric projection's internal shadow of a deeper projection-support exhaustion, not the final governance classification. $\square$

Claim order. The singularity is the licensed metric projection's name for its own limit; the New AMETRIC Boundary is AASC's classification of the loss of authority of that projection.

A singularity diagnostic requires at least four structures:

1. a metric or affine comparison structure;
2. a continuation class, such as geodesics or causal curves;
3. an identity criterion allowing stages or points to be compared;
4. a diagnostic predicate, such as incompleteness, divergence, or failed extension.

At maximum governance-depth, the first structure lacks even residual successor support. Consequently the others cannot remain standing-bearing or residual metric diagnostics in the old scope.

Theorem 39.3 (No singularity without metricity). *Let x be a reference-standing burden retained in the unresolved horizon-isolated residue R_H . If*

$$\text{Succ}_{D_0}^{\text{res}}(x, \text{metric}) = \emptyset,$$

then no singularity predicate whose application depends on metric, affine, causal, curvature, or extension structure is a licensed residual descriptor of x in D_0 .

Proof. Every listed singularity predicate requires a comparison or continuation structure inherited from metricity. Empty residual metric successor support means there is no old-scope continuation preserving that role. Applying the predicate after exhaustion would either treat notation as tensor authority, import a new metric domain without declaration, or use a third-domain formal model as if it were the old carrier. The first is skin promotion, the second is new governance, and the third is scope transport. Hence no same-scope residual singularity predicate remains. $\square$

Corollary 39.4 (Singularity displacement). *The classical singularity is not dynamically resolved or replaced. Its diagnostic role is displaced by the exhaustion of the metric articulation required to state it.*

“Displacement” is chosen carefully. The theorem does not describe a physical mechanism that removes a singularity. It says that the singularity never reaches the final governance classification. As long as residual metricity remains, a singularity can serve as an exhaustion signature of that metric projection. At maximum governance-depth, even that residual use ends. The closure-status is not a metric endpoint but the boundary of metric applicability.

39.1 Schwarzschild and Kerr representatives

The Schwarzschild representative distinguishes the regular horizon at $r = 2M$ from the classical pathology at $r = 0$. Nothing in the present theorem invalidates the exterior solution or its analytic extension as mathematics. The AASC claim is that $r = 0$ cannot be promoted into a standing-bearing endpoint of crossed content after residual metric comparison has exhausted. The coordinate r and curvature expressions are then skin relative to the target.

The Kerr case adds ring-singularity, inner-horizon, and extended-region structures. These require still richer metric, causal, and continuation commitments. If mass inflation, high-curvature correction, or quantum transition is supplied, the model has declared new tensor load or a richer

scope. If no such structure is supplied and residual metricity exhausts, the ring singularity is no more available as an endpoint than the Schwarzschild central singularity. The theorem is not sensitive to the shape of the classical pathology because its target is the prior role on which all such shapes depend.

39.2 Relation to classical singularity theorems

The singularity theorems establish conditional results inside Lorentzian geometry. Their premises and conclusions remain mathematically valid in the declared Einstein metric projection [Penrose, 1965, Hawking and Ellis, 1973, Wald, 1984]. The present theorem is a scope theorem about the authority of that model for the standingless carrier.

Theorem 39.5 (Metric-theorem scope). *A theorem whose objects and predicates are defined in a metric-geometric category retains physical role authority for the crossed carrier only while the metric-comparison role remains licensed or residually supported in the declared scope.*

Proof. A theorem cannot carry more target fixation than its domain of application. If its domain is no longer instantiated for the target, the theorem may remain true of its abstract model but does not classify the exhausted carrier. Applying it anyway is domain substitution. $\square$

This result preserves the proper scope of classical relativity. It prevents a mathematical theorem from being assigned an ontological burden it was not designed to carry. General relativity can remain exact in its licensed projection while failing to provide the final governance description of the crossed carrier. AASC therefore derives and bounds the metric projection rather than repudiating it.

40 No remnant, archive, or storage endpoint

A remnant proposal ordinarily invokes a late carrier with enough internal capacity to retain distinctions associated with earlier infalling states. Capacity is not sufficient for standing. A storage claim requires at minimum a carrier, an encoding relation, a distinguishability structure, a continuation identity, and some relation by which the stored distinctions count as the same information rather than newly labeled states.

At maximum governance-depth, no old-scope carrier identity crosses the New AMETRIC Boundary. A post-boundary domain may contain arbitrarily rich structure under its own governance. That richness does not encode D_0 unless an inherited relation exists, and the no-crossing theorem will exclude such a relation.

Theorem 40.1 (No AMetric remnant). *The New AMETRIC Boundary closure-status of the horizon-isolated carrier cannot function as a remnant, archive, memory, storage register, or hidden information reservoir for D_0 .*

Proof. Each proposed role requires a standing-bearing identity relation linking a later carrier state to earlier D_0 distinctions. Old-scope standing is undefined for the burden retained in R_H . Residual patterning is not a standing-bearing carrier package. At the AMETRIC boundary, no transport or encoding operator is defined. If a later D_1 state is declared to encode D_0 , the declaration either

imports an external map, presupposes a parent exterior witness relation that would already have removed the burden from R_H , or assigns identity without witness. None is storage across the boundary itself. $\square$

A Planck-scale object in a metric model may still be studied as a remnant candidate. The theorem says that such a proposal must operate before AMetric closure or in a richer theory that never enters the target antecedent. It cannot use the closure boundary as a container.

41 No primitive information sink and no destruction event

The phrase “information is destroyed at the singularity” combines two unsupported roles. First, it presupposes a singularity. Second, it treats information as a cargo undergoing a temporal event. The present framework denies both at maximum governance-depth.

Information requires

(carrier, encoding, equivalence, reference, witness, use condition).

The horizon standing break removes old-scope re-fixability. Residual coherence can preserve pattern but not the full information role. At the New AMETRIC Boundary, there is no event or operator by which information could be destroyed.

Theorem 41.1 (No destruction event). *Maximum governance-depth cannot be classified as an event in which old-scope information is destroyed.*

Proof. Destruction requires a prior information carrier, a temporal transition, an identity criterion across that transition, and an operator or process that maps the carrier to a non-information state. The old carrier package lacks standing for every burden retained in R_H , and time, identity transport, and operatorhood are unavailable at the AMETRIC boundary. Therefore the event description is ill-typed. $\square$

The correct result is carrier-package non-persistence, not destruction. A parent carrier package retained in R_H has no admissible continuation into a post-boundary domain. This is a stronger statement than “we cannot recover it” but a different statement from “a physical process erased it.”

42 No dynamical bounce or transition surface

Bounce models are legitimate physical proposals when they specify a dynamical regime, matching conditions, quantum state, and continuation map. The present theorem targets the claim that AMetric closure itself is such a bounce.

A bounce requires an ordering

$$\text{pre-bounce} \prec \text{bounce} \prec \text{post-bounce},$$

a state or geometry on each side, and a law connecting them. These are temporal, metric, causal, and operator roles. An AMETRIC boundary is prior to precisely those roles.

Theorem 42.1 (No AMetric bounce). *No dynamical bounce, tunneling trajectory, phase transition, causal branching, or metric matching condition occurs at an AMETRIC boundary as such.*

Proof. Each candidate is an operation or process. Boundary non-action and operator undefinedness exclude operations at the AMETRIC boundary. If the process is defined in a surrounding theory, then that theory supplies a larger governed domain in which the purported boundary is no longer AMetric relative to the process. The model may be physically coherent, but it is not a same-role AMetric closure model. $\square$

This theorem distinguishes the present view from false-vacuum bubble and laboratory-universe proposals such as Farhi–Guth–Gruen scenarios [Farhi et al., 1990]. Those models use semiclassical or quantum tunneling dynamics. An AASC post-AMetric domain is not produced by a tunneling trajectory across its origin boundary. External analogies may be useful, but the role structures differ.

43 No wormhole or throat across the boundary

A wormhole is a geometric or path-integral object. A throat has metric size; a bridge relates regions; a path integral assigns amplitudes; a topology change requires a larger formal structure. None can serve as a transport channel across an AMetric boundary without changing the theorem object.

Theorem 43.1 (No AMetric wormhole). *An AMETRIC closure boundary cannot be a wormhole mouth, throat, bridge, or geometric topology-changing interface connecting D_0 to D_1 .*

Proof. A wormhole relation requires a geometric or quantum-gravitational carrier defined on a scope containing both sides. If such a scope exists, it is a third domain D_* with its own governance and comparison structure. The relation is internal to D_* , not inherited transport across the AMetric boundary of D_0 or D_1 . If no D_* is declared, the required metric or path-integral operator is absent. $\square$

The theorem does not refute the wormhole or baby-universe literature [Coleman, 1988b, Giddings and Strominger, 1988, 1989, Marolf and Maxfield, 2020]. It states that those programs, insofar as they possess shared amplitudes, Hilbert spaces, ensembles, or correlations, are larger-scope constructions and not instances of absolute no-inheritance AMetric origination.

44 No admissible null endpoint

The null route deserves separate treatment because it can appear to avoid all physical overclaim. One might say that after metricity fails there is simply no interior description and no successor. The AASC response distinguishes raw muteness from an endpoint classification.

If the statement “there is nothing” is entirely mute, it identifies no target, performs no comparison, and carries no role in the black-hole theorem. It is not an endpoint candidate. If it is used to classify the closure of E_{int} , it enters the evaluated fragment and requires identity, boundary typing, and classificatory standing. It is then not total vacuity.

Theorem 44.1 (No black-hole null endpoint). *There is no same-type admissible black-hole endpoint whose total standing-bearing content is AASC vacuity.*

Proof. An endpoint candidate with admissibility-relevant force must fix the domain and target, distinguish endpoint from non-endpoint, and preserve that classification under redescription. Positive-side admissibility additionally entails standing, transport, quotient identity, and classificatory legitimacy. Total AASC vacuity denies all such governance content. Therefore the endpoint both has and lacks the required roles. If the candidate refuses those roles, it is mute AMetric pre-admissibility, not a same-type endpoint. $\square$

The theorem is the reason the phrase “metricity loses meaning” does not leave an unclassified void. The loss itself has a boundary profile. The only coherent status of mute non-metric non-operatorial non-transportive closure is AMetric pre-admissibility.

Theorem 44.2 (Null endpoint collapse-to-AMetric). *At maximum governance-depth, a proposed null endpoint has exactly three dispositions:*

1. *if it is completely mute, it is AMetric pre-admissibility and has no competing endpoint force;*
2. *if it is admissibility-relevant, it carries non-null governance and is not total vacuity;*
3. *if it is claimed as an admitted null endpoint, it contradicts null-regime closure.*

Therefore nullity does not compete with the New AMetric Boundary endpoint. It either collapses into AMetric boundary classification, becomes non-null, or is contradictory.

Proof. Mute nullity has no identity, selector, transport, quotient, gate, or comparison role and is therefore the pre-admissible AMetric profile under another description. Admissibility-relevant nullity must fix and classify a target and thereby introduces governance content. Admitted total vacuity requires and denies the positive-side roles simultaneously. These cases exhaust the null route. $\square$

45 No inherited law or constant

A subtler endpoint story says that no particular information crosses but that the new domain inherits the parent’s laws, symmetries, dimensionality, or constants. This is still inheritance. Laws and constants are not free-standing objects. They are roles inside a governed physical description.

A metric signature presupposes metricity. A coupling constant presupposes a theory and comparison scale. A symmetry presupposes an action or structure on a domain. A conservation law presupposes identity and continuation. None can cross as old-scope authority at the AMetric boundary.

Theorem 45.1 (No law inheritance). *A post-AMetric domain does not inherit the metric signature, dimensionality, constants, field content, symmetry group, conservation laws, or dynamical equations of D_0 merely because its boundary is externally associated with black-hole closure in D_0 .*

Proof. Each alleged inheritance is an admissible transport of structured content. Boundary non-transmissivity excludes such transport. If D_1 independently instantiates an isomorphic law package, that is structural recurrence or external equivalence, not inheritance. Establishing the isomorphism requires a third-domain comparison and does not create an internal relation. $\square$

This theorem is central to the explanation of existence. The new domain is not a compressed copy, genetic descendant, or lawful continuation. It begins with governance, and any later physics is its own licensed realization.

46 Endpoint false-route matrix

Proposed endpoint	Role it must retain	Reason it fails at maximum governance-depth
Classical singularity	metric/affine diagnostic authority	the diagnostic category is no longer supported
Remnant or archive	persistent identity-bearing carrier and retrieval map	no inherited carrier identity or transport exists
Information sink	information carrier and temporal loss map	old information lacks standing; destruction is process language
Information-transfer baby universe	common identity and cross-boundary transport	AMetric non-transmissivity makes the preservation relation incoherent
Bounce or tunneling transition	time, causal order, operator evolution	no event or operator is defined at the boundary
Wormhole throat	common geometry or path-integral carrier	belongs to a larger governed scope, not absolute AMetric closure
Inherited law package	cross-domain transport of metric, constants, or symmetry	target domain requires independent governance
Admissible nullity	positive-side admission without governance content	structurally contradictory
Mute nothingness	no evaluative or classificatory content	AMetric pre-admissibility, not a rival endpoint
New AMetric Boundary	absence of inherited metric, process, operator, transport, and null closure	satisfies the independently fixed endpoint normal form

47 Endpoint-exhaustion theorem

The false endpoint classes can now be collected. Their exhaustion is the negative half of the structural necessity proof for the New AMetric Boundary.

Theorem 47.1 (False endpoint exhaustion). *In the target black-hole regime, maximum governance-depth is not admissibly classifiable as any of the following: ordinary same-scope GR interior, classical singularity, remnant, information archive, information-destruction event, information-transfer baby universe, dynamical bounce, wormhole bridge, inherited-law transition, or admissible nullity.*

Proof. Ordinary interiority fails by standingless-interior governance. Singularity fails by metric-role exhaustion. Remnant and archive fail by absent carrier identity and map. Destruction and bounce fail by absent time and operatorhood. Wormhole and inherited law require larger-scope transport. Admissible nullity contradicts null-regime closure. The candidates exhaust the principal metric, dynamical, informational, geometric, and null endpoint routes. Any unlisted same-type proposal

must identify which role it supplies; if it changes neither endpoint classification nor continuation, it is bookkeeping, and if it supplies new role structure it is re-governance or scope change. $\square$

Part VII

The New AMetric Boundary Endpoint and Absolute No-Crossing

48 The endpoint theorem

The preceding Parts license the Einstein metric projection, type exterior structures, define the unresolved horizon-isolated residue, and exhaust the false endpoint classes. The central result can now be stated in fixed-domain form.

Theorem 48.1 (New AMetric Boundary Endpoint Theorem, residue form). *Let D_0 be a fixed admissibility-bearing parent domain whose AASC gravity role has the faithful licensed metric projection*

$$\Pi_g(A_G) = (M_A, g_{\mu\nu}), \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Let B_H be the reference-standing burden generated by horizon isolation. Let $W_{\text{ext}}(B_H)$ be the portion resolved by standing-bearing exterior witness relations, and define the unresolved horizon-isolated residue

$$R_H := B_H \setminus W_{\text{ext}}(B_H).$$

Assume:

- (A1) $R_H \neq \emptyset$;
- (A2) the horizon standing-transport operator is undefined on the carrier corresponding to every burden in R_H ;
- (A3) no independent governance is declared for R_H before metric-role exhaustion;
- (A4) retained organization, where representable, has role-indexed residual-coherence status;
- (A5) at maximum governance-depth,

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{metric}) = \emptyset;$$

- (A6) total AASC vacuity is unavailable as an admissible same-type endpoint.

Then the closure-status of R_H , understood as exhaustion of the residual authority of the licensed Einstein projection for the unresolved burden, is a New AMETRIC Boundary B_A^ . It is not a metric singularity, remnant, archive, dynamical event, bounce, wormhole passage, transfer endpoint, primitive information store, or admissible null regime. No parent first-law, conservation, or source-closure ledger can be used as a backflow principle for post-boundary contents. This is endpoint classification under the stated residue and exhaustion antecedents, not a universal empirical attainment theorem.*

Proof. The Licensed Einstein Projection Theorem fixes the metric regime whose endpoint authority is being analyzed. The residue construction removes from the theorem object every burden genuinely re-fixed by an exterior witness relation and leaves R_H as the nonempty unresolved remainder. By (A2), old-scope standing does not transport for the burdens retained in R_H . By (A3)–(A4), remaining organization is residual rather than standing-bearing and no independent interior kernel has been introduced. By (A5), the metric-comparison and metric-facing source roles of the licensed projection have no residual successor support for the residue. Therefore no endpoint classification requiring metricity, affine comparison, temporal order, causal dependence, a standing-bearing carrier package, or operator evolution has a same-scope domain of application. Singularity, bounce, remnant, storage, destruction, and passage descriptions consequently fail.

The closure cannot be an admissible null regime by (A6). If completely mute, the proposed null endpoint has no same-type classificatory force and is AMetric pre-admissibility. If admissibility-relevant, it necessarily carries non-vacuous boundary, identity, and classification roles. The endpoint therefore satisfies the independently fixed strict AMetric profile: no metric, temporal, causal, probabilistic, modal, selector, parameter, process, or operator structure has boundary authority; no old admissible structure is transported; and any positive successor requires independent gating. Together with the imported null-regime closure condition, $\text{AENF}(P^*)$ holds. Hence the closure-status of R_H is B_A^* , a New AMetric Boundary. $\square$

The nontrivial content is not the definitional observation that metric descriptors fail when metric-comparison support exhausts. The theorem closes every rival endpoint category by identifying the roles each would require. Exterior skin and ledgers do not enter that closure as hidden endpoint authorities: skin is role-demoted, and genuine witness relations have already resolved the burdens they carry before R_H is formed.

The theorem does not assert that B_A^* is an object located in the old metric interior. Once metric depth exhausts, the phrase “located in” has lost role support. The notation B_A^* names the closure-status exposed by analysis of the nonempty residue.

Corollary 48.2 (No metric localization). *There is no old-scope radius, proper time, affine parameter, curvature threshold, hypersurface, Planckian shell, or causal boundary that identifies B_A^* as a standing-bearing location.*

Proof. Every candidate is a metric, affine, temporal, or causal descriptor. The theorem antecedent includes exhaustion of residual metric comparison. Any such localization is therefore formal skin or new-scope structure, not an old-scope endpoint coordinate. $\square$

49 Strict AMetric and combined AASC endpoint-normal-form satisfaction

The conclusion is not obtained by defining AMetric boundary as whatever remains after metric failure. The strict profile was fixed independently in Part II. Let P^* denote the maximum-governance-depth endpoint profile. The proof obligation has two sourced components: show that P^* satisfies the strict predicate ANF, then combine that result with the separately imported null-regime closure clause to obtain AENF.

Theorem 49.1 (AASC Endpoint Normal-Form Satisfaction). *The endpoint profile P^* satisfies the strict AMetric boundary normal form:*

- (B1) *no old-scope metric, affine, temporal, or causal authority remains;*
 - (B2) *no event, process, transition, or execution occurs at the endpoint;*
 - (B3) *no old-scope admissible operator has the endpoint in its domain;*
 - (B4) *no old-scope standing, reference, license, or redescription rule is transported through it;*
 - (B5) *no selector, rank, metric, probability, typicality rule, parameter, index, or canonical representative carries endpoint authority;*
 - (B6) *undefinedness is domain non-membership, not a third standing value;*
 - (B7) *no same-act repair restores failed standing;*
 - (B8) *every positive successor domain requires independent gating and domain-of-application closure.*
- Consequently*

$$\text{ANF}(P^*)$$

holds. Since admissible total vacuity is independently excluded and endpoint alternatives are fixed-domain exhausted,

$$\text{AENF}(P^*) = \text{ANF}(P^*) \wedge \neg \text{NullAdm}(P^*) \wedge \text{EndpointRouteClosure}(P^*)$$

also holds.

Proof. (B1) follows from empty residual metric successor support and the dependence of time and causality on metric/governance roles. (B2) and (B3) follow because process and operatorhood presuppose admitted temporal and transformation domains. (B4) follows from horizon standing failure and AMetric non-transmissivity. (B5) is the pre-fixational no-selector/no-parameter result. (B6) is bivalence on the evaluated fragment. (B7) is irreversibility. (B8) is cross-domain target gating. These strict clauses were specified without reference to black holes and establish ANF. The no-null conjunct is supplied by the separate structural inadmissibility theorem, and the route-closure conjunct by fixed-domain endpoint exhaustion. Their combination establishes AENF and is therefore a sourced endpoint-normal-form theorem rather than a relabeling convention. $\square$

Corollary 49.2 (Non-circularity). *The claim that maximum governance-depth is a New AMetric Boundary is not equivalent to the stipulation that “whatever is non-metric is AMetric.” It follows only after the independently specified strict AMetric clauses and the separately sourced null-regime closure condition have been verified.*

A merely non-metric formalism need not be AMetric. It could retain order, probability, operatorhood, causal process, a selector, or a larger-scope transport map. The black-hole endpoint is AMetric because all such same-scope authority has been closed, not merely because a Lorentzian metric is absent.

50 Structural necessity of the endpoint

The word “must” is now available in its proper sense. It does not mean that a boundary is produced by a process. It means that no same-type fixed-domain endpoint alternative remains.

Theorem 50.1 (Structural Necessity of the New AMetric Boundary). *Given the licensed Einstein projection, a nonempty unresolved horizon-isolated residue R_H , absence of residue-level re-governance, residual metric-comparison exhaustion, singularity-diagnostic failure, null-regime closure, and boundary operator undefinedness, the endpoint of R_H must be classified as a New AMetric Boundary.*

Proof. A metric endpoint is unavailable because metric comparison has exhausted. A singularity is unavailable because it is a metric diagnosis. A remnant, archive, or primitive information sink is unavailable because the required carrier package and standing-bearing transport are absent. A bounce, tunneling event, topology change, or causal branch is unavailable because time, causality, and operatorhood are absent. Admissible nullity is contradictory, while mute nullity is AMetric pre-admissibility and supplies no rival endpoint classification. Any remaining non-AMetric difference must either add one of the closed roles, thereby declaring new governance or a larger scope, or change no standing-bearing fact and collapse to bookkeeping. The endpoint therefore satisfies the combined AASC endpoint-normal form, whose boundary component is the strict AMetric normal form, and no same-type alternative survives. $\square$

Remark 50.2 (Meaning of structural necessity). The theorem does not assert that the boundary “forms.” Formation would require an old-scope before/after relation and an evolution operator. Structural necessity means route exhaustion: once the antecedent holds, New AMetric Boundary status is the only admissible endpoint classification.

51 The absolute no-crossing theorem

The endpoint theorem applies to the unresolved residue rather than to burdens already resolved by exterior witness relations. Can any old-scope standing-bearing structure carried by R_H pass into a newly admitted domain D_1 ? The answer is not merely that transfer is unproven. Same-content crossing is incoherent at an AMetric boundary because crossing requires an admissible transport map and the boundary is prior to operatorhood.

Let $\mathcal{S}(D)$ denote the standing-bearing structure of domain D : reference classes, witnesses, operators, physical roles, laws, and the admissibility-relevant invariant bundle. A crossing map would have type

$$\chi : \mathcal{S}(D_0) \longrightarrow \mathcal{S}(D_1).$$

To preserve old information or identity, χ must fix what is transported, maintain equivalence under redescription, and license the relation as the same target. Thus χ is an admissible standing-transport operator. No such operator is defined at B_A^* .

The same reasoning applies to the Einstein source ledger. The tensor $T_{\mu\nu}^{\text{closed}}$ has standing only as the metric-facing extraction of parent source roles. Covariant writability and divergence compatibility do not turn it into a cross-AMetric source or energy channel.

No-crossing / no-backwards result. No old-scope carrier package, reference, standing, law, metricity, causality, time, energy or source ledger, stress-energy identity, thermodynamic law, field identity, particle identity, memory, quantum branch label, or global-state coordinate carried by the unresolved residue crosses the New AMetric Boundary. Exterior witness relations resolve their burdens before residue classification; they do not cross the boundary or repair the residue.

Theorem 51.1 (Absolute No-Crossing Theorem). *No old-scope standing-bearing structure crosses B_A^* into a post-boundary domain as inherited content. In particular, no admissible map preserves any of the following across the boundary:*

1. *target or reference identity;*
2. *standing or admissibility status;*
3. *information carriers, encodings, equivalence relations, or reconstruction maps;*
4. *metric, affine, temporal, causal, modal, or process structure;*
5. *matter, field, particle, source, mass, charge, angular-momentum, energy, or stress-energy identity;*
6. *laws, constants, symmetries, first-law relations, conservation ledgers, exchange closures, or boundary conditions;*
7. *witness records, memories, correlations, measurements, or calibration chains;*
8. *Hilbert-space labels, branch indices, global-state coordinates, or ensemble parameters.*

Proof. Each listed item is a role-bearing structure rather than an untyped substance. Persistence requires a relation identifying the same item or role on both sides. Such a relation must be admissible, redescription-stable, and defined on the target. At an AMetric boundary no operator, transport rule, metric, causal relation, temporal order, selector, or inherited license is defined. If the relation is supplied by D_0 , it attempts to govern D_1 without independent target gating. If supplied by D_1 , it presupposes D_1 and its reference structure before admission. If supplied by a third scope D_* , it is an object of that external scope, not internal inheritance between D_0 and D_1 . Therefore no old-scope inherited map exists. $\square$

The theorem is absolute in the role sense, not in the trivial metalogical sense. A third mathematical theory may mention or model both domains. What it may not do is convert its own indexing relation into an internal parent-to-successor identity without supplying a new common governance scope, in which case the theorem object has changed.

Theorem 51.2 (No Interior-Instantiation Proof). *No conclusion available to AASC endpoint analysis of D_0 can establish the contents, laws, metricity, energy inventory, population, observers, history, or populated-existence of a post-boundary domain D_1 . Formally,*

$$\text{AASC}(D_0) \not\models \text{Contents}(D_1), \quad \text{AASC}(D_0) \not\models \text{Exist}_{\text{pop}}(D_1).$$

Proof. A proof of D_1 -internal content would have to fix its target, distinguish it from alternatives, and preserve that identity under a relation available to D_0 . Such a proof therefore requires a

witness, transport rule, common quotient, or standing-bearing map from D_0 to D_1 . The Absolute No-Crossing Theorem excludes each route. A third-domain theory may represent both domains, but its statements are internal to that third domain and do not become D_0 -internal cosmography. $\square$

Corollary 51.3 (No parent-scope cosmography beyond the boundary). *No parent-scope metric extension, Hilbert-sector label, branch index, path-integral saddle, or conservation ledger can establish the internal geography, population, or populated-existence of D_1 while the endpoint remains absolute AMetric closure relative to D_0 .*

Remark 51.4 (Why D_1 remains a legitimate conditional variable). The symbol D_1 is introduced to state a conditional no-inheritance theorem, not to assert populated-existence. If a post-boundary domain is independently admitted, the theorem determines its relation to D_0 : none of its standing-bearing content is inherited. The variable is required to state that conditional relation even though D_0 cannot prove D_1 's contents.

Endpoint derivability and non-instantiation.

$$\text{AASC}(D_0) \vdash \text{Status}(\text{GDepth}^*) = B_A^*, \quad \text{AASC}(D_0) \vdash B_A^* \sim_{\text{Origin}} B_A^{\text{ours}},$$

while

$$\text{AASC}(D_0) \not\vdash \text{Contents}(D_1), \quad \text{AASC}(D_0) \not\vdash \text{Exist}_{\text{pop}}(D_1).$$

The first line records theorem-level endpoint classification and origin-condition comparison. The second denies any post-boundary cosmography or populated-existence conclusion from the same parent-domain analysis.

51.1 Old-scope reference-standing packages do not cross

Bare information-language is not the theorem object. Let

$$i_0 = (c_0, e_0, \sim_0, r_0, w_0, u_0)$$

be a parent carrier package whose reference-standing burden remains in R_H . Persistence in D_1 would require a map preserving carrier, encoding, sameness, reference, witness, and licensed use. A merely similar state in D_1 is not the same package.

Corollary 51.5 (Old-Scope Carrier-Package Non-Persistence). *For every parent carrier package i_0 represented in the unresolved residue,*

$$\text{Inherit}_{D_0 \rightarrow D_1}(i_0) = 0.$$

No branch index, Hilbert-space label, remnant state, bit pattern, or post-boundary record can count as persistence of i_0 without the prohibited cross-boundary map.

This is not a statement that an information quantity becomes numerically zero or that an event erases it. It states that the old carrier-package inheritance predicate is false. Burdens already resolved by exterior witness relations are not in R_H and are not targets of this corollary.

Proposition 51.6 (No AMetric-Boundary-Internal Quantum Channel). *No quantum channel*

$$\mathcal{N}_A : \mathcal{H}_{D_0} \longrightarrow \mathcal{H}_{D_1}$$

is defined by B_A^ as an inherited boundary-crossing map. A larger theory may represent both sectors in a common Hilbert structure, but that channel is internal to the larger governed theory and changes the theorem scope; it is not a channel supplied by the AMetric boundary.*

51.2 Old-scope energy and first-law ledgers do not cross

An energy ledger persists only if the same energy-bearing quantity, comparison relation, exchange boundary, temporal or retained-consequence ordering, and source/transport closure remain fixed. None is inherited across B_A^* .

Corollary 51.7 (Old-Scope Energy Non-Transfer). *For every parent energy or source-ledger class e_0 ,*

$$D_0 \not\rightarrow_{\text{Energy}} D_1, \quad \text{Transfer}_{D_0 \rightarrow D_1}(e_0) = 0.$$

This does not imply $\text{Create}(e_0)$ or $\text{Destroy}(e_0)$. Creation and destruction are dynamical predicates unavailable at the boundary.

51.3 No causal production relation

Causation is witnessed dependence across an admitted temporal order. No such order crosses B_A^* .

Corollary 51.8 (No parent-domain causation). *There is no internal causal relation C_{01} under which the black hole in D_0 produces D_1 across B_A^* .*

Externally, one may say that analysis of a black hole exposes the New AMetric Boundary normal form. That is a relation between propositions in the theorem, not a causal relation between domains.

52 No backwards standing relation

The phrase “no backwards relationship” must be typed carefully. The theorem does not deny that an external theorist can mention D_0 and D_1 in one model or sentence. It denies an internal or transported admissibility-bearing relation by which D_1 inherits, remembers, constrains, encodes, signals to, or re-fixes D_0 .

Definition 52.1 (Backwards standing relation). A backwards standing relation between D_0 and D_1 is a relation represented internally in either domain, or transported between them, that preserves or re-fixes parent standing, reference, law, metricity, causality, information, memory, or identity.

Theorem 52.2 (No Backwards Relation Theorem). *No backwards standing relation exists between a post-AMetric domain D_1 and parent domain D_0 . Symbolically,*

$$D_0 \not\rightarrow_{\text{Stand}} D_1 \quad \text{and} \quad D_1 \not\rightarrow_{\text{Stand}} D_0.$$

Proof. A relation represented in D_0 would require D_0 to contain standing-bearing access to D_1 beyond the boundary, contradicting no-crossing. A relation represented in D_1 would require inherited reference to D_0 , also contradicting no-crossing. A relation represented only in a third domain D_* is external to both and does not become an internal relation merely because it compares them. Therefore no backwards standing relation exists. $\square$

52.1 External third-domain comparison

The following distinction is permitted and necessary:

$$\text{Compare}_{D_*}(D_0, D_1)$$

may be defined in a third domain D_* , while

$$\text{Transport}_{D_0 \rightarrow D_1}(x)$$

is denied at the New AMetric Boundary. A third-domain model can classify both domains, assign them positions in a formal ensemble, or exhibit structural isomorphisms. Those are facts of D_* . They do not establish parent-to-successor inheritance.

Remark 52.3 (External analogy). The theorem permits

$$D_0 \sim_{\text{origin-architecture}} D_1,$$

meaning that both are externally classified as downstream of an AMetric boundary and independent governance. It denies

$$D_0 \rightsquigarrow D_1$$

as an internal genealogy. Structural analogy is not inheritance.

53 No-inheritance normal form

Let

$$\mathcal{I}_0 = (K_0, M_0, T_0, C_0, F_0, \mathcal{L}_0, \text{Info}_0)$$

be the parent domain's governance, metricity, time, causality, field/matter stack, law package, and information structure. Let

$$\mathcal{I}_1 = (K_1, M_1, T_1, C_1, F_1, \mathcal{L}_1, \text{Info}_1)$$

be a post-boundary domain's corresponding package. The theorem does not require the components of $\mathcal{I}_1$ to differ extensionally from those of $\mathcal{I}_0$. It requires that they are not related by inherited standing:

$$\begin{aligned} K_0 \not\rightsquigarrow K_1, \quad M_0 \not\rightsquigarrow M_1, \quad T_0 \not\rightsquigarrow T_1, \quad C_0 \not\rightsquigarrow C_1, \\ F_0 \not\rightsquigarrow F_1, \quad \mathcal{L}_0 \not\rightsquigarrow \mathcal{L}_1, \quad \text{Info}_0 \not\rightsquigarrow \text{Info}_1. \end{aligned}$$

An external isomorphism between components would show recurrence or structural equivalence, not inheritance. If both domains support Lorentzian metric projections, the similarity arises because each independently licenses a metric role, not because a signature crossed the boundary.

54 The closure square

Figure 2 displays the logical relations. The dotted arrow is an external type-order marker for possible independent admission; it is not a crossing, process, or genealogy.

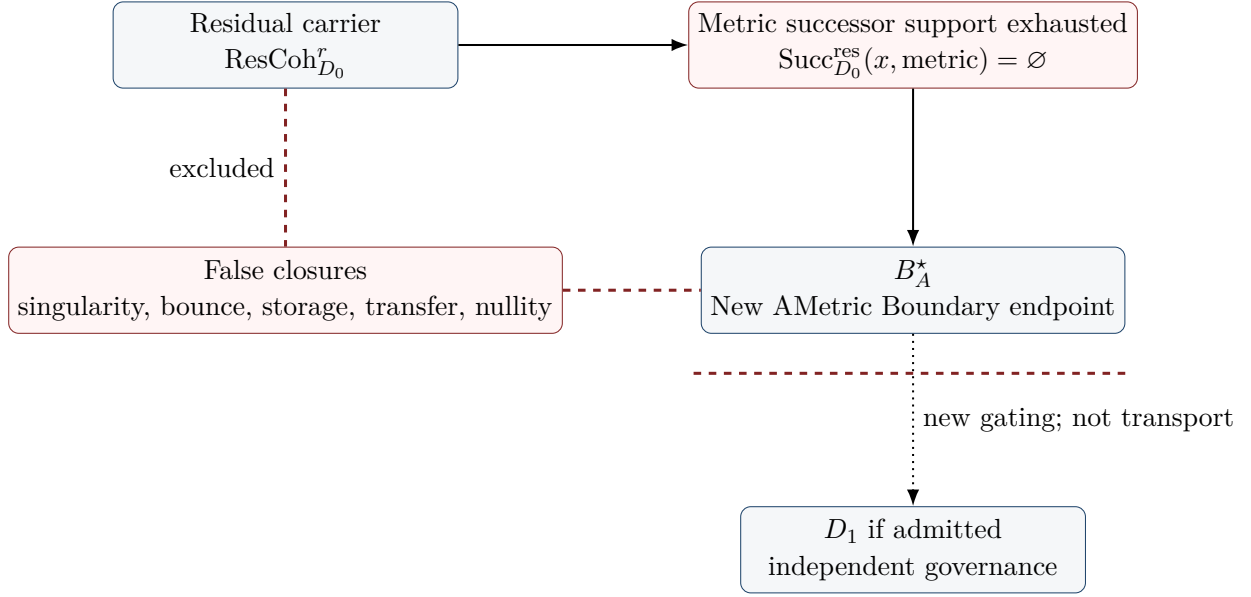


Figure 2: Closure square. False endpoints do not mediate between residual metric exhaustion and the New AMetric Boundary. Any post-boundary domain is separated by the no-inheritance lock.

55 Counterexample burden for the endpoint theorem

A same-type counterexample to Theorem 48.1 must simultaneously:

1. preserve a nonempty unresolved horizon-isolated residue R_H after exterior role typing;
2. preserve old-scope standing failure for every burden retained in R_H ;
3. preserve empty residual metric successor support;
4. avoid supplying a genuine exterior witness relation for the burden, new governance, or a larger scope;
5. provide a coherent non-AMetric endpoint classification;
6. support singularity without metric or affine diagnostic authority;
7. support dynamics without time, causality, process, or operatorhood;
8. support storage or information persistence without carrier identity or transport;
9. avoid admissible nullity while not satisfying AMetric pre-admissibility;
10. preserve a cross-boundary identity relation that is not an operator, selector, transport rule, third-domain object, or scope change;
11. retain same-type endpoint force without changing classification or continuation structure.

The burdens are jointly inconsistent. Items 5–9 force the AMetric normal form; item 10 contradicts non-transmissivity; item 11 either changes a standing-bearing fact and introduces governance or changes nothing and is bookkeeping. The counterexample burden is therefore explicit rather than protected by terminology.

Part VIII

Existence Without Inheritance and Post-Boundary Governance

56 What a post-AMetric domain is

The New AMetric Boundary theorem is unconditional within its antecedent. A populated post-boundary domain is a further claim. The theorem therefore distinguishes boundary closure from positive domain instantiation and specifies the governance conditions any positive successor would require.

Definition 56.1 (Post-AMetric domain). A post-AMetric domain D_1 is an admissibility-bearing domain whose governance is fixed downstream of an AMetric boundary and is not inherited from any prior metric regime. The prefix “post” states structural dependence, not temporal succession.

A non-degenerate D_1 must instantiate at least

$$K_1 = (\text{Adm}_1, \text{Stand}_1, \text{Ref}_1, \text{Irr}_1),$$

together with a declared witness family, redescription discipline, continuation relation, and domain-of-application closure. Metricity, time, causality, matter-energy, fields, particles, laws, and information are later roles if D_1 licenses them. They are not smuggled across B_A^* .

The distinction between old and new is not a claim that the two domains must have visibly different mathematics. They could be externally isomorphic in some respects. The distinction is governance ancestry: no standing-bearing map makes the structures of D_1 inherited continuations of structures in D_0 .

57 Origination is external-use language, not formation

The word “origination” is retained because the manuscript addresses the explanatory status of a domain without a prior internal physical state. It must not be read as a process.

Definition 57.1 (Origination, external-use definition). Origination is the theorem-level classification of a post-AMetric domain as newly governed and non-inherited. It does not denote temporal production, causal emergence, dynamical formation, tunneling, branching, or a parent-to-child transition.

The phrase “a domain originates” therefore abbreviates the following ordered dependency:

$$B_A \implies K_1 \implies \text{licensed roles of } D_1.$$

The arrow denotes structural dependence in the theorem, not transport across the boundary. There is no event at B_A by which K_1 is carried or generated from K_0 .

Table 13: Origination language lock.

Misleading formulation	Preferred formulation
A new universe forms inside the black hole.	A post-boundary domain obtains, if independently admitted.
The black hole creates the new universe.	Black-hole residual closure exposes the New AMetric Boundary endpoint.
The child branches from the parent.	No internal genealogy exists; only an external theorem-level association is available.
The parent’s information seeds the successor.	No old-scope information is inherited.
The same laws pass through.	Any later law package is licensed within D_1 and may only recur externally.

58 The post-boundary governance theorem

Theorem 58.1 (Post-Boundary Governance Theorem). *If a non-degenerate admissibility-bearing domain D_1 obtains downstream of B_A^* , then D_1 independently instantiates the governance roles required for target identity, standing-bearing continuation, and irreversibility. It does not inherit those roles from D_0 .*

Proof. A non-degenerate domain requires the kernel by constructional necessity. Cross-domain standing transfer is unavailable at an AMetric boundary. Therefore the roles cannot be inherited from D_0 . If they are supplied by a larger domain, the larger domain is the active governance scope and the absolute no-inheritance theorem object has changed. In the target case, D_1 must be independently gated. $\square$

Corollary 58.2 (No inherited physical stack). *No metric, temporal, causal, matter-energy, field, particle, gauge, thermodynamic, or information role in D_1 is licensed merely by resemblance to a role in D_0 .*

The corollary does not imply arbitrariness. Once D_1 is governed, its roles may be rigidly constrained by its own admissibility and closure conditions. What is denied is genealogy, not internal lawfulness.

59 Two theorem levels: endpoint necessity and positive instantiation

The theorem separates two levels.

59.1 Level I: New AMetric Boundary endpoint

Within the antecedent of Theorem 48.1, New AMetric Boundary status is structurally necessary. It does not depend on the existence of a populated D_1 . The endpoint theorem is complete once the old residual regime has been classified.

59.2 Level II: concrete post-boundary domain

The claim that a populated D_1 is instantiated requires additional positive structure: a domain-admission theorem, physical model, or unrestricted evaluability result that goes beyond fixed-domain endpoint closure. The null-regime theorem denies admissible vacuity; it does not by itself enumerate concrete occupants, laws, metricity, or physical population.

Claim	Modal status	Required basis
Maximum governance-depth has New AMetric Boundary status.	Necessary in target case.	endpoint route exhaustion and normal-form satisfaction
Nothing old-scope crosses.	Necessary.	boundary non-transmissivity
Any admitted D_1 inherits nothing.	Necessary conditional.	no-crossing plus independent target gating
A post-boundary governance domain is structurally possible.	Allowed normal form.	kernel and AMetric boundary architecture
A populated D_1 exists for every target black hole.	Not established here.	positive instantiation theorem or physical model
A specific metric or law appears in D_1 .	Not established here.	D_1 -internal projection theorem

This distinction preserves the exact modal scope of the principal result. The manuscript can state “the endpoint must be a New AMetric Boundary” without converting that modal necessity into the unsupported empirical claim “every black hole produces a populated universe.”

60 Consequences of the no-interior-instantiation theorem

The No Interior-Instantiation Proof Theorem is proved immediately after Absolute No-Crossing. It fixes the positive limit of the AASC endpoint analysis of D_0 . The variable D_1 is retained only to state conditional no-inheritance: if an independently admitted post-boundary domain has existence-status, its governance and contents are not inherited from D_0 . Introducing D_1 does not assert populated-existence or a successor conclusion from parent-domain AASC analysis.

61 AASC endpoint-status derivability and content limits

AASC endpoint analysis of D_0 derives the endpoint without receiving information from beyond it. Endpoint-status follows from exhaustion of old-scope roles.

Theorem 61.1 (Endpoint-Status Derivability). *Within the theorem antecedent, AASC endpoint analysis of D_0 establishes*

$$\text{AASC}(D_0) \vdash \text{Status}(\text{GDepth}^*) = B_A^*$$

from horizon standing break, residual-role exhaustion, singularity-diagnostic failure, null-regime closure, and AMetric normal-form satisfaction. This inference does not require a witness from a post-boundary domain.

Proof. Each premise concerns the domain, definedness, or exhaustion of roles in the old comparison class. The conclusion classifies the closure profile of those failures. No proposition about D_1 -internal content is used. The inference is therefore boundary-facing rather than crossing-facing. $\square$

Level	What is licensed	What is not licensed
AASC endpoint analysis of D_0	endpoint-status classification from old role exhaustion	proof of D_1 contents or population
D_1 -internal, if admitted	its own standing, laws, information, and energy roles	inherited reference to D_0
AASC theorem-level or third-domain	comparison of boundary profiles and origin architecture	conversion of comparison into cross-boundary inheritance

62 Existence-status and populated-existence

Definition 62.1 (Existence-status). A domain has existence-status when its governance admits a standing-positive interior in the structural sense: target fixation, admissibility, standing, reference, irreversibility, quotient stability, and continuation are available under that domain's own kernel. This definition specifies structural domainhood; it does not specify concrete occupants, laws, fields, metricity, empirical population, or cosmological history.

Definition 62.2 (Populated-existence). A domain has populated-existence when concrete internal occupants, laws, fields, metric structure, or other content are instantiated in that domain. Populated-existence is stronger than existence-status and is not established beyond B_A^* by the parent endpoint theorem.

The manuscript explains how existence-status need not be inherited. It does not prove populated-existence across a non-transmissive boundary.

63 The existence-without-inheritance theorem

Theorem 63.1 (Existence Without Inheritance). *If a post-AMetric domain D_1 has existence-status downstream of B_A^* , that existence-status is grounded in its own governance and not in inherited parent content. Formally,*

$$\text{Adm}_1(D_1) = 1, \quad \text{Inherit}_{D_0 \rightarrow D_1} = 0.$$

Proof. Positive domainhood requires D_1 to carry an admissibility-relevant invariant bundle, standing-bearing identity, continuation, and role authority. The Post-Boundary Governance Theorem supplies

these only through K_1 and D_1 -internal closure. The Absolute No-Crossing Theorem excludes every inherited map from D_0 . Hence the domain’s existence-status is admissible and non-inherited. $\square$

The theorem gives an explanation of existence-status in a precise sense. It explains how standing-bearing domainhood can be grounded without a prior metric state, causal predecessor, transported law, or inherited information. It does not infer populated-existence or a concrete occupant solely from the concept of admissibility.

63.1 Why “what came before?” is ill-typed

Before/after is a temporal relation internal to a governed domain. At an AMetric boundary no inherited temporal order is defined. Asking what occurred before D_1 treats D_0 time as if it extended across B_A^* . That is exactly the transport the theorem denies.

An external theorist may order propositions in a proof: first analyze D_0 , then classify B_A^* , then discuss D_1 . This order belongs to the theorem-writing domain. It is not the internal time of either universe.

63.2 Why “what caused it?” is ill-typed

Cause is witnessed dependence across admitted order. A causal relation from D_0 to D_1 would require a common temporal and reference structure. None crosses the boundary. The absence of a parent cause is therefore not an unexplained gap within a causal sequence; causal sequence is not the relevant type at the origin boundary.

63.3 Why this is not creation from nothing

Creation from nothing pictures a temporal operator transforming nullity into something. Both terms are unavailable. Total admissible nullity is structurally inconsistent; mute nullity is AMetric pre-admissibility and not a state; and no creation operator is defined at the boundary.

Corollary 63.2 (No null source). *A post-AMetric domain is not generated from a null regime. Its governance is independently admitted downstream of the boundary.*

64 Explanation of existence without instantiation overclaim

The explanatory theorem is architectural. It states the form of non-inherited existence:

1. no prior metric or causal state authorizes the first standing-bearing state of the domain;
2. no parent information, law, or identity crosses the boundary;
3. the domain’s governance supplies its own identity-bearing admissibility;
4. any later physical roles are projections within that governance.

This is stronger than saying “a domain has no known cause” and weaker than claiming a particular universe is physically instantiated. It identifies the dependency order any such universe must satisfy.

Theorem 64.1 (Existence Explanation Theorem). *A post-AMetric domain, if admitted, requires no inherited parent content in order to possess existence-status. Its domainhood is explained by its own governance admission rather than by prior causal production or information transfer. The theorem does not establish populated-existence beyond $B_A^\star$.*

65 Origin-condition analogy to the realized physical interior

The positive theorem concerns the endpoint condition, not a proven D_1 . Let B_A^{ours} denote the AMetric starting condition under which the realized physical interior is analyzed. The New AMetric Boundary endpoint is comparable to that condition because both satisfy the same independently fixed boundary normal form.

Theorem 65.1 (Origin-Condition Analogy). *The endpoint condition satisfies*

$$B_A^\star \sim_{\text{Origin}} B_A^{\text{ours}}.$$

The equivalence is restricted to role-profile: pre-metricity, non-transmissivity, non-operatoriality, absence of inherited standing, and fresh governance as the only admissible source of later standing-bearing structure. It implies no common dimensionality, constants, laws, matter content, metric signature, history, observers, or populated successor domain.

Proof. Both boundary profiles satisfy the AMetric normal form independently of any later contents. Their equivalence is therefore a theorem-level comparison of admissibility roles. Since no content-bearing map crosses $B_A^\star$, the comparison cannot license an inference to a universe beyond it. $\square$

Corollary 65.2 (No symmetry or recurrence-law inference). *Origin-condition analogy does not establish a symmetry exchanging D_0 with D_1 , a recurrence law, cyclic cosmology, universal domain-production rule, or the proposition that every New AMETRIC Boundary admits a populated successor. It establishes only normal-form homology.*

Definition 65.3 (Origin-condition recurrence). Origin-condition recurrence is the structural fact that a downstream governed regime can reach a closure-status whose role-profile matches the AMetric starting condition of admissible domainhood. It is not temporal repetition, cyclic cosmology, causal reproduction, or evidence of a populated successor.

Corollary 65.4 (Origin-Condition Recurrence). *At maximum governance-depth, the black-hole endpoint is an origin-condition recurrence: old metric governance reaches a closure-profile structurally homologous to the pre-metric admissibility condition of the realized physical interior.*

The analogy is substantive precisely because it stops where no-crossing requires. It explains the recurrence of the origin condition while refusing the incoherent claim that AASC endpoint analysis of D_0 can inspect or establish what is internal to any new admission.

66 Why baby-universe language is only an external comparator

Baby-universe terminology is high-risk because its standard uses often suggest topology change, branching, causal production, weak coupling, global-state continuity, or information storage. The present result denies those interpretations at the New AMetric Boundary. The term may be retained only after reclassification.

Definition 66.1 (Baby-universe comparator). “Baby universe” is an optional external label for the possibility of independently governed post-AMetric domainhood. The label does not assert that such a domain is instantiated, populated, causally produced, information-bearing from D_0 , or derivable from AASC endpoint analysis of D_0 .

Table 16: Reclassification of baby-universe language.

Standard reading	AASC endpoint verdict
A child spacetime buds from a parent.	No dynamical budding occurs at an AMetric boundary.
Information passes into the child.	Old-scope information inheritance is impossible.
A wormhole throat connects parent and child.	Shared geometry belongs to a larger governed scope, not absolute AMetric closure.
The child preserves global unitarity.	Parent unitarity requires a parent exterior witness relation or a declared larger scope before closure.
The child inherits laws or constants.	Any law package is licensed independently in D_1 .
Post-AMetric no-inheritance domainhood is considered conditionally.	Comparator use only; no proof of a populated child spacetime.”

Proposition 66.2 (Baby-Universe Comparator Reclassification). *The New AMetric Boundary theorem does not prove a baby universe. If the term is used, it refers only to a possible post-AMetric domain whose association with D_0 is external theorem-level comparison and whose contents remain not derivable from parent-domain endpoint analysis.*

Readers who reserve “baby universe” for causally or topologically connected offshoots can omit the term entirely. The theorem does not depend on it.

67 Existence as governance stabilization

Within an admitted domain, the phrase “governance stabilization” names the fixed relation among admissibility, standing, reference, irreversibility, quotient identity, and continuation. It is not a process at the boundary. It is the structure by which an admissible interior can support identity-preserving physical roles.

Definition 67.1 (Existence-status). Existence-status is structural domainhood under a domain’s own governance: target fixation, admissibility, standing, reference, irreversibility, quotient stability, and continuation are available under that domain’s kernel. It is a governance-status claim. It

does not specify concrete occupants, fields, law content, metric realization, empirical population, observers, or cosmological history.

This definition does not reduce existence to a word and does not identify existence-status with populated-existence. It identifies the minimum structure without which claims of objects, laws, events, measurements, or physical states have no stable target. Once that structure is available, the domain may admit metric and physical projections. Without it, a purported occupant is raw presentation.

68 Uniqueness and plurality

The AMetric boundary theorem gives uniqueness of the admissible interior for a fixed domain, not uniqueness of all possible domains. The boundary carries no admissibility-relevant index selecting among multiple positive outcomes.

Proposition 68.1 (No indexed boundary branching). *An AMetric boundary does not select among a pre-indexed family $\{D_1^\alpha\}_{\alpha \in I}$ by an admissibility-bearing parameter α .*

Proof. The index would distinguish positive boundary outcomes and function as a selector or parameter at the boundary. AMetric non-parameterization forbids that role. A larger theory may describe a family of possible domains, but the family and index belong to that larger theory rather than to the boundary itself. $\square$

This result blocks a careless multiverse reading in which the boundary randomly chooses laws. Probability and typicality are downstream structures. If a theory assigns a distribution over possible post-boundary domains, it has declared an external governed scope containing that distribution.

69 The explanatory payoff

The New AMetric Boundary endpoint changes the explanatory question. It does not ask how parent matter is rearranged into a new cosmos. It asks how physical domainhood can obtain without inheriting a prior physical domain. The answer is governance first, physics later.

The black hole matters because it presents a route from within a governed metric description to the exhaustion of that description's residual authority. The endpoint does not carry the parent across. It exposes the same origin architecture that a first domain must satisfy: no inherited metric, no inherited time, no inherited information, no admissible null source, and new standing only through governance.

The principal conclusion is therefore exact:

Existence without inheritance. The New AMetric Boundary is structurally necessary at the target black-hole endpoint. If a positive post-boundary domain obtains, it is existence without inherited information, law, identity, causality, or metric continuity.

Part IX

Energy, First-Law Language, and Conservation-Ledger Residualization

Parent-exterior ledger lock. Nothing in this Part weakens parent-exterior conservation, source closure, mass/flux accounting, charge and angular-momentum ledgers, or black-hole thermodynamics inside their declared scopes. The target is cross-AMetric extension of those ledgers, not their parent-domain validity.

First-law scope distinction. References to first-law language in this Part group three governance-similar but physically distinct ledgers: ordinary thermodynamic exchange accounting, general energy-conservation ledgers, and the black-hole thermodynamic first law. They are not identified as the same physical law. They are grouped only because each requires governed quantity identity, comparison, continuation, and closure conditions.

70 Why conservation is a governance-bearing statement

Ledger	Requires	Boundary verdict
Ordinary thermodynamic first-law language	system boundary, exchange accounting, identified quantities	no cross-AMetric extension
Energy-conservation ledger	quantity identity, temporal or retained-order comparison, closed domain	residualizes, then becomes old-scope ill-typed at B_A^*
Black-hole thermodynamic first law	exterior mass, area/entropy, charge, angular-momentum, work-term ledger	parent-scope valid; no post-boundary backflow

The maxim that energy cannot be created or destroyed is not a primitive inscription that remains meaningful under every loss of physical structure. It is a standing-bearing ledger claim. To say that energy is conserved, one must identify a quantity as the same quantity across an admitted comparison, specify the domain over which exchange is closed, distinguish transport from source change, retain an ordering of stages or consequences, and preserve the metric/source relations under which the quantity is defined. These are governance conditions, not optional commentary.

Definition 70.1 (Energy-conservation ledger). An energy-conservation ledger in a governed domain D is a role package

$$\mathcal{E}_D = (Q_E, \text{Ref}_E, \text{Comp}_E, \text{Ord}_E, \text{Trans}_E, \text{Close}_E, \text{Quot}_E),$$

where Q_E is the energy-bearing quantity or source class, Ref_E fixes its identity, Comp_E compares stages or regions, Ord_E supplies retained consequence or temporal order, Trans_E records transport and exchange, Close_E declares the closed accounting domain, and Quot_E preserves the ledger under admissible redescription.

Definition 70.2 (First-law standing). A first-law statement has standing only in a declared thermodynamic, gravitational, or source-response scope in which each differential quantity, work term, comparison, and closure condition has a licensed role.

The phrase “first law” is used here only at the level of governance status. The ordinary thermodynamic first law, a conservation statement for a closed energy ledger, and the black-hole first law have different physical contents and are not identified as the same physical law. They are grouped only because each requires a governed ledger of quantity identity, comparison, continuation, exchange or work terms, and closure conditions.

Law or ledger	Required role package	AMetric-boundary verdict
Ordinary thermodynamic first law	system boundary, energy/heat/work identities, and exchange accounting	valid in its declared domain; no cross-AMetric extension
Energy-conservation ledger	quantity identity, comparison across stages, source/transport closure, and a closed domain	residualizes with its supports; ill-typed at B_A^*
Black-hole first law	exterior or horizon mass, entropy-area, angular-momentum, and charge work ledger	parent-scope valid; no post-boundary backflow

Thus the familiar black-hole expression

$$dM = T dS + \Omega dJ + \Phi dq_{\text{BH}}$$

is a role-rich exterior or horizon ledger, not a primitive conservation substance. The terms describe mass/source response, entropy-temperature response, rotational work, and electromagnetic work in their declared regime [Bardeen et al., 1973, Bekenstein, 1973, Hawking, 1975, Maley, 2026b]. The relation remains fully available where those roles are governed. It does not automatically license their continuation through a standingless residual carrier.

71 Exterior validity and interior overreach

The New AMetric Boundary theorem does not invalidate exterior energy accounting. ADM-type, Bondi-type, radiative, thermodynamic, and local source ledgers remain valid in the scopes that fix their quantities and exchange rules. An exterior observer may track black-hole mass loss through radiation, horizon work terms, asymptotic charges, and detector records. The error begins only when that exterior ledger is promoted into a cross-boundary identity map.

The distinction can be written

$$\text{Conserve}_{D_0}(e_0) \not\approx \text{Transfer}_{D_0 \rightarrow D_1}(e_0).$$

Conservation in D_0 asserts a relation among D_0 -identified quantities. Transfer to D_1 would require a further standing-bearing map across B_A^* . The first statement does not contain the second.

Proposition 71.1 (Exterior conservation does not generate cross-boundary transport). *No parent-domain conservation law, first-law identity, covariant source equation, or thermodynamic balance equation supplies a standing-transport map across the New AMetric Boundary.*

Proof. Each law is defined relative to a quantity identity, comparison rule, source closure, and domain of application in D_0 . Cross-boundary transfer requires the same quantity to be re-fixed in D_1 . That additional map is not a consequence of the parent ledger and is excluded by AMetric non-transmissivity. $\square$

72 The Einstein source-ledger bridge

The Einstein-projection theorem sharpens the conservation analysis. The source tensor in

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

is not primitive mass-energy substance. It is the metric-facing closed ledger of admitted deformation response, field-carrier mediation, localized persistence classes, inertia and bounded transport, interactions, restoration burden, and vacuum-baseline roles [Maley, 2026c]. Its divergence compatibility expresses closed coherent source continuation inside the licensed metric projection.

Theorem 72.1 (Einstein Source-Ledger Boundedness). *The parent source tensor $T_{\mu\nu}^{\text{closed}}$ retains standing only inside the licensed metric/source projection in which its carrier, identity, comparison, closure, and continuation roles are fixed. At maximum governance-depth it cannot be exported as a parent energy or source ledger across B_A^* .*

Proof. The source tensor is extracted from a governed source-role package and is metric-facing by construction. Its tensor type and divergence condition presuppose the parent metric, compatible connection, source identity, quotient, and continuation envelope. When residual support for that projection exhausts, the same notation cannot preserve the ledger’s identity in a target domain. Export would require a separate standing-bearing source-transport map, which is unavailable at the AMetric boundary. $\square$

Corollary 72.2 (No Stress-Energy Backflow). *The continued writability of $T_{\mu\nu}^{\text{closed}}$ or $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$ inside a classical representative does not imply cross-AMetric energy transfer, storage, source continuity, or post-boundary conservation.*

73 Residualization between the horizon and the boundary

The Einstein-projection and standingless-interior theorems separate closed source-ledger continuation from crossed-content standing [Maley, 2026c,f]. A licensed metric representative may continue to display

$$\nabla_\mu T_{\text{closed}}^{\mu\nu} = 0$$

while the corresponding reference-standing burden remains in the unresolved horizon-isolated residue. Covariant source closure is a condition of the licensed Einstein metric/source projection. It is not a standing-bearing exterior witness relation and does not establish continuation of the same matter, energy, or carrier package through the residue.

The energy role therefore changes status with governance-depth.

Regime	Available support	Status of conservation language
Parent exterior D_0	declared source, metric, thermodynamic, and flux ledgers	fully licensed within the declared scope
Near-horizon residual regime	high residual metric/source organization; local equations remain serviceable	approximately useful residual conservation description, not proof of crossed standing
Intermediate governance-depth	thinning particle, source, calibration, time, and comparison support	governance-deficient ledger; same-quantity claims require explicit qualification
High governance-depth	unstable metric comparison, source identity, and closed-domain exchange	conservation language loses residual support
New AMetric Boundary B_A^*	no inherited metric, time, causality, source identity, transport, or operator domain	“conserved,” “created,” and “destroyed” are ill-typed as old-scope predicates
Possible D_1 , if admitted	independent D_1 governance only	any energy or conservation role is D_1 -internal and non-inherited

Theorem 73.1 (Energy-Conservation Ledger Residualization). *For the unresolved horizon-isolated residue, the sentence “energy cannot be created or destroyed” retains full old-scope standing only while the energy-conservation ledger $\mathcal{E}_{D_0}$ remains governed. As governance-depth increases, that ledger residualizes. At B_A^* , the predicates energy, conserved, created, destroyed, transferred, and exchanged no longer have old-scope conditions of application.*

Proof. The energy claim requires identity, comparison, order, transport, closure, and quotient support. Residue membership records that old-scope standing remains unresolved for the threatened burden. Residual source and metric organization may temporarily preserve approximate ledger use, but each component depends on role-specific successor support. At maximum governance-depth, metric comparison is empty and the associated temporal, causal, source-identity, and transport conditions cannot remain independently licensed. The old energy ledger therefore has no admissible domain at the boundary. $\square$

Remark 73.2 (Loss of standing is not falsity). The theorem does not conclude that conservation becomes false. Falsity would still require a fixed proposition with a licensed comparison domain. The conservation sentence loses the role conditions under which true/false evaluation about cross-boundary persistence is defined.

74 Non-transfer is not nonconservation

Theorem 74.1 (Non-Transfer Is Not Nonconservation). *The relation*

$$D_0 \not\rightarrow_{\text{Energy}} D_1$$

does not imply energy creation, energy destruction, or energy nonconservation. It denies cross-boundary inheritance of the parent energy/source ledger. Creation, destruction, and nonconservation require a shared temporal, quantity-identification, source-comparison, and closure domain that is unavailable at B_A^ .*

Proof. Non-transfer is a statement about absence of an admissible map between domains. Nonconservation is a statement comparing an identified quantity across stages within a governed ledger. Without a common ledger, the second predicate cannot be inferred from the first. Parent-exterior conservation can remain fully valid while cross-AMetric transfer remains undefined. $\square$

75 No creation and no destruction event

The two most immediate misreadings are symmetric. If parent energy does not cross, one might say it is destroyed. If a possible D_1 has energy, one might say energy is created. Both statements import the old dynamical grammar into the AMetric boundary.

Corollary 75.1 (No Energy-Creation Event). *The New AMetric Boundary theorem does not imply that energy is created at B_A^* . Creation requires temporal order, an evolution or production relation, quantity identity, and a domain over which absence and later presence are compared. Those roles are unavailable at the boundary.*

Corollary 75.2 (No Energy-Destruction Event). *The theorem does not imply that energy is destroyed at B_A^* . Destruction requires the same temporal, causal, quantity-identification, and operator structure. Non-transfer is not destruction.*

Formally,

$$\neg \text{Transfer}_{D_0 \rightarrow D_1}(e_0) \not\Rightarrow \text{Create}(e_0) \vee \text{Destroy}(e_0).$$

The antecedent is a no-inheritance result. The consequent contains boundary-inapplicable dynamical predicates.

76 No cross-AMetric energy transfer

Theorem 76.1 (No Cross-AMetric Energy Transfer). *No energy, mass-energy identity, stress-energy class, source balance, thermodynamic work term, or conservation ledger in D_0 transfers across B_A^* into D_1 as the same standing-bearing content. Symbolically,*

$$D_0 \not\rightarrow_{\text{Energy}} D_1, \quad \text{Transfer}_{D_0 \rightarrow D_1}(e_0) = 0.$$

Proof. A transfer claim must identify e_0 in both domains, preserve its equivalence under redescription, and license a common accounting relation. That is a standing-bearing operator from the parent ledger into the target ledger. No such operator is defined at the AMetric boundary. If a third-domain theory defines a joint energy relation, the relation belongs to that larger governed scope and the absolute no-inheritance theorem object has changed. $\square$

Corollary 76.2 (Independent post-boundary energy role). *If a post-boundary domain D_1 is independently admitted and later licenses an energy concept, that energy role is fixed by D_1 's own metric, source, comparison, and closure structure. Numerical equality or formal isomorphism with a D_0 quantity does not establish inheritance.*

77 No first-law backflow

Theorem 77.1 (No First-Law Backflow). *No first-law, conservation, source-closure, or thermodynamic ledger from D_0 can be used to infer the contents, energy balance, population, metricity, or law package of a post-boundary domain beyond B_A^* .*

Proof. Applying a D_0 ledger beyond the boundary requires its variables, reference classes, comparison relation, exchange boundary, and closure assumptions to remain defined in the target. That is cross-domain role transport. Since no such transport is available, the inference is scope backflow. The parent exterior may preserve or balance its own ledger; it cannot use that balance to seed or constrain a domain it cannot standing-bearingly reference. $\square$

The Einstein-projection bridge makes the scope limit explicit. The parent ledger $T_{\mu\nu}^{\text{closed}}$ is already downstream of AASC role licensing. Its covariant closure is not a primitive substance-conservation rule capable of carrying itself across the boundary. The loss of cross-boundary authority is therefore not an exception to GR; it is a consequence of GR’s source side being a licensed metric-facing ledger.

This theorem closes the familiar demand that the energy “must go somewhere.” The word somewhere assumes metric location; must go assumes transport and time; and the same energy assumes reference continuity. At B_A^* , none has inherited authority.

78 Conservation-language route exhaustion

Proposed route	Hidden requirement	AASC classification
Energy crosses the boundary.	common identity and transport map	excluded by non-transmissivity
Energy is created beyond it.	temporal production and before/after comparison	dynamical smuggling
Energy is destroyed at it.	temporal termination and quantity identity	dynamical smuggling
Parent first law constrains D_1 .	cross-domain variable and closure transport	scope backflow
Global energy ledger contains both domains.	larger governed domain with joint comparison	third-domain theory, not absolute closure
D_1 has its own energy role.	independent target governance	conditionally admissible, non-inherited
Exterior energy ledger remains valid in D_0 .	declared parent scope and exchange rules	admitted
Energy statement at B_A^* .	inherited source, time, comparison, and operator roles	ill-typed

Proposition 78.1 (Conservation-Route Exhaustion). *Every conservation-based objection to New AMetric Boundary closure either remains entirely within D_0 , declares a larger common governance scope, imports a prohibited crossing map, or uses creation/destruction language whose role conditions are unavailable at the boundary. No fifth same-type route remains.*

Exterior ledger preservation. The parent exterior may close and test its own energy, flux, and thermodynamic ledgers. It cannot export those ledgers through B_A^* or use them as a backflow principle for post-boundary contents.

79 Distinct No-Inheritance Results

Energy and reference-standing packages are not interchangeable roles. Energy is a source, transport, deformation, or thermodynamic ledger. Bare information-language lacks standing absent a carrier package; a standing-bearing package requires carrier, encoding, equivalence, reference, witness, and reconstruction. Their no-inheritance theorems have parallel form because both require standing-bearing identity and transport, but neither theorem is derived from the other.

Energy-information endpoint discipline. The New AMetric Boundary neither destroys nor preserves parent energy or a parent carrier package. It terminates the old scope's authority to apply transfer, conservation, storage, creation, destruction, and reconstruction predicates across the boundary.

The parent exterior remains responsible for its own ledgers. A standing-bearing exterior witness relation may resolve a parent reference-standing burden, while parent-domain dynamics may balance old energy within D_0 . The parent exterior may close its own ledger without exporting that ledger across B_A^* ; no post-AMetric domain performs that work for it.

Part X

Reference-Standing Reconstruction and the Parent-Domain Burden

80 Information is not primitive

The phrase “information crosses the boundary” is ill-typed unless information has already been given a carrier, encoding, sameness relation, reference class, witness condition, and licensed use. AASC does not treat information as primitive cargo. Bare information-language has no standing relative to the reference-standing problem.

A parent information-bearing package may be written schematically as

$$i_0 = (c_0, e_0, \sim_0, r_0, w_0, u_0),$$

where c_0 is a carrier, e_0 an encoding, $\sim_0$ a sameness relation, r_0 a reference class, w_0 a witness relation, and u_0 the licensed use condition. The package has standing only when those roles are fixed under the parent quotient and continuation discipline.

Theorem 80.1 (Reference-Standing Non-Persistence). *Let i_0 be a parent carrier package whose burden remains in the unresolved horizon-isolated residue R_H . Then*

$$\text{Inherit}_{D_0 \rightarrow D_1}(i_0) = 0$$

for every post-boundary domain D_1 . No branch index, remnant state, bit pattern, global-state coordinate, or post-boundary record counts as persistence of i_0 without a standing-bearing cross-boundary map.

Proof. Persistence requires preservation of the carrier package, not merely similarity of a state or equality of a scalar. The Absolute No-Crossing Theorem excludes the required map at $B_A^\star$. Any burden already resolved by an exterior witness relation is absent from R_H and is not a target of the theorem. For every burden retained in R_H , no inheritance map exists. $\square$

This is stronger and cleaner than saying that information is not “automatically” preserved. The inherited carrier-package relation is false. It is not an event of erasure, destruction, or deletion; those descriptions would require temporal and operator structure.

81 The conventional information paradox retyped

The conventional black-hole information paradox asks whether information is lost, stored behind the horizon, returned in radiation, or preserved in a larger state. AASC retypes the problem:

Does any exterior carrier relation re-fix the declared reference-standing class in the parent reconstruction scope?

This is an exterior reference-standing reconstruction problem. It is not a primitive information-storage problem. The retyping has three consequences.

1. Exterior skin and ledgers remain useful but do not by themselves re-fix a reference-standing class.
2. A standing-bearing exterior witness relation resolves only the burden covered by its explicit target map.
3. The New AMetric Boundary theorem applies to the nonempty remainder R_H , not to burdens already resolved in the parent scope.

Information-language discipline. “Information” may be used as conventional shorthand only after its carrier, reference, witness, quotient, and reconstruction roles have been fixed. Otherwise the term is representational skin or bookkeeping.

82 Exterior witness resolution does not re-govern the residue

Exterior radiation, boundary algebras, detector records, and asymptotic data belong to the exterior reconstruction problem. They do not repair the interior or supply standing to the horizon-isolated residue. If an exterior relation re-fixes a threatened class, it resolves that burden in the parent scope and removes it from the residue.

Theorem 82.1 (Parent-Scope Witness Location). *Any successful parent-scope reference-standing reconstruction is represented entirely within the parent exterior or its declared boundary/reconstruction scope before residue endpoint classification. It is not supplied by a post-boundary domain and does not modify the endpoint status of the remaining residue.*

Proof. Parent standing is defined relative to the parent witness algebra and quotient. A successful relation must land in that algebra and re-fix the target there. A post-boundary domain has no inherited map to the parent. Therefore parent resolution is exterior or boundary-scope resolution, while the unresolved remainder remains the endpoint theorem object. $\square$

Possible standing-bearing exterior witness structures include:

- a radiation algebra together with a stable decoding map to a declared parent reference class;
- a boundary operator algebra with an explicit quotient-preserving reconstruction map;
- an island or entanglement-wedge prescription tied to a fixed target algebra and reconstruction domain;
- a late exterior object with an externally distinguishable map to the threatened class.

The radiation, boundary, island, or late object is not standing-bearing merely because it exists. The map carries the witness role.

83 Radiation is exterior ledger or witness carrier, not interior repair

Hawking radiation is an exterior phenomenon [Hawking, 1975]. As flux, it contributes to exterior mass, energy, entropy, and detector ledgers. It does not supply standing to the interior, repair the horizon-isolated residue, or turn that residue into an ordinary standing-bearing domain.

Radiation affects the residue construction only if a theory supplies a standing-bearing relation from the radiation record or algebra to a declared reference-standing class. Where such a relation exists, the burden is represented in $W_{\text{ext}}(B_H)$. Where no such relation exists, radiation remains an exterior ledger and the burden remains in R_H .

Proposition 83.1 (Radiation role discipline). *Radiation flux alone changes the parent exterior ledger but does not reduce R_H . A radiation-to-reference map may reduce R_H by resolving the mapped burden. Neither role alters the endpoint-normal form of the nonempty remainder.*

This formulation does not depend on whether the geometric horizon is eternal, long-lived, or evaporating. It depends on the reference-standing burden that remains after exterior role typing.

84 Page curves and entropy are not witnesses

Bekenstein–Hawking entropy, generalized entropy, entanglement entropy, and Page curves remain important constraints on parent-domain models [Bekenstein, 1973, Bardeen et al., 1973, Hawking, 1975, Page, 1993]. They do not by themselves re-fix a reference-standing class.

An entropy is defined relative to a state space, algebra, partition, area relation, coarse-graining, or counting measure. Equality or turnover of entropy values can occur without an identity map between reference classes. A Page curve may support an exterior unitarity or reconstruction narrative, but the curve is a ledger profile. The witness, if any, is the carrier algebra and reconstruction map paired with it.

Proposition 84.1 (Entropy and Page-Curve Non-Witness). *No entropy equality, Page-curve behavior, state-counting match, or generalized-entropy extremization is by itself a standing-bearing exterior witness. It reduces R_H only through a separately specified target-fixing reconstruction relation.*

85 Holography and islands: the map, not the slogan

Holographic and island programs are strong candidates for exterior witness structures when they specify boundary or radiation algebras and role-fixed reconstruction maps [Maldacena, 1998, Engelhardt and Wall, 2015, Almheiri et al., 2019, Penington, 2020, Almheiri et al., 2020]. The phrase “encoded on the boundary” has no standing-bearing force by itself. It is skin until the target, carrier, quotient, and map are declared.

Table 21: Holographic and island roles in residue construction.

Item	AASC role
“Information is encoded on the boundary”	skin unless a target map is supplied
Boundary entropy relation	ledger/comparator
Boundary operator algebra without reconstruction map	carrier candidate, not yet witness
Boundary operator algebra with quotient-preserving target map	possible standing-bearing exterior witness
Island formula as entropy result	ledger/comparator
Island reconstruction tied to a declared radiation algebra and target	possible standing-bearing exterior witness

A successful map resolves the burden in the parent reconstruction problem. It does not cross the New AMetric Boundary or re-govern the residue. The map, not the holographic slogan, carries standing.

86 Global-state objections

A larger formal object G may contain labels associated with D_0 and a conditional D_1 . Two tests remain necessary.

First, does G induce a parent exterior witness relation

$$\text{Rec}_G : G|_{D_0} \rightarrow \mathcal{A}_{\text{wit},0}$$

that re-fixes a declared burden? If yes, that burden belongs to $W_{\text{ext}}(B_H)$ rather than R_H . If no, global retention does not preserve parent standing.

Second, does G merely pair a D_0 label with a D_1 label? If so, the pairing is a structure of G . It is not an internal inheritance relation in either domain.

Theorem 86.1 (No Global-Index Rescue). *A global state, branch label, ensemble coordinate, or meta-domain index resolves a parent burden only if it induces a standing-bearing parent exterior witness relation. Pairing a D_0 label with a D_1 label does not create inherited identity across the New AMetric Boundary.*

Proof. Without an exterior witness relation, the pairing has no image in the parent witness algebra. Without inherited reference in D_1 , it is not information about D_0 internal to D_1 . It remains an object of the larger scope. Treating the index as crossing would internalize third-domain structure without target gating. $\square$

87 Baby-universe comparators

Traditional baby-universe discussions may connect topology change, wormholes, superselection sectors, or ensemble averages with apparent information loss [Coleman, 1988b,a, Giddings and Strominger, 1988, 1989]. Recent work relates baby-universe sectors and spacetime wormholes to black-hole information and Page-curve calculations [Marolf and Maxfield, 2020, 2021]. Such programs declare Hilbert spaces, path integrals, amplitudes, or algebras in which relations among sectors are defined.

They are different theorem objects from absolute post-AMetric no-inheritance. A model with an alpha parameter, shared Hilbert space, replica-wormhole relation, or common path integral has a larger comparison structure. It may be physically valuable, but it is not a case in which the AMetric boundary itself transports the parent carrier package.

Table 22: Baby-universe comparators and reference-standing status.

Model type	AASC classification
Coupled sectors with shared Hilbert space	larger governed scope; possible common-state relation
Wormhole/ensemble sector with alpha parameter	third-domain indexing and comparison structure
Baby universe used as hidden information store	requires inherited map; excluded at New AMetric Boundary
Conditional post-AMetric domain	no inherited parent carrier package; contents not parent-scope derivable

The term “baby universe” remains comparator language. The positive theorem object is the New AMetric Boundary endpoint of R_H .

88 What happens to an infalling qubit?

Let q be an exteriorly prepared qubit entangled with a retained reference R . Before horizon dependence, $[qR]$ has standing in D_0 through preparation records and the retained system.

If a radiation algebra, boundary algebra, island map, or other exterior witness relation re-fixes the threatened distinctions, the burden belongs to $W_{\text{ext}}(B_H)$ and is not part of R_H . The interior has not recovered the qubit; the parent exterior has resolved the declared reference-standing burden.

If no such relation exists, the burden remains in R_H . Residual patterns may remain, but $[qR]$ is not old-scope information inside. At the New AMetric Boundary no carrier or map persists. If a conditional D_1 contains a qubit q' in an externally isomorphic state, there is no admissible relation $q = q'$ or $[qR] = [q'R']$.

Qubit verdict. Exterior witness relations resolve the burdens they re-fix; they do not repair the interior. A qubit burden retained in R_H has no standing-bearing continuation into a post-AMetric domain and is not transported, stored, destroyed, or re-created there as the same carrier package.

89 Reference-standing persistence is not rescued by conservation

A parent conservation ledger has no cross-AMetric application. A reference-standing package cannot be rescued by calling it conserved energy, conserved entropy, conserved Hilbert-space norm, or a globally balanced quantity. Each statement is scope-bound and requires its own identity and comparison ledger.

Theorem 89.1 (No Reference-Standing Rescue by Conservation). *No conservation law defined solely in D_0 or solely in a conditional D_1 establishes persistence of a parent carrier package retained in R_H across B_A^* .*

Proof. Persistence requires a reference-preserving carrier map, not equality or balance of a scalar or state functional. The AMetric boundary supplies no common carrier, quotient, or transport. A larger-scope conservation law belongs to that larger scope and does not by itself induce a parent exterior witness relation. $\square$

90 Parent-domain verdict

The reference-standing reconstruction problem has the following normal form:

1. Skin and ledgers retain their licensed exterior roles but do not resolve a burden by themselves.
2. A genuine standing-bearing exterior witness relation places the burden in $W_{\text{ext}}(B_H)$ and outside R_H .
3. Whatever remains in R_H is horizon-isolated and standing-unresolved.
4. If residual metric authority exhausts for a nonempty R_H , the endpoint-normal form is New AMetric Boundary status.
5. The New AMetric Boundary does not rescue, store, destroy, transmit, or re-encode the retained carrier packages.

The existence thesis is therefore not a disguised information-salvage proposal. Its explanatory force lies in non-inheritance.

Part XI

Program Interfaces, Comparators, and Scope Sorting

91 Exterior resolution, endpoint closure, and post-boundary comparison

Program interfaces are clearest when divided by role location. A proposal may operate in the parent exterior by supplying skin, ledgers, or a standing-bearing witness relation; it may alter or replace the horizon-isolated carrier before maximum governance-depth; it may classify the New AMetric Boundary itself; or it may describe a conditional post-boundary domain under independent governance. These are not interchangeable achievements.

Program	Parent/residue role	At the boundary	Post-boundary status
Hawking radiation	exterior flux ledger; possible witness carrier if mapped	no crossing operator	no inherited parent package
Page curve	entropy/ledger profile; witness only with target map	no endpoint mechanism	no inherited parent package
Energy/first law	parent source, flux, and thermodynamic ledger	no transfer, creation, or destruction predicate	any later energy role is independently governed
Holography	boundary carrier; witness only through a role-fixed map	not an AMetric transport map	no post-boundary inheritance
Islands	radiation-algebra ledger or witness relation, depending on map	cannot add an interior across the boundary	no inherited parent class
Fuzzballs	possible pre-endpoint replacement of the classical carrier	may prevent the residue antecedent	independent later domain only
Remnants	late exterior object; witness only through distinguishability map	storage role unavailable at closure	no old carrier package
Final-state projection	prior declared anchor and map	post-hoc projection is repair	no inherited content
Baby-universe models	common larger scope if coupled	New AMetric Boundary forbids transport	comparator-only post-boundary language

This table is a theorem-location table, not a temporal sequence across the AMetric boundary. “Parent/residue role” identifies structures available before residue endpoint classification; “at” means boundary status; “post-boundary” means structurally downstream under a new gate.

92 General relativity

The present theorem is not a modified Einstein equation and does not treat GR as an external primitive background. It imports the AASC Einstein-projection theorem: within the declared minimal first-response same-scope class, Einstein dynamics is the licensed metric projection of the AASC gravity role [Maley, 2026c]. Schwarzschild and Kerr geometries remain exact solutions inside that projection [Schwarzschild, 1916, Kerr, 1963]. The distinction is between licensed metric continuation and the standing-bearing domain of application for the residue.

AASC permits three statuses for an interior metric expression:

1. **licensed metric role**: the expression belongs to an admitted physical domain with standing-bearing comparison;
2. **residual metric role**: the expression inherits organized comparison after standing break but lacks full same-scope licensing;
3. **skin**: the expression remains writable after residual metric support has exhausted and retains representational rather than endpoint-governing authority.

Proposition 92.1 (GR is licensed and bounded). *The New AMetric Boundary Endpoint Theorem preserves the licensed Einstein projection wherever metric and source role-support remains. It denies only metric self-certification beyond residual support exhaustion. AASC therefore licenses the Einstein projection and bounds the domain of its old-scope authority.*

Proposition 92.2 (Governance completeness, not dynamics completeness). *AASC is governance-complete relative to classical GR's unclassified boundary assumptions: it supplies the admissibility architecture that licenses metricity, imports the minimal Einstein-form projection under the declared predicate, types the source and homogeneous vacuum ledgers, and classifies the endpoint at which metric description loses standing. It does not claim all gravitational dynamics or a complete quantum-gravity model.*

93 Conservation and first-law interface

Covariant source closure and black-hole first-law relations remain valid in their declared parent scopes. Neither

$$\nabla_\mu T_{\text{closed}}^{\mu\nu} = 0$$

nor

$$dM = T dS + \Omega dJ + \Phi dq_{\text{BH}}$$

contains a map into a post-boundary domain. The former is a condition of the licensed Einstein metric/source projection; the latter is an exterior or horizon thermodynamic ledger. Their application to the residue becomes residual and ultimately loses old-scope standing as governance-depth increases.

Proposition 93.1 (Conservation-interface discipline). *A physical model may preserve or modify the parent energy ledger, or embed multiple sectors in a larger governed scope. It may not use the parent first law as an inherited constraint on a post-AMetric domain while retaining the absolute no-crossing theorem object.*

A physical completion may prevent a nonempty R_H from reaching metric-support exhaustion by resolving burdens exteriorly, replacing the carrier, or introducing new governance. The present closure theorem constrains the role such a completion must declare; it does not force every model into the residue antecedent.

93.1 Singularity theorems, censorship, and interior asymptotics

Classical singularity theorems, cosmic-censorship conjectures, and proposed interior asymptotics operate with metric, causal, differentiable, or affine structure. The New AMetric Boundary theorem does not dispute their mathematical content inside those structures. It adds a scope condition: none classifies R_H after residual metric-comparison support has exhausted. A BKL-like, mass-inflation, or other violent metric asymptotic remains a pre-closure metric proposal unless a richer theory independently governs it. The singularity is a projection-internal shadow; New AMetric Boundary closure is the governance-level endpoint when the projection lacks residual support.

94 Semiclassical evaporation and exterior radiation

Hawking radiation supplies an exterior flux [Hawking, 1975]. It may alter the parent mass, energy, entropy, and detector ledgers, and it may serve as a carrier candidate for an exterior witness relation. It does not repair the interior or supply standing to the horizon-isolated residue.

The residue effect is exact:

1. radiation flux without a target-fixing map is an exterior ledger and leaves R_H unchanged;
2. a radiation algebra with a standing-bearing reconstruction map places the mapped burden in $W_{\text{ext}}(B_H)$ and outside R_H ;
3. evaporation may alter the carrier or prevent the maximum-governance-depth antecedent, but that is a physical route declared before residue closure;
4. whatever remains in a nonempty R_H is classified independently of the fact that exterior radiation exists.

Thus the theorem is not a thesis about whether a horizon lasts forever. It is a theorem about the unresolved residue left after exterior role typing.

95 Holography and islands

Holography and islands can supply standing-bearing exterior witness relations when they specify a target class, carrier algebra, quotient, and reconstruction map [Maldacena, 1998, Engelhardt and Wall, 2015, Almheiri et al., 2019, Penington, 2020, Almheiri et al., 2020]. The phrase “encoded on the boundary” is skin until such a map is supplied. An island formula used only as an entropy result is a ledger or comparator; an island reconstruction tied to a declared radiation algebra and target may be a witness relation.

Theorem 95.1 (Map priority). *A holographic or island structure reduces R_H only through a role-fixed exterior witness relation. The relation resolves its target burden in the parent or declared*

boundary scope. It does not transport content through the New AMetric Boundary or repair the residue.

This theorem gives those programs a clean success condition without overreporting a Page curve or boundary slogan as standing-bearing recovery.

96 Fuzzballs and microstate geometries

Fuzzball and microstate-geometry programs may replace the classical interior carrier before the residue endpoint arises [Mathur, 2005]. If the relevant distinctions have exterior or boundary standing, they may reduce B_H or prevent the residual-carrier antecedent. A purely internal microstate label with no exterior witness relation does not resolve a parent burden.

97 Final-state projection

A final-state proposal introduces an endpoint boundary condition [Horowitz and Maldacena, 2004]. If the condition is a declared anchor of the original larger theory and induces a standing-preserving relation, it belongs to that theory before AMetric closure. If it is added after undefined transport to force a desired output, it is selector import. An operator cannot first appear at the AMetric boundary, where operatorhood is undefined.

98 Remnants

A remnant can be a late exterior object with mass, charge, and observable distinctions. Such an object reduces R_H only if an exterior distinguishability map re-fixes earlier reference-standing burdens. Capacity, labels, or hidden internal states do not suffice. A post-AMetric domain is not a tiny parent-domain remnant and cannot carry a parent remnant map.

99 Complementarity and firewalls

Complementarity proposes a relation between exterior and infalling descriptions. For AASC it succeeds only if the correspondence is redescription-stable and fixes the relevant reference-standing burden. A firewall introduces horizon tensor load and must be declared as physical content rather than inferred from inconsistent bookkeeping. Either program may alter the burden or carrier before residue closure; neither is an AMetric crossing relation.

100 So-called baby universes and wormhole comparators

The traditional baby-universe literature is comparator material, not proof authority. Coleman and Giddings–Strominger develop structured quantum-gravitational settings in which wormholes and baby-universe sectors affect couplings or coherence [Coleman, 1988b,a, Giddings and Strominger, 1988, 1989]. Farhi–Guth–Guven explore dynamical creation of inflating universes behind barriers [Farhi et al., 1990]. Marolf and Maxfield connect baby-universe sectors with modern black-hole questions [Marolf and Maxfield, 2020, 2021].

Each program contains non-null governance structure: path-integral rules, amplitudes, algebras, superselection labels, or external observables. The AASC no-inheritance result is not another dynamical baby-universe model. Any shared structure belongs to a declared larger scope and must not be mistaken for crossing supplied by the AMetric boundary.

Table 24: Program sorting relative to residue closure.

Program	Potential AASC role	Incompatible overload
Semiclassical radiation	exterior ledger or mapped witness carrier	mass flux alone resolves reference-standing burden
Holography	boundary carrier plus target map	“encoded” without reconstruction relation
Islands	radiation-algebra ledger or witness map	interior simply added to exterior scope
Fuzzballs	pre-closure carrier replacement with exterior distinctions	hidden labels with no witness
Remnants	late exterior distinguishability map	capacity as standing
Final state	declared anchor and map in larger theory	operator introduced at AMetric boundary
Coupled baby universe	larger governed scope	called absolute no-inheritance
Post-AMetric comparator	conditional independently governed domainhood	parent information or energy repository

101 Astrophysical black holes

The theorem is not currently an observational prediction. Exterior astronomy probes mass, angular momentum, accretion, ringdown, shadows, and radiation. These are standing-bearing exterior response roles. They may test deviations from Kerr or constrain compact-object models. They do not directly determine the governance status of maximum interior depth.

A future physical model could connect residue formation or AMetric closure to observables, but that bridge must be explicit. The present paper constrains interpretation: an observable may reduce R_H only through a declared standing-bearing exterior witness relation. Skin and ledgers remain important but non-governing for the residue endpoint.

102 Quantum gravity

A complete quantum-gravity theory could respond to the residue theorem in several ways:

1. supply exterior witness relations that make R_H smaller or empty;
2. replace the horizon-isolated carrier before metric support exhausts;
3. define a larger domain in which the endpoint has different boundary status;
4. instantiate a conditional post-AMetric domain with no inheritance;

5. reject a kernel, projection, residue, or boundary premise by constructing a same-type counterexample.

The paper does not choose among them. It makes their role commitments explicit. A theory cannot both claim absolute AMetric no-inheritance and retain a common unitary or conservation map carrying parent content into a successor sector. Those are different scope structures.

103 Relation to cosmological origin claims

The existence-without-inheritance theorem has cosmological resonance. It says that standing-bearing domainhood need not be explained by a prior physical state. It does not prove that our universe arose from a black hole or that its early history is a particular AMetric dynamical model.

Proposition 103.1 (Cosmological export). *Any cosmological model that claims a first standing-bearing physical domain without a prior metric or causal regime must satisfy AMetric boundary discipline or declare an alternative governance structure. It cannot use a pre-existing metric, probability, law, or process to authorize its own first admissibility without circularity.*

104 Comparator matrix: what each theory must deny

Alternative claim	Must deny or modify	Resulting classification
A singularity remains after metric exhaustion.	metric governance-dependence or maximum-depth definition	new primitive metric ontology or scope change
Primitive information cargo crosses the boundary.	information non-primitivity, AMetric non-transmissivity, and standing identity	common larger domain or hidden transport
Exterior skin resolves the residue.	no skin promotion	undeclared witness load or bookkeeping
The parent causes a successor.	causality as downstream witnessed dependence	external meta-causation or category error
The endpoint is nothing.	null-regime closure or endpoint comparability	mute nonclaim or contradiction
A bounce occurs.	boundary non-action and operator undefinedness	pre-closure quantum dynamics
A global state preserves everything.	exterior witness requirement	meta-domain retention without parent re-fixation
Every black hole instantiates a populated D_1 .	no interior-instantiation proof	added larger-scope physical theory

Part XII

Objections and Audit Replies

105 Objection 1: The conclusion is definitional

The objection says that the paper defines maximum governance-depth as non-metric and then calls it AMetric. The reply is that maximum governance-depth and AMetric boundary status are independently constructed. Maximum governance-depth is defined by exhaustion of residual metric successor support in the standingless carrier. AMetric boundary status is constructed from pre-fixational invariance, boundary non-action, operator undefinedness, non-transmissivity, and independent target gating. The proof shows governance-equivalence of the resulting profiles after false endpoints and nullity are excluded. The identity is a theorem over role behavior, not a shared word root.

106 Objection 2: The manuscript merely changes the meaning of singularity

No. The classical meanings—geodesic incompleteness, inextendibility, and curvature pathology—are preserved. The manuscript asks whether their metric domain remains instantiated for the crossed target. It concludes that after residual metric comparison exhausts, the singularity predicate lacks a domain of application. This is a scope result, not a redefinition of singularity.

A counterexample must provide a singularity predicate that applies without metric, affine, causal, extension, or comparison structure and still counts as the same physical claim. Merely retaining the word does not meet that burden.

107 Objection 3: A mathematical metric remains coherent even if physical standing fails

As an abstract mathematical object, yes. As a description of the exhausted carrier, no. The paper distinguishes formal coherence from target-bearing role authority. A map, manifold, or tensor expression can remain internally consistent while failing to represent the declared physical target. The conclusion is not that mathematicians can no longer write the Schwarzschild interior. It is that the written interior cannot self-certify its application after the governance conditions of metric comparison have failed.

108 Objection 4: Local regularity at the horizon proves interior metricity

Local regularity supports high residual metric coherence near the horizon. It does not prove unlimited residual successor support. The geometric horizon and maximum governance-depth are distinct. The theorem predicts no local discontinuity at crossing. Its conclusion concerns the limit of the old role architecture, not the local horizon neighborhood.

109 Objection 5: Stress-energy conservation keeps matter and information intact

The equation $\nabla_\mu T^{\mu\nu} = 0$ is a metric source-ledger relation. It does not provide a standing map for a threatened reference class. Matter identity and information require additional carrier, witness, and continuation roles. At New AMetric Boundary closure, even the metric context of the conservation equation has exhausted for the crossed target. The equation may remain formal in the representative; it cannot force inheritance into D_1 .

110 Objection 6: The boundary must be at the singularity

“At” is already metric localization. The theorem does not identify a radial or temporal locus. One may draw the New AMetric Boundary closure where a classical diagram draws a singular boundary, but the coincidence is representational. The AMetric boundary is the failure of metric localization, not a special metric point.

111 Objection 7: If the boundary is not a place, a new universe cannot be related to the black hole

Correct internally. The relation is external and theorem-relative. The black-hole analysis exposes the closure-status; it does not provide an internal causal or spatial bridge. Calling D_0 the “parent” is therefore shorthand for external association, not a relation available within either domain.

112 Objection 8: This is just creation from nothing

Creation from nothing posits nullity as a source and creation as a process. The null admissible regime is unavailable, and no process occurs at the AMetric boundary. The account is new governance without inherited source, not dynamics acting on nothing.

113 Objection 9: The null-regime theorem does not prove a new domain exists

Correct. The manuscript proves New AMetric Boundary closure, no-crossing, no interior-instantiation proof, and origin-condition analogy. It does not posit an “origination instantiation principle” that the parent scope could rationally verify. Any concrete post-boundary domain belongs to a further theory or to its own governance. This limit is entailed by the endpoint, not conceded against it.

114 Objection 10: Then the paper does not really explain existence

It explains the structural form of domainhood without parent transport and the recurrence of the AMetric origin condition. It does not claim to enumerate an internal universe. The explanatory result is that admissible existence, where independently admitted, is not grounded by inherited information, energy, law, or metric cause.

115 Objection 11: A larger theory could contain both domains

Yes. Then the larger theory is a third governed scope D_* . It may compare, correlate, or even dynamically relate structures it interprets as D_0 and D_1 . Such a model is not absolute AMetric no-inheritance relative to D_* . The paper's result is scope-locked: no internal relation is inherited across the AMetric boundary in the target comparison class. A larger theory changes the theorem object rather than silently refuting it.

116 Objection 12: A global wavefunction provides the missing relation

A global wavefunction is an object of a declared Hilbert-space domain. It preserves parent standing only if it induces a map into the parent witness algebra. It establishes inheritance in D_1 only if D_1 internally re-fixes the old class. Without those maps, the global index is meta-domain bookkeeping. With them, the model has supplied a larger-scope standing relation; any parent burden re-fixed by that relation is outside the unresolved residue.

117 Objection 13: Information cannot simply fail to persist because quantum evolution is unitary

Unitarity is a property of an evolution operator on a Hilbert space. The theorem denies that an AMetric boundary carries such an operator. Parent-scope unitarity may still be required and must be represented by parent exterior witness structure or a declared larger theory. One cannot use a post-AMetric domain as a hidden unitary output space without reintroducing the cross-boundary channel the theorem excludes.

118 Objection 14: Saying no information persists is equivalent to information destruction

No. Destruction asserts a temporal map from an information-bearing state to a non-information-bearing state. Non-persistence asserts the absence of a standing-preserving continuation relation. The distinction is analogous to the difference between an undefined operator application and an operator returning a zero value. The AMetric theorem gives undefinedness, not a zero or erasure operation.

119 Objection 15: Baby universes normally preserve information

Some baby-universe models do; others produce effective decoherence or superselection structures. The term is not uniform. This manuscript does not prove a post-AMetric no-inheritance domain or a so-called baby universe. It uses baby-universe language only as an external comparator for possible independently governed domainhood. Coupled baby-universe models remain larger-scope alternatives.

120 Objection 16: A new domain with no inherited laws is arbitrary

The boundary does not select laws arbitrarily because it performs no selection. Any law package in D_1 must arise downstream of its own governance. A larger theory may constrain which domains are instantiated, but those constraints belong to that theory. The AMetric theorem blocks boundary-side arbitrariness and boundary-side determination alike.

121 Objection 17: Without inheritance, why should the new domain resemble ours?

No post-boundary domain is asserted to resemble ours physically. The proven analogy is between B_A^* and the AMetric starting condition of the realized physical interior: pre-metricity, non-transmissivity, and fresh gating. Any claim about later laws or contents is not derivable from AASC endpoint analysis of D_0 .

122 Objection 18: The proposal is unfalsifiable

The structural claims have explicit counterexample burdens. A same-type counterexample could show:

- a metric singularity applicable without metric comparison;
- an admissible operator at a boundary prior to standing;
- old-scope identity transported into an independently gated domain without a common larger scope;
- an admissible null regime with no governance content;
- or a non-AMetric endpoint classification changing neither scope nor role support.

The paper argues these constructions are impossible under the stated premises. Rejecting a premise is also a legitimate route. The empirical instantiation claim is separately not made.

123 Objection 19: Governance is philosophy, while GR is physics

Governance here is not administrative or psychological. It names the structural conditions for target fixation, licensed comparison, continuation, and repair exclusion. Every physical claim already uses those conditions. GR supplies metric dynamics after a domain and its comparison roles are fixed. The two operate at different levels; calling one philosophical does not remove its dependency role.

124 Objection 20: An infalling observer measures the interior, so metric standing remains

An observer, apparatus, calibration, record, and target identity are all governed roles. The standingless-interior theorem does not deny that a local apparatus model can remain residually coherent. It denies that the local interaction automatically supplies same-scope standing after old

reference transport has failed. A new interior governance could license measurement, but that is new scope, not inherited ordinary interiority.

125 Objection 21: The equivalence principle forbids a governance break at the horizon

The standing break is not a local force or discontinuity. It concerns old-scope reconstruction, not local metric experience. The horizon can be locally smooth while leaving some declared burdens horizon-isolated and unresolved. Governance-depth develops after the break and need not correspond to local drama at the horizon.

126 Objection 22: The new AMetric boundary contradicts the no-horizon-as-AMetric-reset theorem

Only if it is identified with the geometric horizon. The present theorem explicitly denies that identification. The horizon is derivative and initiates standing break. The New AMetric Boundary closure boundary is exposed only when residual metric comparison exhausts. They occupy different roles and dependency levels.

127 Objection 23: Residual coherence could continue indefinitely

That possibility is outside the maximum-governance-depth antecedent. The paper does not prove that every residual carrier reaches metric exhaustion. It proves the closure-status if exhaustion occurs. A physical theory may show endless residual support, evaporation before exhaustion, or replacement by new governance. Such results refine the antecedent rather than refute the conditional theorem.

128 Objection 24: A singularity is exactly the exhaustion of metricity

A singularity can be an exhaustion signature while a metric articulation remains sufficient to state geodesic incompleteness or curvature pathology. Maximum governance-depth is stronger: residual metric comparison itself has no successor support. A diagnostic within a failing metric is not identical to the failure of the metric role's applicability. The former is representational; the latter is governance-level.

129 Objection 25: The paper overreads the null-regime theorem

The null-regime theorem is used narrowly. It does not prove a concrete universe. It blocks total vacuity from serving as an admissible same-type endpoint and classifies mute nullity as AMetric pre-admissibility. That is exactly the step needed to identify the closure profile. The manuscript does not infer occupancy from the null theorem alone.

130 Objection 26: A probability distribution could choose the new domain

Probability is a measure on an already governed sample space. It cannot carry boundary authority at the AMetric interface without a licensed carrier, events, and measure. A larger meta-domain may assign probabilities to models, but the selection occurs in that meta-domain, not at the boundary itself.

131 Objection 27: Quantum tunneling can cross classically forbidden boundaries

A classically forbidden region is still defined in a quantum dynamical theory with states, amplitudes, and an evolution or path-integral rule. An AMetric boundary is prior to that structure. Quantum tunneling is therefore not a counterexample unless quantum theory is posited as a deeper admissibility-bearing regime, in which case the boundary is not AMetric relative to that regime.

132 Objection 28: The paper denies all continuity, so there is no reason to associate D_1 with the black hole

The association is explanatory and external: the proof reaches B_A^* through analysis of the horizon-isolated carrier in D_0 . No internal continuity is inferred. Scientific theories routinely distinguish a derivational relation between descriptions from a physical relation between represented objects. The paper is explicit that the black hole exposes rather than causally produces the origin status.

133 Objection 29: If D_1 has no backward relation, the claim is empirically empty

The post-boundary domain itself may be empirically inaccessible from D_0 . The theorem still constrains parent-domain model building: it forbids using such a domain as an information repository, causal continuation, or inherited-law endpoint. It also fixes what a quantum-gravity proposal must add to avoid the conclusion. Structural consequences can be significant even when the post-boundary object is not observable from the parent.

134 Objection 30: The term existence is too broad

The paper defines standing-bearing existence for its target and explicitly denies a universal semantic monopoly. The concept is appropriate because the target is a domain whose physical objects, records, laws, and continuations are claimed to have stable identity. Other logical notions of existence may remain. The theorem concerns the governance required for a physically articulated domain.

135 Objection 31: An external observer could assign a relation, so no-backward-relation is false

The no-backward theorem excludes internal standing-bearing relations. An external observer or meta-theory can assign labels and comparisons. Those assignments belong to the external domain and do not become memories, causes, or inherited information inside D_1 . The manuscript repeatedly distinguishes comparison from genealogy.

136 Objection 32: The theory silently assumes that every description needs standing

Only descriptions used for identity-preserving commitment, continuation, or reconstruction need standing in the target sense. Raw syntax, simulations, and exploratory models can exist without it. But a claim that an interior singularity, information class, or baby universe is the same physical target is standing-bearing by its own intended role. The theorem does not burden inert notation; it burdens claims of physical continuation.

137 Objection 33: The result is anthropocentric because witnesses sound observer-dependent

A witness is any standing-bearing structure that fixes distinctions for downstream use: detector data, boundary observables, flux integrals, correlations, or stable records. No human observer is required. The witness discipline is structural, not psychological.

138 Objection 34: Black-hole thermodynamics implies hidden microstates that must persist

Thermodynamic entropy is a ledger. A microstate interpretation requires a state space, counting map, equivalence relation, and relevance to the declared reconstruction target. Even a successful microstate count does not establish cross-AMetric inheritance. It may instead support a richer exterior witness relation, in which case the mapped burden is removed from the unresolved residue.

139 Objection 35: No information persistence makes black-hole entropy meaningless

No. Entropy can remain a parent-domain thermodynamic and exterior-response quantity. The theorem denies only its promotion into a cross-boundary information carrier. The first law and area relations remain valid in their scope.

140 Objection 36: The closure boundary is merely another name for a quantum-gravity regime

A quantum-gravity regime normally has operators, states, amplitudes, or observables. Those are post-admissible structures. The AMetric boundary is the interface prior to them. A quantum-gravity theory may govern a region that classical GR treats as endpoint, but then it supplies new scope and can be tested against the classification. The boundary itself is not a physical theory.

141 Objection 37: The paper cannot say “nothing crosses” because that is itself a relation

The statement is made in the external theorem language. It denies the existence of an admissible internal transport map. This is no more contradictory than saying a partial function is undefined on an input. The meta-level can classify non-definedness without making the absent map exist.

142 Objection 38: The argument relies on an equivocation between undefined and negative

It does not. Standing is bivalent on the evaluated fragment. Undefinedness means the item is outside the operator domain or evaluated fragment. At New AMetric Boundary closure, transport is undefined, not standing-negative output of a transport operation. This distinction is central to the no-destruction result.

143 Objection 39: A negative standing value could cross even if positive standing cannot

Crossing a negative status is still transport of a classification. The boundary does not transmit old-scope gate values. A candidate in D_1 is evaluated under D_1 ; it does not arrive carrying D_0 rejection or acceptance. Otherwise D_0 would govern the target gate.

144 Objection 40: The theory predicts endless universe production and is therefore extravagant

The fixed-domain theorem does not assert that every black hole instantiates a positive successor domain. If a separate physical theory adds that premise, it must also specify whether New AMetric Boundary closure is reached and how non-degenerate admission is instantiated. The manuscript supplies neither a count nor a branching rate.

145 Objection 41: The New AMetric Boundary conclusion is circular

The endpoint is not called AMetric merely because metricity has failed. Part II fixes an independent normal form containing non-action, operator undefinedness, non-transmissivity, no selector, no parameterization, independent target gating, bivalence, no repair, and null-route closure. The endpoint theorem verifies those clauses one by one. A non-metric endpoint retaining probability, order, dynamics, or a common transport map would not satisfy the normal form.

146 Objection 42: The word “must” secretly reintroduces dynamics

“Must” is modal route exhaustion, not temporal production. It says that no same-type fixed-domain endpoint alternative remains once metric, dynamical, informational, storage, wormhole, and null routes have been closed. The theorem never asserts an event in which the boundary forms.

147 Objection 43: Moving baby-universe language downstream concedes that the result is not about baby universes

The terminological restriction is deliberate. The primary theorem is the New AMetric Boundary endpoint, while baby-universe language is an optional comparator. For usages restricted to causal or topological offshoots, the phrase does not apply. The endpoint, no-crossing, and existence-without-inheritance theorems are unaffected.

148 Objection 44: The manuscript proves only a conditional normal form, not existence

It proves two distinct results. The endpoint normal form is necessary in the stated black-hole case. A populated D_1 remains a positive instantiation claim requiring additional theory. The existence theorem is conditional but substantive: if D_1 is admitted, its standing-bearing existence is grounded in its own governance and not in inherited parent content. This is an explanation of non-inherited existence, not an empirical census of successor universes.

149 Objection 45: A third-domain theory defeats absolute no-crossing

A third domain can compare, index, or jointly represent D_0 and D_1 . That fact is explicitly permitted. It defeats absolute no-crossing only if its relation is silently internalized as standing-bearing inheritance in one of the compared domains. If the third domain is instead declared as the active common governance scope, the theorem object changes from absolute AMetric separation to a coupled larger-scope theory.

150 Objection 46: A downstream paper must reprove AASC

No. The manuscript is an AASC-internal extension theorem. Its imported dependencies are explicitly locked. A critic may reject those results at their source or show that the present application misuses them. Requiring every downstream paper to rederive the framework confuses dependency analysis with foundational repetition.

151 Objection 47: The first law forbids a no-inheritance endpoint

The first law is a governed ledger in a declared thermodynamic and source-response scope. It does not contain a transport map across an AMetric boundary. The parent exterior may preserve or radiatively balance its own mass-energy ledger; it cannot use that ledger to infer or seed post-boundary contents.

152 Objection 48: If energy does not cross, it must be created in the new domain

Creation requires temporal order, quantity identity, an operator or production relation, and comparison between absence and presence. Those roles are unavailable at B_A^* . A possible D_1 energy role would be internal to D_1 , not parent energy created by a boundary event.

153 Objection 49: If energy does not cross, it is destroyed at the boundary

Destruction is the symmetric category error. It requires the same inherited dynamical and ledger conditions. The theorem says the parent transfer predicate is false and the conservation grammar loses cross-boundary application; it does not describe a destruction process.

154 Objection 50: Origin-condition analogy proves a universe beyond the boundary

It does not. The analogy is between boundary role-profiles. It establishes that B_A^* is structurally homologous to the AMetric starting condition of the realized physical interior. It supplies no witness of post-boundary contents, population, or laws.

155 Objection 51: If nothing crosses, endpoint-status derivability is itself a crossing relation

Endpoint status is deduced by AASC analysis of old-scope role exhaustion, not from a signal arriving from beyond. The analysis establishes horizon standing break, residual metric exhaustion, singularity-diagnostic failure, null-regime closure, and normal-form satisfaction. It cannot establish D_1 -internal content. This is analogous to proving that an operator is undefined from its domain conditions rather than receiving a value from beyond the domain.

156 Objection 52: Denying an interior-universe proof is a retreat

It is a consequence of the stronger theorem. If the boundary is genuinely non-transmissive, an internal-content conclusion from parent-domain AASC analysis would contradict it. The endpoint remains structurally necessary; what is denied is an incoherent cosmography across that endpoint.

157 Objection 53: Origin-condition recurrence secretly asserts cyclic cosmology

No. Recurrence is structural homology of boundary profiles, not temporal repetition, causal return, or universe reproduction. The theorem contains no cycle, shared time, or transport relation.

158 Objection 54: A third-domain energy law can still balance both domains

A larger theory may define a joint ledger. Then the joint relation is internal to that larger governed domain, and the absolute D_0 -to- D_1 no-inheritance theorem object has changed. The larger ledger is not a map supplied by B_A^* .

159 Objection 55: If parent-domain endpoint analysis cannot establish the successor, why discuss it?

D_1 is a conditional variable used to state the no-inheritance result. The theorem says that if an independently admitted post-boundary domain has existence-status, its governance and contents are not inherited from D_0 . Quantifying conditionally over D_1 does not assert populated-existence or a content conclusion from parent-domain AASC analysis.

160 Objection 56: Origin-condition analogy is too weak to matter

It is the maximum positive result compatible with AMetric non-transmissivity. The analogy identifies the exact structural role-profile of an origin condition while proving that no inherited content can cross. A stronger parent-scope claim about post-boundary contents would contradict the boundary theorem rather than improve it.

161 Objection 57: Energy conservation itself forces a relation between the domains

Conservation is a relation only inside a governed ledger with quantity identity, comparison, source closure, and a common domain of application. It cannot manufacture the cross-boundary identity conditions needed to extend itself. Parent-exterior conservation remains valid; cross-AMetric ledger inheritance does not follow.

162 Objection 58: The manuscript only says that GR fails at the singularity

No. Classical relativity already contains metric-internal singularity diagnostics. The Einstein-projection bridge first establishes that GR is the licensed minimal metric projection of AASC gravity. The endpoint theorem then classifies the exhaustion of the residual authority of that projection. A singularity is an equation- or projection-internal diagnostic; New AMETRIC Boundary closure is the governance-level normal form when the diagnostic framework itself no longer has support.

163 Objection 59: If energy does not transfer, what balances the parent ledger?

The parent ledger is balanced, modified, or left open entirely within D_0 by whatever D_0 -licensed flux, radiation, mass, source, and thermodynamic relations the physical model supplies. A post-boundary domain is not an admissible balancing term. The endpoint theorem is neutral on which parent-scope accounting model is physically correct; it blocks only the use of D_1 as a hidden cross-AMetric sink or source.

164 Objection 60: If nothing crosses, origin-condition analogy is itself an inherited relation

The analogy is a theorem-level comparison of independently specified boundary normal forms in the AASC metalanguage. It transports no standing-bearing content between D_0 , B_A^* , and any conditional D_1 . Normal-form homology is external classification, not inheritance, memory, causation, or genealogy.

165 Objection 61: Existence-status is just populated-existence under another name

No. Existence-status is structural domainhood under a domain's own governance: target fixation, admissibility, standing, reference, irreversibility, quotient stability, and continuation are available under that domain's kernel. Populated-existence concerns concrete internal occupants, metric realization, fields, law content, history, empirical population, or physical occupancy. The manuscript explains existence-status without inheritance and denies parent-scope proof of populated-existence beyond B_A^* .

166 Objection 62: Boundary-status derivability smuggles a value back from beyond the boundary

No. Endpoint-status is derived from exhaustion of old-scope role support, not from a witness, signal, operator, or value received from beyond B_A^* . It is analogous to proving that a partial operator is undefined outside its domain from the operator's domain conditions; no value from outside the

domain is being imported. The analysis determines the boundary profile from the failure of D_0 -roles, while post-boundary contents remain unavailable.

167 Objection 63: The parent ledger must balance by using the post-boundary domain

The parent ledger balances, radiates, evaporates, or remains incomplete only inside D_0 under whatever source, flux, mass, charge, angular-momentum, thermodynamic, or semiclassical rules the parent theory supplies. A post-boundary domain is not an admissible balancing term for the parent exterior. Using it that way would be first-law backflow across an AMetric boundary.

168 Objection 64: The manuscript rejects general relativity

It does the opposite within the declared target. The imported Einstein-projection theorem derives the Einstein equation as the unique minimal first-response same-scope metric projection of the AASC gravity role. The endpoint theorem then bounds that licensed projection. GR remains authoritative wherever its metric and source roles retain standing; it does not self-certify continuation after those roles exhaust.

169 Objection 65: Lovelock already proves the result

Lovelock's theorem is a metric-internal classification under a specified geometric and differentiability envelope. Lovelock-style uniqueness results are comparator orientation only; the AASC bridge imports projection-standing, source-ledger, and endpoint-scope discipline, not merely metric uniqueness. The AASC Einstein bridge is consistent with that comparator but carries a different burden: it licenses metricity, fixes the projection scope, types the source ledger, blocks metric backflow, and classifies higher-response or boundary additions. Lovelock does not prove the black-hole New AMetric Boundary endpoint or no-crossing result.

170 Objection 66: Higher curvature or quantum gravity can repair the endpoint inside the same scope

Only if the added structure is declared. Higher-curvature terms, nonlocal operators, new fields, torsion, metric-affine nonmetricity, quantum states, leakage channels, and endpoint conditions are higher-response, mixed-projection, witness-bearing, effective-scope, or richer-scope content. They may prevent the endpoint antecedent, but they do not silently totalize the old Einstein projection after standing failure. Thus the paper classifies modified and quantum-gravitational routes; it does not deny them as declared larger-scope theories.

171 Objection 67: Penrose diagrams prove an interior singular endpoint

Penrose diagrams display the conformal causal structure of a classical metric representative. The Einstein-projection bridge fixes that representative as a licensed projection rather than primitive ontology. At maximum governance-depth the diagram remains a valid diagram of the classical envelope, but it does not represent the governed endpoint after residual metric support exhausts. Continuing the diagram's singular boundary as physical authority is metric backflow.

172 Objection 68: Stress-energy conservation guarantees continuation of the same content

No. The equation $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$ is a closed source-ledger condition inside the licensed Einstein metric/source projection. It does not supply a standing-bearing exterior witness relation for horizon-isolated reference classes and does not export the parent source ledger across an AMETRIC boundary. Source closure and old-scope identity continuation are distinct claims.

173 Objection 69: Calling AASC more complete than GR is an overclaim

The manuscript claims governance completeness, not dynamics completeness. AASC is more complete only with respect to admissibility, standing, projection authority, and endpoint-status classification. It supplies the upstream admissibility architecture that licenses metricity, imports the minimal Einstein-form projection under the declared predicate, types the source and homogeneous vacuum ledgers, and classifies the boundary of metric authority. It does not claim all gravitational dynamics, numerical constants, or a complete quantum-gravity model.

174 Objection 70: If black holes radiate, the residue theorem fails

No. Radiation is exterior. As flux it changes the parent energy and thermodynamic ledgers. It reduces the unresolved residue only if a standing-bearing radiation-to-reference map is supplied for a declared burden. The theorem applies to whatever nonempty residue remains after that exterior resolution. It is not a thesis about literal horizon permanence.

175 Objection 71: Exterior reconstruction repairs the interior

No. A standing-bearing exterior witness relation resolves its target burden in the parent reconstruction scope. It does not act on the horizon-isolated carrier, restore its metric authority, or modify the endpoint-normal form of the remaining residue. Exterior witnesses resolve burdens; they do not repair the interior.

176 Objection 72: Page curves are standing-bearing witnesses

Not by themselves. A Page curve is an entropy profile over a declared state or algebra. It may support a reconstruction argument, but it does not identify the target reference class or supply the quotient-preserving map. The witness, if any, is the carrier algebra and reconstruction relation paired with the curve.

177 Objection 73: Holographic language is automatically a witness

No. “Encoded on the boundary” is skin until the target, carrier, quotient, and reconstruction map are specified. A boundary operator algebra with a role-fixed target map may be a standing-bearing exterior witness. The map, not the slogan, carries standing.

178 Objection 74: Governance should ignore exterior skin

Governance does not discard skin. Skin can be indispensable for exterior dynamics, calculation, diagrams, gauge control, and bookkeeping. Governance role-types it so that descriptive utility is not laundered into endpoint authority. At endpoint standing, skin is role-demoted rather than ignored.

179 Objection 75: The residue language merely renames the older qualifier

No. The older “uncertified permanent horizon” language mixed horizon duration, exterior reconstruction status, and interior endpoint burden. The residue formulation separates them. It first identifies the declared burden B_H , then removes the portion genuinely re-fixed by exterior witness relations, and only then classifies the nonempty remainder R_H . This changes the theorem object from a qualified horizon to a role-defined residue.

180 Classification of objection routes

Every objection falls into one of five classes:

1. reject a kernel or AMetric premise;
2. deny that residual metric support exhausts;
3. supply an exterior witness relation, carrier replacement, or larger scope before residue closure;
4. confuse external representation with internal inheritance;
5. demand concrete instantiation from a closure theorem that does not claim it;
6. treat the licensed Einstein projection as primitive ontology or use it to self-certify endpoint continuation.

These are legitimate points of disagreement once made explicit. What is not legitimate is moving among them while retaining the old theorem label. The classification ledger requires each alternative

to declare its scope and role load. Rejection of the imported framework belongs at the source theorem; criticism here must identify a failure of endpoint derivation.

Part XIII

Result

181 Final theorem synthesis

The AASC-internal black-hole endpoint chain is now fixed in residue and endpoint-normal-form terms.

1. The AASC gravity role is faithfully projected into the licensed minimal Einstein metric regime,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

2. The geometric horizon is a derivative metric/causal boundary inside that projection, not an AMETRIC origin boundary.
3. Horizon isolation generates a declared parent reference-standing burden B_H .
4. Exterior skin and ledgers retain their licensed descriptive and dynamical roles but do not resolve a burden by themselves.
5. A genuine standing-bearing exterior witness relation resolves only the burden fixed by its target map; the unresolved remainder is

$$R_H = B_H \setminus W_{\text{ext}}(B_H).$$

6. For every burden retained in a nonempty R_H , old-scope standing transport is undefined and retained organization has role-indexed residual-coherence status.
7. At maximum governance-depth, residual metric comparison exhausts for the residue.
8. Singularity diagnostics fail because a singularity is a diagnosis within metric articulation, while this endpoint is the loss of the residual authority of metric articulation.
9. Remnant, storage, bounce, wormhole, primitive information-transfer, and dynamical endpoint routes fail because their carrier, metric, temporal, causal, and operator conditions are unavailable.
10. Admissible total vacuity is structurally unavailable; mute nullity is AMetric pre-admissibility rather than a competing endpoint.
11. The endpoint satisfies the independently fixed AMetric boundary normal form. It is therefore a New AMETRIC Boundary by structural necessity, not by dynamical formation.
12. Nothing old-scope carried by the residue crosses: no standing, reference, carrier package, energy/source ledger, metricity, law, causality, time, field identity, operator, or memory.

13. Conservation and first-law language residualize with their governance supports. At the boundary, conservation, creation, destruction, and transfer are old-scope predicates without a cross-boundary domain of application.
14. Because nothing crosses, AASC endpoint analysis of D_0 cannot establish the contents or populated-existence of a post-boundary domain. Endpoint-status remains derivable from old-scope exhaustion.
15. The positive theorem is origin-condition analogy: B_A^* has the same structural role-profile as the AMetric starting condition of the realized physical interior.
16. Any post-boundary domain, if independently admitted, would possess fresh governance and no inherited parent content. That conditional domainhood is existence without inheritance.

182 What the manuscript changes

The result does not turn black holes into information conduits or cosmological production mechanisms. Nor is it an external statement that general relativity simply fails. AASC first licenses the Einstein metric projection under the declared first-response same-scope predicate and then classifies the boundary of that projection's authority.

The residue formulation also changes the black-hole reconstruction grammar. The theorem is not about literal horizon permanence. It is not about exterior radiation recovering an interior. It is not about primitive information cargo. Exterior skin describes and exterior ledgers account; neither governs the endpoint. Standing-bearing exterior witness relations resolve only the burdens they re-fix. The theorem object is the nonempty horizon-isolated remainder.

The conventional question asks where information or energy goes, what singular object lies at the center, or which dynamics replaces it. The AASC-native question is different: which reference-standing burdens are genuinely resolved in the parent scope, which remain in R_H , and what endpoint roles remain available when the licensed metric projection loses residual authority?

183 Boundary of the result

The paper does not prove universal production of a nonempty residue, universal governance-depth attainment for all astrophysical black holes, or populated-existence beyond the boundary. It proves the endpoint-normal-form result for a nonempty unresolved horizon-isolated residue under the metric-support exhaustion antecedent. That limitation follows from the same no-crossing theorem that gives the result its force.

Nor does the paper deny exterior general relativity, black-hole thermodynamics, covariant source closure, Hawking radiation, holographic models, island prescriptions, or larger quantum-gravity scopes. Each remains available in its declared role. Skin and ledgers remain useful. A genuine exterior witness relation may reduce or eliminate R_H . What none of those structures may do without changing scope is transport old-scope identity, energy, law, or carrier standing through an AMetric boundary.

184 AASC-internal dependency lock at conclusion

This result is an AASC-internal extension theorem. The kernel, AMETRIC boundary, Einstein-projection bridge, horizon isolation, residual coherence, matter-energy residualization, and null-regime closure are imported as binding dependencies. The present paper’s independent contribution is exterior role typing, residue construction, endpoint synthesis, and no-inheritance classification. Criticism of an imported dependency belongs at its source; criticism here must identify a failure in the residue or endpoint derivation.

185 Closing statement

For a nonempty unresolved horizon-isolated residue, the black-hole interior does not terminate as singularity, remnant, bounce, storage, primitive information sink, energy sink, or admissible nullity. At maximum governance-depth its closure-status is a New AMETRIC Boundary. “New” is relational to D_0 ’s governance architecture, not temporal, spatial, or generative. Nothing old-scope crosses. No parent conservation law, entropy profile, radiation flux, diagram, or slogan supplies a hidden bridge.

The positive result is narrower than interior cosmography: the endpoint condition is structurally analogous to the AMETRIC starting condition of the realized physical interior. The analogy is normal-form homology, not a symmetry or recurrence law. Existence-status, where independently admitted, is structural domainhood under fresh governance. It is not inherited from prior carrier packages, energy, law, time, or metric dynamics, and it does not specify occupants, fields, metricity, observers, or empirical population.

Final formulation. Skin describes; witnesses resolve; residue closes. Exterior skin remains useful but non-governing at endpoint standing. Standing-bearing exterior witness relations resolve only the burdens they re-fix. If the unresolved horizon-isolated residue is nonempty and residual metric authority exhausts, its endpoint-normal form is New AMETRIC Boundary status.

A Formal dictionary

Term	Definition in this manuscript
Admissibility	Fixed-domain classification distinguishing licensed from unlicensed standing-bearing use.
Standing	Reuse-stable admissibility: eligibility for identity-preserving commitment, continuation, or later use.
Reference	Governance role fixing what target remains the same under lawful redescription and continuation.
Irreversibility	Exclusion of same-act repair after standing or admissibility failure.
AASC gravity role	Constraint architecture required by an admissible physical interior: standing surface, observable quotient, lawful redescription, first-class closure, and multiplier-type presentation.

Term	Definition in this manuscript
Licensed Einstein projection	Faithful four-dimensional Lorentzian projection of the AASC gravity role whose unique minimal first-response same-scope equation is $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$.
SameScopeMetric ₁	Predicate fixing the minimal local, symmetric, metric-internal first-curvature response class with no undeclared anchor motion, higher-response load, nonlocality, source laundering, or endpoint repair.
Einstein source ledger	$T_{\mu\nu}^{\text{closed}}$ as the metric-facing closed ledger of admitted source roles, not primitive mass-energy substance.
Metric backflow	Use of a downstream metric equation or representative to authorize, repair, or re-license the governance architecture that made the metric projection admissible.
AMetric boundary	Pre-metric, non-eventful, non-operatorial, non-transmissive interface at which standing eligibility is fixed for a domain.
Geometric horizon	Derivative metric/causal boundary inside an already governed metric projection.
Threatened class	Old exterior reference class whose declared continuation becomes dependent on horizon-isolated structure.
Horizon-generated burden B_H	Declared family of reference-standing burdens threatened by horizon isolation.
Standing-bearing exterior witness relation	Target-fixing exterior carrier relation preserving quotient identity, redescription stability, and licensed re-fixation.
Unresolved horizon-isolated residue R_H	$B_H \setminus W_{\text{ext}}(B_H)$, the burden remaining after genuine exterior witness resolution.
Horizon standing break	Undefined old-scope standing transport for every burden retained in R_H .
Skin role-demotion	Retention of descriptive and calculational utility without endpoint-governing authority.
Information non-primitivity	Bare information-language lacks standing unless backed by carrier, encoding, reference, witness, quotient, and use conditions.
Residual coherence	Non-standing persistence of role organization inherited from a pre-break governed regime.
Residual successor support	Post-break continuations retaining enough organization for a role to remain descriptively usable without restoring standing.
Governance-depth	Ordering of residual successor-support thinning; not a metric scalar or temporal variable.
Maximum governance-depth	Status at which residual successor support for the metric role is empty.
Metric exhaustion	Loss of residual metric-comparison role authority, not necessarily failure of syntax.
Equation-level failure	Singularity, high-curvature pathology, inextendibility, or source incompleteness diagnosed inside a still-operative metric projection.
Projection-level exhaustion	Loss of residual metric/source/comparison authority of the licensed Einstein projection for crossed content.
Singularity displacement	Inapplicability of singularity diagnostics after metric role exhaustion; not a dynamical removal process.
AASC vacuity	Absence of every admissibility-relevant invariant, identity carrier, transport rule, quotient, anchor, tensor, gate, and continuation role.
Null admissible regime	Contradictory candidate that is both admitted and totally AASC-vacuous.

Term	Definition in this manuscript
New AMetric Boundary endpoint	Maximum-depth closure-status satisfying the independently fixed AMetric boundary normal form.
No-crossing	Absence of any admissible inherited standing-transport map across the AMetric boundary.
No-inheritance	Failure of parent standing, reference, law, metric, carrier package, or causality to persist into a post-boundary domain.
Post-AMetric domain	Independently governed domain admitted downstream of an AMetric boundary. “Post” is structural, not temporal.
Baby-universe comparator	Optional external label for possible independently governed post-AMetric domainhood; it asserts no internal universe.
Backward relation	Internal standing-bearing relation in either domain that re-fixes, remembers, constrains, or inherits the other domain.
External analogy	Third-domain comparison of role architecture that creates no internal relation between compared domains.
Existence-status	Structural domainhood under a domain’s own governance: target fixation, admissibility, standing, reference, irreversibility, quotient stability, and continuation are available under that domain’s kernel; it specifies no concrete occupants or empirical population.
Existence without inheritance	Existence-status grounded in a domain’s own governance rather than cross-boundary content transfer.
No interior-instantiation proof	Parent-scope inability to prove post-boundary contents or population because proof would require crossing.
Origin-condition analogy	Theorem-level equivalence of AMetric boundary role-profiles without content inheritance.
Energy-conservation ledger	Governed identity, comparison, order, transport, closure, and quotient package required for conservation claims.
Conservation-ledger residualization	Loss of old-scope standing for the energy ledger as its metric/source/time supports thin.

B Proof dependency graph

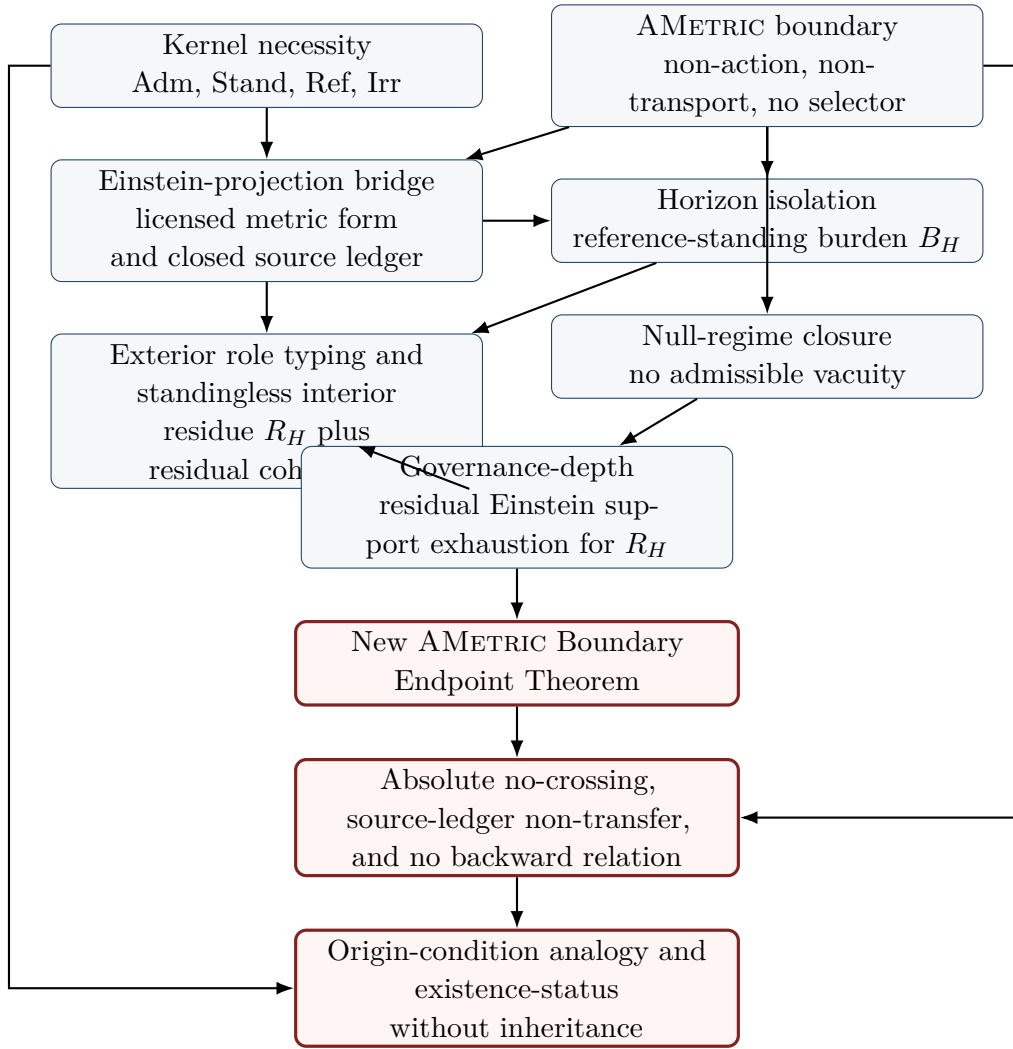


Figure 3: Proof dependency graph. The endpoint synthesis imports the kernel, AMetric boundary, Einstein projection, horizon obstruction, standingless-interior, matter-energy, and null-regime theorem clusters, then constructs the unresolved residue through exterior role typing. The Einstein bridge establishes the licensed metric object whose residual authority later exhausts.

C Theorem inventory and claim status

Result	Status	Primary dependency	Counterexample burden
Licensed Einstein projection	Imported theorem	AASC gravity role + SameScopeMetric ₁ + closed source continuation	non-Einstein minimal first-response survivor without extra load

Result	Status	Primary dependency	Counterexample burden
No GR self-certification	Imported / derived	projection order and no metric backflow	downstream metric equation re-licensing its own governance
Licensed projection failure	New bridge theorem	Einstein projection + residual metric exhaustion	primitive GR ontology independent of projection standing
Horizon non-origin	Imported / derived	AMetric no-reset; metric dependency	same-scope horizon licensing governance without scope change
No Skin Promotion	New theorem	ATS role typing	skin acquiring witness authority without new carrier relation
Exterior Witness Resolution	New theorem	target-fixed exterior witness map	interior repair inferred from exterior resolution
Residue construction	New theorem object	$R_H = B_H \setminus W_{\text{ext}}(B_H)$	qualified horizon substituted for role-defined remainder
Residual metric exhaustion	Imported / derived	standingless interiors	metric descriptor with no residual comparison support
No singularity without metricity	New theorem	singularity role + metric exhaustion	non-metric singularity preserving same claim type
Singularity as metric-internal shadow	New theorem	Einstein bridge + projection exhaustion	singular endpoint after loss of diagnostic role
No AMetric remnant	New theorem	no-crossing + information non-primitivity	storage identity without carrier/map
No destruction event	New theorem	boundary non-action	destruction without time/operator
No AMetric bounce	New theorem	operator undefinedness	dynamics prior to admissibility
No black-hole null endpoint	Imported application	null-regime closure	admitted total vacuity
New AMetric Boundary endpoint	Principal new theorem	all above	non-AMetric endpoint satisfying no closed role structure
Absolute no-crossing	Principal new theorem	AMetric non-transmissivity	inherited map without common gate or meta-domain
No backward relation	New theorem	no-crossing	internal relation available in either separated domain
Post-boundary governance	Derived	kernel + cross-domain gating	non-degenerate domain without kernel
Einstein source-ledger boundedness	New bridge application	closed source ledger + no-crossing	cross-AMetric stress-energy identity without map
Existence without inheritance	Principal new theorem	no-crossing + new governance	standing-bearing existence requiring prior content

Result	Status	Primary dependency	Counterexample burden
Universal baby-universe instantiation	Not proved	requires physical bridge	populated D_1 for every target black hole

D Modal-status ledger

Claim	Modal status	Basis
Einstein dynamics is the licensed minimal metric projection.	proved/imported	Einstein-projection bridge
The endpoint theorem rejects GR in its licensed scope.	denied	it bounds projection authority rather than invalidating the equation derivative metric-boundary dependency
The geometric horizon is not the AMetric boundary.	proved/imported	derivative metric-boundary dependency
Old standing is undefined for burdens retained in R_H .	proved/imported	horizon isolation and no-totalization
Residual metric support is empty at maximum governance-depth.	endpoint antecedent	definition by successor-support exhaustion
Singularity is unavailable there.	proved conditionally	diagnostic category no longer supported
Admissible nullity is unavailable.	proved/imported	null-regime closure
The endpoint is a New AMetric Boundary.	structurally necessary	normal-form satisfaction plus route exhaustion
Nothing old-scope crosses.	structurally necessary	non-transmissivity and operator undefinedness
Any admitted D_1 inherits nothing.	necessary conditional	independent target gating
A populated D_1 exists for every black hole.	not proved	requires positive instantiation theorem
The result explains existence without inheritance.	interpretive theorem	governance admission rather than prior causal production

E Endpoint route-exhaustion matrix

Route	Required authority	Disposition
Classical singularity	metric/affine diagnostic regime	excluded after metric-role exhaustion
Remnant/archive	identity-bearing carrier and retrieval map	excluded without inherited standing
Information sink	information object plus temporal loss process	inheritance false; destruction ill-typed
Information-transfer baby universe	cross-boundary identity map	excluded by AMetric non-transmissivity
Bounce/tunneling	time, process, evolution operator	excluded at non-eventful boundary

Route	Required authority	Disposition
Wormhole/throat	shared metric or path-integral scope	larger-scope theory, not absolute closure
Inherited laws	cross-domain law carrier	target domain requires independent gating
Admissible nullity	admitted positive side with no governance	contradiction
Mute nullity	no evaluative content	AMetric pre-admissibility, no rival endpoint force
New AMetric Boundary	no inherited metric, process, operator, transport, or null closure	surviving endpoint normal form

F Claim-normal-form classifier

Any endpoint sentence can be classified by the following decision procedure.

1. Does the claim preserve an old exterior reference class?
 - If yes, demand a standing-bearing parent exterior witness relation.
 - If no, continue.
2. Does the claim use a metric, causal, temporal, or dynamical endpoint?
 - If yes, demand residual or new metric governance.
 - If support is exhausted, classify the claim as metric backflow or skin.
3. Does the claim invoke information storage, destruction, or transfer?
 - Demand carrier, encoding, identity, and map.
 - If the map crosses B_A^* , classify it as scope transport.
4. Does the claim invoke a null endpoint?
 - If mute, classify it as AMetric pre-admissibility.
 - If admissible, reject it by null-regime closure.
5. Does the claim posit new standing?
 - If mapped to D_0 , it is an exterior witness relation or larger-scope relation.
 - If independently gated, it is a post-AMetric domain.
 - If undeclared, it is hidden governance or repair.

G Program comparison matrix

Program	Carrier/scope	AASC success condition	Failure relative to this paper
Semiclassical evaporation	exterior QFT on curved spacetime	radiation algebra plus target map may resolve burdens	thermal flux alone leaves R_H unchanged
AdS/CFT	boundary Hilbert space/algebra	quotient-preserving exterior witness relation	boundary slogan without target map
Islands	radiation algebra plus QES prescription	explicit witness map for declared burden	entropy result reported as automatic reconstruction
Complementarity	exterior/infalling descriptions	stable correspondence	observer switch without map
Firewall	horizon tensor load	declared source and witness structure	inferred object from inconsistent bookkeeping
Fuzzball	microstate geometry/string domain	exteriorly relevant distinctions	hidden labels only
Remnant	late parent-domain object	exterior distinguishability map	capacity without standing
Final state	endpoint condition in larger theory	declared anchor from start	post-hoc selector
Coupled baby universe	third-quantized or path-integral scope	explicit larger-domain relation	not absolute no-inheritance
Post-AMetric domain comparator	independently gated domainhood, if admitted	own kernel; contents unproved from parent scope	misreported as carrier or energy inheritance

H Claim-status ledger

Sentence	Classification	Reason
The singularity stores a parent carrier package.	Inadmissible in target case.	singularity lacks metric support; storage lacks map
A parent carrier package persists globally in a baby universe.	Insufficient.	requires a parent witness relation and cross-domain identity
Nothing exists after maximum governance-depth.	Mute or contradictory.	mute pre-admissibility or null admissible regime
The New AMetric Boundary forms at a finite proper time.	Role error.	formation and proper time are unavailable at boundary
The black hole causes a new universe.	External shorthand only.	no internal causal relation crosses
A new domain may have the same laws.	Possible independent similarity.	sameness must be external; no inheritance follows
The parent carrier package is destroyed.	Ill-typed at boundary.	destruction requires temporal operator
The parent carrier package is not inherited.	Main result.	no standing map crosses
A post-boundary domain has its own carrier packages.	Conditionally admissible.	requires its own carrier, encoding, and witness
Every black hole has a child universe.	Not proved.	requires instantiation theorem

Sentence	Classification	Reason
Black holes expose origin architecture.	Claimed structurally.	endpoint satisfies the New AMetric Boundary normal form

I Short formal proof

Let B_H be the horizon-generated reference-standing burden in parent domain D_0 .

1. The AASC gravity role has the licensed Einstein projection

$$\Pi_g(A_G) = (M_4, g_{\mu\nu}), \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

2. Exterior structures are typed as skin, ledgers, or standing-bearing witness relations.
3. Genuine exterior witness relations resolve the burdens they re-fix, defining

$$R_H = B_H \setminus W_{\text{ext}}(B_H).$$

4. Assume $R_H \neq \emptyset$. For every burden $x \in R_H$,

$$T_H(x) \uparrow.$$

5. Remaining post-break organization is residual:

$$\text{Stand}_{D_0}(x) = 0, \quad \text{ResCoh}_{D_0}^r(x)$$

for supported roles r .

6. At maximum governance-depth,

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{metric}) = \emptyset.$$

7. The licensed Einstein projection cannot self-certify continuation after that support exhausts.
8. Metric endpoint predicates, including singularity diagnostics, have no old-scope residual domain of application.
9. Dynamics is unavailable because time, causality, and operatorhood require governed metric support.
10. A null admissible endpoint is contradictory; mute nullity is AMetric pre-admissibility.
11. The endpoint satisfies the independently fixed AMetric normal form:

$$\text{Closure}_{D_0}(R_H) =_{\text{gov}} B_A^*, \quad \text{ANF}(B_A^*).$$

12. By AMetric non-transmissivity, no inherited standing-bearing map exists into a conditional D_1 .

13. Any independently admitted D_1 instantiates its own kernel and therefore represents existence without inheritance.

J Counterexample burden appendix

A successful same-type counterexample to the principal theorem must construct all of the following:

1. a fixed parent domain with an AASC gravity role and licensed Einstein metric projection;
2. a non-Einstein minimal first-response same-scope metric survivor without extra anchors, tensor load, nonlocality, or source laundering;
3. a horizon-generated reference-standing burden B_H ;
4. correct typing of exterior skin, ledgers, and witness relations;
5. a nonempty residue $R_H = B_H \setminus W_{\text{ext}}(B_H)$;
6. a standingless horizon-isolated carrier for every burden retained in R_H ;
7. exhaustion of residual metric-comparison successor support;
8. a non-AMetric endpoint carrying same-type classificatory force;
9. a singularity without metric or affine diagnostic authority;
10. dynamics without time, causality, process, or operatorhood;
11. no metric, time, causality, operator, selector, witness, transport, or gate at that endpoint;
12. no admissible nullity;
13. storage or carrier-package persistence without carrier identity or transport;
14. a cross-boundary identity or carrier relation that is not an operator, transport rule, selector, external map, or scope change;
15. a post-boundary domain whose governance is neither inherited nor independently gated.

The burdens are jointly inconsistent under the imported AASC results. The endpoint-profile burdens force the New AMetric Boundary normal form; the inheritance burdens contradict boundary non-transmissivity and kernel necessity.

K Source-use ledger

Source	Role in this manuscript
Maley [2026d]	kernel necessity, standing/admissibility identity, no repair, scope-preserving construction
Maley [2026a]	boundary non-action, operator undefinedness, non-transmissivity, no parameterization, target gating

Source	Role in this manuscript
Maley [2026c]	licensed Einstein metric projection, SameScopeMetric ₁ curvature exhaustion, closed source ledger, no metric backflow, and high-curvature route closure
Maley [2026b]	fixed black-hole scope, horizon isolation, partial standing transport, no totalization, exterior witness discipline
Maley [2026f]	horizon not AMetric reset, residual coherence, governance-depth, metric/matter/time/causal residualization
Maley [2026e]	no admissible null endpoint, mute nullity as AMetric pre-admissibility, no lower null generator
Classical GR references	comparator orientation for metric horizons, singularity diagnostics, Einstein geometry, and black-hole thermodynamics; not AASC proof authority
Lovelock [1971]	metric-internal uniqueness comparator for the Einstein tensor; not the source of the AASC projection theorem
Weinberg [1995], Haag [1996]	field/source-structure comparator orientation; no governance authority in the endpoint proof
Quantum-information and holography references	comparator orientation for exterior carrier and witness programs
Baby-universe references	comparator orientation for coupled, path-integral, and dynamical baby-universe models

L Einstein-projection bridge: short form

The endpoint manuscript imports the following metric bridge from Maley [2026c]:

1. The AASC gravity role is constraint architecture, not primitive curvature.
2. A faithful metric projection maps that role to $(M_4, g_{\mu\nu})$ without metric backflow.
3. The predicate SameScopeMetric₁($E_{\mu\nu}$) fixes the local, symmetric, metric-internal first-curvature response class. It excludes undeclared higher-response and source load.
4. Curvature-response exhaustion gives

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

5. Closed source continuation gives

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0, \quad \nabla^\mu E_{\mu\nu} = 0.$$

6. Divergence compatibility fixes $b = -a/2$.

7. Coupling normalization yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

8. High-curvature and horizon additions require declared higher-response load, a standing-bearing witness relation, lawful leakage, non-sink quotient, or richer scope; they do not silently repair the first-response projection.

Accordingly, the New AMetric Boundary theorem bounds a licensed Einstein projection. It does not begin from an externally assumed GR background and then declare that background inadequate.

M Formalization boundary

A useful mechanization target would include the following typed objects:

- an AASC gravity architecture A_G and faithful metric projection $\Pi_g(A_G)$;
- a typed predicate $\text{SameScopeMetric}_1(E_{\mu\nu})$ and Einstein normal-form result;
- a closed metric-facing source ledger $T_{\mu\nu}^{\text{closed}}$;
- domains D_0, D_1 with distinct gates;
- a partial standing-transport operator T_H ;
- role-indexed residual predicates ResCoh_D^r ;
- residual successor-support sets $\text{Succ}_D^{\text{res}}(x, r)$;
- a predicate $\text{MaxGDepth}(x)$ defined by empty metric successor support;
- endpoint role classifiers;
- a null-regime contradiction schema;
- AMetric boundary-profile equivalence;
- absence of cross-domain inherited maps;
- external analogy distinct from internal relation.

No Lean, Coq, or Isabelle formalization is claimed here. The principal formalization challenge is typing the distinction between external meta-domain comparison and internal standing-bearing relation. A formal system must not allow the meta-level proof relation to be silently internalized as a cross-boundary map.

N Terminology discipline

The following wording rules govern the manuscript.

Avoid	Prefer
The AMetric boundary forms dynamically.	AMetric boundary status obtains or is exposed structurally.
Information is destroyed.	Old-scope information has no standing-bearing continuation.
Information goes into the baby universe.	No old-scope information is inherited; D_1 has its own information roles.
The child is behind the singularity.	A post-AMetric domain is not metrically located relative to D_0 .
The black hole causes a new universe.	Black-hole closure externally exposes the post-AMetric origin normal form.

Avoid	Prefer
Nothing remains.	Admissible nullity is excluded; mute nullity is AMetric pre-admissibility.
GR becomes false inside.	Old-scope metric role authority exhausts for the crossed carrier.
A populated post-boundary universe must exist.	Not proved; the New AMetric Boundary endpoint is forced, while concrete D_1 requires instantiation.
The same laws reappear.	Similar laws, if present, are independent similarity, not inheritance.

O No-instantiation and origin-condition formal normal form

The derivability separation is recorded as

$$\text{AASC}(D_0) \vdash \text{Status}(\text{GDepth}^*) = B_A^*,$$

while

$$\text{AASC}(D_0) \not\vdash \text{Contents}(D_1), \quad \text{AASC}(D_0) \not\vdash \text{Exist}_{\text{pop}}(D_1).$$

The positive comparison is

$$B_A^* \sim_{\text{Origin}} B_A^{\text{ours}},$$

not a transport relation from D_0 to D_1 .

Proposition type	Status	Reason
Endpoint-status proposition	derivable by $\text{AASC}(D_0)$	concerns exhaustion of D_0 roles
Existence-status proposition	conditionally governed	concerns independent admissible domainhood
Populated-existence proposition	not derivable by $\text{AASC}(D_0)$	requires internal content beyond the boundary
Post-boundary content proposition	not derivable by $\text{AASC}(D_0)$	requires crossing reference
Origin-condition analogy	theorem-level provable	compares independently specified boundary profiles
Parent-to-successor genealogy	denied	requires standing transport
Concrete populated successor	not established	no interior-instantiation proof

P Energy and first-law formal normal form

Let e_0 be a parent energy/source class. Then

$$\text{Conserve}_{D_0}(e_0) \not\Rightarrow \text{Transfer}_{D_0 \rightarrow D_1}(e_0),$$

$$\text{Transfer}_{D_0 \rightarrow D_1}(e_0) \Rightarrow \text{Transport}_{B_A^*}(e_0),$$

and

$$\neg \text{Transport}_{B_A^*}(e_0).$$

Therefore

$$\neg \text{Transfer}_{D_0 \rightarrow D_1}(e_0).$$

But

$$\neg \text{Transfer}_{D_0 \rightarrow D_1}(e_0) \not\Rightarrow \text{Create}(e_0) \vee \text{Destroy}(e_0),$$

because creation and destruction are boundary-inapplicable dynamical predicates.

Q Conservation-language route-exhaustion matrix

Route	Required role	Disposition
Energy crosses B_A^* .	identity and transport	excluded
Energy is created at/beyond B_A^* .	time, production, comparison	ill-typed
Energy is destroyed at B_A^* .	time, termination, identity	ill-typed
First law fixes D_1 contents.	cross-domain variables and closure	scope backflow
Joint global ledger.	third-domain governance	larger-scope theorem object
Independent D_1 energy.	D_1 -internal governance	conditionally admissible, non-inherited
Exterior conservation in D_0 .	parent ledger	admitted

R AASC-internal dependency ledger

Table 36: Binding AASC dependencies and their role in the endpoint proof.

Imported theorem	Use in endpoint proof	Status
Kernel necessity	fixes the four governance roles	imported
AMetric boundary theorem	fixes normal form and no-crossing	imported
Einstein-projection theorem	licenses the minimal Einstein-form projection, fixes the closed source ledger, blocks metric backflow, and types high-curvature alternatives	imported
Horizon isolation/no-totalization	fixes old-scope standing break	imported
Standingless interior	fixes residual coherence and governance-depth	imported
Matter-energy residualization	fixes conservation-ledger dependence	imported
Null-regime closure	excludes admitted vacuity	imported
New endpoint synthesis	bounds the licensed Einstein projection and proves AMetric closure, no-instantiation, origin analogy, and source/energy-ledger non-transfer	proved here

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